

# TFPT — Architecture and the $E_8$ Compiler

The two axioms  $\{c_3, g_{\text{car}}\}$ , the derivation map, and the Coxeter–cyclotomic  $D_5 \times A_3 \rightarrow E_8$  construction

Stefan Hamann

Alessandro Rizzo

July 29, 2026 · v5.4 (rev 474)

## What this document is about

The architecture layer: how the two axioms  $c_3 = \frac{1}{8\pi}$  and  $g_{\text{car}} = 5$  build the Coxeter–cyclotomic compiler — the carrier  $C^+ = D_5$ , the family geometry  $\mathbb{P}^1 \setminus \mu_4 = A_3$ , the  $\mu_4$  glue  $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ , the EM fixed point  $\alpha^{-1}$  (with its ablation), and the whole number alphabet 16, 40, 41, 48, 240, 248 as carrier traces.

## The TFPT document set — what is treated where

**Plain language:** TFPT (Topological Fixed-Point Theory) is a small discrete compiler. Two inputs — the seam constant  $c_3 = \frac{1}{8\pi}$  and the carrier rank  $g_{\text{car}} = 5$  — build  $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$  and read off the Standard-Model skeleton, the constants and the scale grammar. The series is **the introduction plus five numbered papers and five companions**, best read in order:

0. **introduction** — reading guide, compiler closure, predictions, the dependency DAG and the single proof ledger (`verification/status_ledger.csv`). (*entry point*)
1. **tfpt\_1\_architecture\_e8** — the two axioms, the derivation map, the EM fixed point  $\alpha^{-1}$ , the  $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$  construction, the QBL theorem chain.
2. **tfpt\_2\_standard\_model** — the SM in one  $\varphi_0$ -ladder formula, flavor from parabolic transport (sheet diamond, dual anchor,  $Q_+$  cohomology), the worked closures.
3. **tfpt\_3\_e8\_audit\_bootstrap** — the seven  $E_8$  slices as an audit raster, the cascade bridge, the Möbius bootstrap.
4. **tfpt\_4\_frontier** — honest status of  $\eta_B$ ,  $m_p/m_e$ , Koide, dark matter, full QG.
5. **tfpt\_5\_redteam** — the adversarial audit: declared attacks, what survives, what each reduces to, and the kill tests.

**Companions:** **R.** **tfpt\_research\_contracts** — the open interfaces  $(v_{\text{geo}}, G_{\text{net}}, F_{\text{transfer}})$  as numbered contracts; **H.** **tfpt\_horizon\_readouts** — Appendix H (unit-system reframe): one seam constant as the universal horizon thermal code, the Nariai/anchor surface, the resummed seam clock; **O.** **origin\_theory** — why no free fundamental number remains (exact core + one honestly-typed cyclic interpretation); **S.** **tfpt\_safeguards** — the verification discipline: every mechanism that defends a claim against coincidence and numerology (status calculus, no-free-pattern, the over-determination map, the firewall, frozen predictions + null model, the independent Wolfram and Lean paths, the red team); **P.** **tfpt\_prime\_front** — research documentation of the prime / zeta line: the seven load-bearing modules on the  $E_8$  glue census (Hecke from geometry through the transport ledger), the sandbox localization program that follows them, and the kill list. Documentation, not a result: it makes no claim about the Riemann Hypothesis.

**You are reading document 1 (Architecture).**

**TFPT status at a glance — read this card the same way in every document**

**Two inputs, one compiler.** From the boundary constant  $c_3 = \frac{1}{8\pi}$  (axiom P1) and the five-slot carrier  $g_{\text{car}} = 5$  (axiom P2) — jointly the elementary symmetric data of one anchor  $a = (1, 1, 2)$  — the discrete compiler  $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$  is *built*, and the Standard-Model skeleton (gauge group, three families, hypercharges, the flavor matrix), the dimensionless constants ( $\alpha^{-1} = 137.0359992$ ) and the scale grammar are read off as fixed points. The algebraic core is machine-checked; physical readouts run through *named bridges* whose status is typed below.

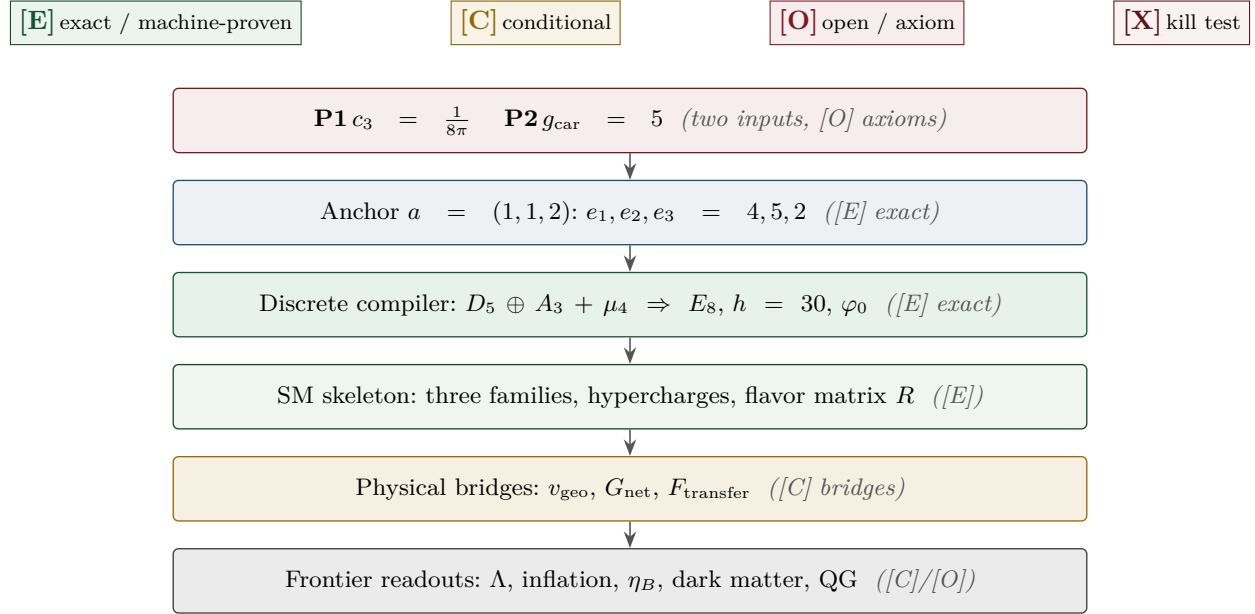
**What is closed, and what genuinely remains.** The discrete/algebraic layer is closed ([E]); the everything-open residual reduces to *three named interface problems*:

| Interface             | Question                                                                                        | Status        |
|-----------------------|-------------------------------------------------------------------------------------------------|---------------|
| $v_{\text{geo}}$      | the one metrology unit ( $= 1/\sqrt{G} = m/\mu$ ); No-Unit Thm: no compiler scale               | primitive [O] |
| $G_{\text{net}}$      | simple-current extension $(D_5)_1 \otimes (A_3)_1 \rtimes \langle (1, 1) \rangle \cong (E_8)_1$ | [C]           |
| $F_{\text{transfer}}$ | one functor, four typed interfaces (Koide, $\eta_B$ , axion, $m_p/m_e$ )                        | [C]           |

**Gravity is parameter-free.** The classical metric-sector field equation is no longer only an  $R+R^2$  readout: the fixed-volume entanglement equilibrium (Jacobson; Faulkner et al.), run with TFPT's atoms, gives the *full* covariant Einstein equation  $G_{ab} + \Lambda g_{ab} = c_3^{-1} T_{ab}$  with *both* coefficients fixed —  $c_3^{-1} = 8\pi$  (no free dimensionless Newton dial; the thermodynamic origin  $2\pi/\eta$  *coincides* with the geometric one  $|\mathbb{Z}_2|2\pi\chi$  via  $|\mu_4| = |\mathbb{Z}_2|\chi(S^2) = 4$ , so  $c_3$  is triply over-determined — anchor, geometry, thermodynamics, v358) and  $\Lambda$  from  $\alpha$  ( $\rho_\Lambda = (3/4\pi^2)e^{-2\alpha^{-1}}$ , v60); the Einstein tensor (not Ricci) is forced by Lovelock, so matter conservation is an output (v359). The residual here is only the equation-of-state status (an external candidate action — Bianconi's entropic action, arXiv:2408.14391 — is quantified in v473 without changing the typing) and the one unit  $v_{\text{geo}}$ ; the global measure (C7 = QG.AMB.01) is now a [C] redundancy (v369), a certification object rather than missing dynamics, conditional on SEAM.EQUIV.01 and Bisognano–Wichmann.

The seam-net structural theorem  $G_{\text{net}}$  (= SEAM.EQUIV.01: the raw seam *is* the holomorphic  $(E_8)_1$  net) now carries two named routes (route split 2026-07-22, no status change in substance): the MMST route SEAM.EQUIV.MMST.01 is *closed modulo cited theorems*; the twistor route SEAM.EQUIV.TWISTOR.01 (Costello–Li, prepared by CELEST.SEAM.01) and the unconditional parent stay [O]. On the MMST route: the explicit lattice model (v367/v368) and the  $S_3$  closure stack pin the target at every computable level — central charge  $c = 8$  (v376), the  $(E_8)_1$  character with 248 currents and one primary (v377), genus-one torus GSD = 1 (v378) and reflection positivity (v379) — and it is Lean-formalised as FORM.SEAM.MMST.01: the collar's MMST hypotheses are kernel-proved, the MMST scaling-limit and Adamo–Moriwaki–Tanimoto OS-reconstruction theorems enter as named cited axioms, and the #print axioms check is clean. The *only* residual of that route that stays [O] is the abstract continuum existence of the scaling limit (v336) — its 128-spinor extension leg since certified at net level by the peer-reviewed crossed-product package ( $h_s=16/16=1 \in \mathbb{Z}$ , the Longo–Rehren locality criterion; Böckenhauer; KLM  $\mu=4/2^2=1$ ; AGT/AMT the independent second witness) with the realisation input reduced to invariant level and the parallel Lean route seamResidualClosed' (v469); so the MMST route is *closed modulo cited theorems*, not solved. QGEO.SYM.01 (the  $\mu_4$  deck acting geometrically) is its *corollary*: a conformal net's vacuum is rotation-invariant by axiom, so the deck follows with no extra premise (v335, Lean FORM.QGEO.BW.01). The nonperturbative quantum-gravity measure QG.AMB.01 is no longer carried as an open item: it is a [C] redundancy (v369), a certification object rather than missing dynamics, conditional on SEAM.EQUIV.01 and Bisognano–Wichmann (and TFPT is moreover rigorously gap-decoupled from the general Euclidean-QG conformal-factor problem, margin  $1.648 > 0$ , v76/v330). So the honest residual is the one unit  $v_{\text{geo}}$  plus the typed  $F_{\text{transfer}}$  interfaces, above the cited-theorem ceiling on the seam and with QG.AMB.01 a [C] redundancy. Everything carries a claim ID resolving to a row of verification/status\_ledger.csv (the single source of truth); **if the text and the ledger ever disagree, the ledger wins.** The historical labels ( $U_{\text{wall}}, G_{\text{metric}}, F_{\text{frontier}}$ ) are retained only for ledger continuity — the live residual is  $v_{\text{geo}} \oplus F_{\text{transfer}}$ , with the seam  $G_{\text{net}}$  closed modulo cited theorems on its MMST route (parent [O]) and QG.AMB.01 a [C] redundancy.

Marker key: [E] exact / machine-proven · [C] conditional (holds under named hypotheses) · [O] open / axiom · [X] falsifiable kill test. Per-claim fine type in verification/status\_ledger.csv.



## Contents

|          |                                                                                                                  |           |
|----------|------------------------------------------------------------------------------------------------------------------|-----------|
| <b>I</b> | <b>Architecture and the derivation map</b>                                                                       | <b>6</b>  |
| <b>1</b> | <b>How to read this document</b>                                                                                 | <b>7</b>  |
| <b>2</b> | <b>Headline: the Pascal compiler on five carrier slots</b>                                                       | <b>10</b> |
| 2.1      | $g_{\text{car}} = 5$ as the Pascal closure . . . . .                                                             | 11        |
| 2.2      | One master moment generates 40, 41, 240, $\delta_2$ . . . . .                                                    | 12        |
| 2.3      | Compact $E_8$ compression and the role of $\nu^c$ . . . . .                                                      | 12        |
| <b>3</b> | <b>The <math>D_5 \times A_3</math> compiler: <math>E_8</math> as a closure, not an input</b>                     | <b>13</b> |
| 3.1      | Carrier = $D_5 = \mathfrak{so}_{10}$ (proved) . . . . .                                                          | 14        |
| 3.2      | Family = $A_3 = \mathfrak{su}_4$ from $\mathbb{P}^1 \setminus \mu_4$ (proved, with the metric settled) . . . . . | 14        |
| 3.3      | The $\mu_4$ glue and the $5/4 + 3/4 = 2$ discriminant theorem (proved) . . . . .                                 | 15        |
| 3.4      | Glue-norm arithmetic, and three readings of 41, 60, 120 . . . . .                                                | 28        |
| 3.5      | What the compiler outputs for the Standard Model and our reality . . . . .                                       | 30        |
| <b>4</b> | <b>Grammar I — the carrier grammar (3 + 2)</b>                                                                   | <b>31</b> |
| 4.1      | Dual-root dictionary (everything from two charges) . . . . .                                                     | 31        |
| 4.2      | Carrier arithmetic (proved identities) . . . . .                                                                 | 32        |
| <b>5</b> | <b>Grammar II — the seed grammar (one decoder <math>u</math>)</b>                                                | <b>33</b> |
| <b>6</b> | <b>Grammar III — the action grammar (exponentials of <math>\alpha_\star^{-1}</math>)</b>                         | <b>33</b> |
| 6.1      | The EM closure: $\alpha_\star^{-1}$ as an explicit fixed point (made explicit here) . . . . .                    | 34        |
| 6.2      | The decuple bridge (new, exact headline) . . . . .                                                               | 38        |
| 6.3      | The full power tower: $x^1, x^5, x^{10}$ on the carrier graph $K_5$ . . . . .                                    | 38        |
| 6.4      | Hubble is the square root of $\Lambda$ (corollary, exact) . . . . .                                              | 38        |
| 6.5      | The $\sim 123$ orders of the cosmological constant . . . . .                                                     | 39        |
| 6.6      | The ladder is binomial; $2g_{\text{car}} = \binom{g_{\text{car}}}{2}$ singles out $g_{\text{car}} = 5$ . . . . . | 39        |
| 6.7      | Discriminant action theorem: the ladder reads the hypercharge discriminant . . . .                               | 39        |
| 6.8      | A two-dimensional action lattice . . . . .                                                                       | 40        |

|                                                         |                                                                                            |           |
|---------------------------------------------------------|--------------------------------------------------------------------------------------------|-----------|
| <b>7</b>                                                | <b>The Einstein normaliser as a corrected tree value</b>                                   | <b>40</b> |
| <b>8</b>                                                | <b><math>\Lambda</math>-anchored metrology closure, and two anchors</b>                    | <b>40</b> |
| <b>9</b>                                                | <b>The landscape: <math>g_{\text{car}} = 5</math> is overdetermined</b>                    | <b>41</b> |
| <b>10</b>                                               | <b>The <math>E_8</math> carrier atlas — stated unambiguously</b>                           | <b>42</b> |
| 10.1                                                    | Level 1 — original $E_8$ data of the series ([E]) . . . . .                                | 42        |
| 10.2                                                    | Level 2 — new exact counting identities ([E]) . . . . .                                    | 42        |
| 10.3                                                    | Level 2.5 — the even-flip transition atlas (a falsifiable program) . . . . .               | 42        |
| 10.4                                                    | Level 2.7 — the carrier Coxeter theorem (the strongest new bridge) . . . . .               | 43        |
| 10.5                                                    | Level 2.8 — positive-root moment: $ R^+(E_8)  = \text{Tr}_{S^+} X^2 = 5!$ . . . . .        | 44        |
| 10.6                                                    | Level 2.9 — the 2·3·5 Coxeter cyclotomic compiler (meta-theorem) . . . . .                 | 44        |
| 10.7                                                    | Level 3 — conjectural geometric readings ([C]) . . . . .                                   | 44        |
| <b>11</b>                                               | <b><math>\mathbb{Z}_6</math> monodromy, flavor and CP readouts</b>                         | <b>45</b> |
| 11.1                                                    | CP as triple seed variance; a closed observable quadrilateral . . . . .                    | 45        |
| <b>12</b>                                               | <b>Mass hierarchy as a hexagon with one extra turn</b>                                     | <b>46</b> |
| 12.1                                                    | Lepton flavor decouple: the charged-lepton determinant carries $u^{10}$ . . . . .          | 46        |
| 12.2                                                    | Why the top Yukawa is $\approx 1$ : the $L = 0$ anchor (Froggatt–Nielsen ladder) . . . . . | 46        |
| <b>13</b>                                               | <b>Inflation interface</b>                                                                 | <b>47</b> |
| <b>14</b>                                               | <b>Further simple bridges</b>                                                              | <b>47</b> |
| <b>15</b>                                               | <b>Closures that the architecture rests on</b>                                             | <b>48</b> |
| <b>16</b>                                               | <b>Status summary and open audit points</b>                                                | <b>48</b> |
| <b>17</b>                                               | <b>Consolidated picture</b>                                                                | <b>50</b> |
| <b>II</b>                                               | <b>The Coxeter–cyclotomic compiler and the explicit <math>E_8</math> construction</b>      | <b>52</b> |
| <b>Theorem 1</b>                                        |                                                                                            | <b>52</b> |
| <b>Theorem 2</b>                                        |                                                                                            | <b>53</b> |
| <b>Theorem 3</b>                                        |                                                                                            | <b>54</b> |
| <b>Theorem 4</b>                                        |                                                                                            | <b>54</b> |
| <b>Theorem 5</b>                                        |                                                                                            | <b>54</b> |
| <b>Theorem 6</b>                                        |                                                                                            | <b>55</b> |
| <b>Theorem 7</b>                                        |                                                                                            | <b>56</b> |
| <b>Theorem 8 — anchor microcode</b>                     |                                                                                            | <b>60</b> |
| <b>Master statement</b>                                 |                                                                                            | <b>60</b> |
| <b>Appendix: explicit <math>E_8</math> construction</b> |                                                                                            | <b>61</b> |

|            |                                                 |           |
|------------|-------------------------------------------------|-----------|
| <b>III</b> | <b>Appendix: ablation of the EM fixed point</b> | <b>63</b> |
|            | <b>Appendix: computational verification</b>     | <b>64</b> |

## Part I

# Architecture and the derivation map

### Abstract

This document develops the *architecture layer* of TFPT: the two axioms, the derivation map, and the explicit  $E_8$  construction. The thesis is that the bridge layer of TFPT is governed by *three nested grammars*: a **carrier grammar** (from  $3 + 2$  come charges, families, hypercharge,  $\Omega_{\text{adm}} = 48$ ,  $b_1 = 41/10$ ), a **seed grammar** (one seed  $u = \varphi_0^{\text{ret}}$  generates the Cabibbo angle, birefringence, the baryon fraction and the reactor angle), and an **action grammar** (the large scale ratios are exponential actions of  $\alpha_\star^{-1}$  with rungs  $1/g_{\text{car}}$ ,  $1$ ,  $2$ ). Two new exact results sharpen the action grammar: the *decuple bridge*  $\rho_\Lambda/\bar{M}_{\text{Pl}}^4 = \frac{3}{4\pi^2} e^{2\delta_{\text{ph}}} [v_{\text{geo}}/(5\beta_{\text{rad}}^2 \bar{M}_{\text{Pl}})]^{10}$  (the cosmological constant is, in essence, the tenth power of the stripped electroweak scale, with  $10 = 2g_{\text{car}}$ ), and the Hubble corollary  $H_0/\bar{M}_{\text{Pl}} = e^{-\alpha_\star^{-1}}/(2\pi\sqrt{\Omega_\Lambda})$ . We also give the exact corrected tree form of the Einstein normaliser  $\xi = \frac{3}{4}/(1 + 9/128\pi^3)$ , the dual-root dictionary that generates all hypercharges and transport cusps from  $y_\pm = \{1/2, -1/3\}$ , the  $\Lambda$ -anchored metrology closure, the overdetermination of  $g_{\text{car}} = 5$ , and the  $E_8$  carrier atlas in three clearly separated levels. Section 3 adds the strongest structural result: the carrier code is the  $D_5 = \mathfrak{so}_{10}$  half-spinor ( $\text{Aut} = W(D_5)$ , order 1920;  $R_4 = \text{Clebsch graph}$ ), the family geometry  $\mathbb{P}^1 \setminus \mu_4$  is an  $A_3$ , and  $E_8$  is the explicit unimodular glue closure of  $D_5 \oplus A_3$  — with the discriminant-form norms  $q(D_5) = 5/4$  and  $q(A_3) = 3/4$  summing to the  $E_8$  root norm 2 and coinciding with the TFPT constants  $\delta_2/\delta_{\text{top}}^2$  and  $\xi_{\text{tree}}$ . Status markers used throughout: **[E]** exact identity, **[E]** Lie/lattice theorem, **[E]** formalised, **[E]** numerical fixed point, **[C]** physical/conditional, **[O]** axiom/open. (All markers are explicit; the legacy **[M]**/**[C]** compatibility aliases have been removed.)

# 1 How to read this document

**Notation convention.** We write  $D_5 \oplus A_3$  for the *root-lattice* direct sum (used in the glue/lattice statements: discriminant forms,  $\mu_4$  glue,  $E_8$  as an even unimodular lattice), and  $D_5 \times A_3$  for the *group/representation* branching (used for subalgebra embeddings and  $E_8$ -rep decompositions). The objects are the same atoms; the symbol marks whether the statement is lattice-theoretic ( $\oplus$ ) or branching/representation-theoretic ( $\times$ ).

Nothing here introduces a new primitive assumption; it reorganises existing outputs and proves the cross-links. The whole bridge layer rests on two inputs,

$$c_3 = \frac{1}{8\pi} \quad (\text{axiom P1}), \quad g_{\text{car}} = 5 \quad (\text{axiom P2}). \quad (1)$$

## Foundational postulates — the two genuine axioms (hardenable, not derivable)

Everything downstream is a theorem *given* two structural inputs that are not reduced further and are honestly flagged as the theory’s axioms:

(P1) Boundary kernel.

A single oriented seam  $\Sigma$  carries a primitive, reflection-positive boundary kernel with unit winding  $[u_\Sigma] = 1$ ; its self-linking normalisation fixes  $c_3 = 1/(8\pi)$ . *Status:* primitive postulate. *Hardenable by:* the Calderón projector / determinant-character chain (the P1 hardening) and the Lean carrier-rigidity development (`TfptCarrier.*`, boundary-polarization and Calderón interfaces).

(P2) Carrier interface.

This splits into an *axiom* and a *theorem*, which must not be conflated:

- **P2<sub>interface</sub> [O] (axiom):** the seam is read out by a five-slot carrier  $g_{\text{car}} = 5$  (3 colour + 2 weak) — a postulate (the readout interface).
- **P2<sub>algebra</sub> [E] (theorem):** *given* the carrier, hypercharge ( $y_- = -\frac{1}{3}$ ,  $y_+ = \frac{1}{2}$ ), rigidity and the family count follow from the idempotent/lattice axioms — machine-verified in Lean (`Hypercharge.lean`, `LatticeRigidityGeneral.lean`, `Rigidity.lean`).

So P2 is an *interface axiom* whose *algebraic consequences* are a *formalised theorem* — “axiom or theorem?” has the precise answer: interface axiom, algebra theorem.

These two are *inputs*: they can be formalised and stress-tested (more Lean, more interface theorems) but not computed away. Every other number in this note —  $N_{\text{fam}} = 3$ ,  $\Omega_{\text{adm}} = 48$ ,  $b_1 = 41/10$ ,  $\alpha_\star^{-1}$ , the flavor matrix, the  $E_8$  glue,  $A_s$  — is a consequence of (P1),(P2).

**Anchor-first refinement.** The two inputs are themselves the elementary symmetric polynomials of the single parabolic anchor  $a = (1, 1, 2)$  (the exponents-at-infinity of the splitting type  $\mathcal{O}(-2) \oplus \mathcal{O}(-1)^2$ ):  $e_1(a) = 4 = |\mu_4|$ ,  $e_2(a) = 5 = g_{\text{car}}$ ,  $e_3(a) = 2 = |\mathbb{Z}_2|$ , with  $c_3 = 1/(2e_1(a)\pi) = 1/(8\pi)$ . The power sums  $p_n(a) = 2 + 2^n$  then give  $|R(E_8)| = p_1 p_2 p_3 = 240$ ,  $\dim E_8 = 240 + (p_4 - p_3) = 248$  and the binary ladder  $p_{n+1} - p_n = 2^n$ . So the input set reduces to  $\{a = (1, 1, 2), \pi\}$ . [`verification/v23_anchor_generator.py`]

## The current reduction state of P2 (2026-07-22) — text consolidation, markers unchanged

The axiom P2 stays *declared* (AX.P2.01); what has changed is how much of it is now carried by theorems. The honest ledger state: P1 ( $c_3 = 1/(8\pi)$ ) and the *four marks* are [E]-anchored (Gauss-Bonnet  $n = 2\chi = 4$ , [`verification/v216_marks_gauss_bonnet.py`]); given the four marks,  $g_{\text{car}} = 5$  follows from the partition  $4 = 2+1+1$  plus the Deligne/Birkhoff-Grothendieck/stability typing ([`verification/v491_p2_partition_corollary.py`], [`verification/v499_p2_weights_deligne_bg.py`]; residual R1, the module identification, [C]); and the clock’s *order*, the  $\mathbb{Z}_8$  tower, the deck’s *class* (a half-period translation) and the *position* half of the alignment bit are theorems ([`verification/v506_seam_clock_rigidity.py`], [`verification/v507_seam_bit_origin.py`], [`verification/v510_seam_bit_freedom.py`]). What remains as carrier input is *one discrete symmetry-lift bit*: flag transitivity of the four marks (marks *and* their two sides indistinguishable,  $V_4 \rightarrow D_4$ )  $\iff \tau = i$  (equivalently: the clock / the CM-fixed half-period / the non-split extension class on the free seam circle) — on the free seam circle bare mark-transitivity is automatic for *every* configuration (the pair-exchanging  $V_4$ ) and no local jet sees the side bit, so the flag form is the sharp discrete statement of the datum and its negation is concretely measurable (odd-doublet split  $(2/\pi) \ln \cot(\delta/2) \neq 0$ )



(`[verification/v512_seam_tau_flag.py]`, SEAM.TAU.FLAG.01). The bit has since survived *ten* side-blind derivation attacks on that scoreboard — free RP/ $\Theta$  existence (the eighth, v521), the exhausted Wick-computable free-plus-twist state class (the ninth, v525) and the first genuinely interacting seam toy (the tenth, v529) — and is now physically *defined* as the *twist-class choice*: facets #14/#15 extend the web to 15 exact equivalences (both directions each), the flip-axis fraction is a gauge-robust order parameter ( $O = \frac{1}{2}$  at  $m = 4$ , 0 elsewhere), and its  $\eta$  holonomy is in principle interferometrically readable (v528, SEAM.BIT.TWISTCLASS.01; the bit remains formal input — a definition replaces no derivation, no marker moves); and it now carries the first *positive* selection datum: reflection positivity, imposed on the interacting mark family, keeps exactly *one* member alive —  $\delta = \pi/2$  with positive coupling, the carrier of that order parameter (`[verification/v534_seam_straddle_cone.py]`, SEAM.STRADDLE.CONE.01; toy-level evidence for a dynamical origin, not a derivation); plus the [C] identifications R1/R2 of v499. In one sentence: **the compiler's dimensionless inputs reduce to four marks (derived from P1-side topology), one discrete symmetry-lift bit — the twist-class choice (flag transitivity  $\iff \tau = i$ ) — and  $\pi$ .** This is a reduction *state*, not an abolition of the axiom: AX.P2.01 remains the declared interface input, and its dynamical realisation (QGeo.REALIZE.01) stays [C]/[O].

**Seed normal form, the factorial spine, and the degree-2 pattern** [E]/[C]  
(`[verification/v106_review_validation.py]`)

Three review-validated sharpenings of the anchor picture. **(1) Seed normal form** [E]. The seed formula is exactly

$$\varphi_0^{\text{ret}} = \frac{|\mu_4|}{N_{\text{fam}}} c_3 + \Omega_{\text{adm}} c_3^{|\mu_4|} \quad \left( \frac{4}{3} = \frac{|\mu_4|}{N_{\text{fam}}}, \quad 48 = \Omega_{\text{adm}} = N_{\text{fam}} \dim S^+ \right),$$

a *linear seam term* plus a *topological fourth-order correction* over all admissible fermion states — a structured anchor readout, not a fitted expression. **(2) Factorial spine** [E]. With  $X = 6Y$  on one 16-state generation,  $\text{Tr } X = 0$  and  $\text{Tr } X^2 = 120 = 5!$  (computed from the explicit charge table); the *same* factorial moment is  $|R^+(E_8)| = \sum (E_8 \text{ exponents}) = 120$ , doubles to  $|R(E_8)| = 240$ , and scales to  $|W(D_5)| = 2^4 \cdot 5! = 1920$  (the Hawking-power fingerprint) — one moment, four projections (hypercharge, roots, Weyl group, horizon). **(3) The degree-2 inventory.** The compiler repeatedly selects *degree two*: the Pascal closure  $2^{g-1} = \sum_{k \leq 2} \binom{g}{k}$  truncates at  $K=2$  with the *unique* solution  $g = 5$  (scan 1..40); the glue norms add to the root norm,  $\frac{5}{4} + \frac{3}{4} = 2$ ;  $c_3 = \frac{1}{2} \times \frac{1}{4\pi}$  (variational half  $\times$  two-dimensional-slice Gauss–Bonnet); the pair sector  $\binom{5}{2} = 10 = \mathcal{A}_\Lambda$ ; the hypercharge moment is a *second* moment; the seam is codimension *two*. *Named hypothesis* [C] (recorded, not claimed): **Quadratic Boundary Locality** — the seam supports only bilinear data, so the truncation  $K = 2$  is *forced* rather than chosen; if proven, the Pascal *selection* (red-team Target B residual, typed [O]/[C]) upgrades to [E] and the chain locality  $\Rightarrow K=2 \Rightarrow g=5 \Rightarrow 16 \Rightarrow D_5 \Rightarrow E_8$  closes the carrier choice. *Audit*: the even-code transition reading  $240 = 16 \times 15$  is exact arithmetic but non-unique ( $240 = 16 \times 5 \times 3$  is the established glue factorisation) and stays recorded, not promoted.

**The Sheet-Pairing Lemma** (`[verification/v109_sheet_pairing.py]`) — **the first exact anchor of the hypothesis.** Character-exact weight-multiset combinatorics of the  $D_5$  spinor weights  $(\pm \frac{1}{2})^5$  gives: (i) **no scalar within a sheet** — the zero-weight multiplicity of  $S^+ \otimes S^+$  is 0, because the odd slot count  $g = 5$  flips the sign parity of  $-w$  (a parity theorem of the five-slot code); (ii) **the scalar lives across the sheets**:

$$S^+ \otimes S^- = \Lambda^0 \oplus \Lambda^2 \oplus \Lambda^4 \quad (256 = 1 + 45 + 210),$$

exact as weight multisets, with zero-mode grading  $(1, 5, 10)$  = the carrier code; (iii) **sheet-diagonal = odd forms**:  $(S^+ \otimes S^+) \cup (S^- \otimes S^-) = 2\Lambda^1 + 2\Lambda^3 + \Lambda^5$  exactly — within-sheet pairings are odd-form-valued, never scalar; (iv) the scalar-bearing tower **tops at  $\Lambda^{2K}$**  with  $K = 2$  — bilinear seam data reach exactly the Pascal-ladder degree (the source-layer transport theorem; document 3). *Consequence* [C]: the seam's single scalar two-point kernel is necessarily a **sheet pairing** (chirality-off-diagonal — the same  $\mathbb{Z}_2$  as the glue chiralities and the branch-kernel selection); the remaining QBL step is to show the Calderón kernel supplies exactly this pairing and nothing beyond.

**The Calderón-sheet selection theorem** (`[verification/v110_calderon_sheet.py]`) — **which involutions certify a scalar kernel.** On  $H = S^+ \oplus S^-$  the certified bilinear data of a Calderón involution  $\varepsilon$  ( $\varepsilon^2 = 1$ ) are the matrix elements  $\langle f, \varepsilon g \rangle$ . The lemma's singlet counts per sheet block,  $(0, 0, 1, 1)$ , give



the exact selection statement [E]:

$$\text{a scalar two-point datum exists} \iff \varepsilon \text{ is sheet-odd}$$

— a sheet-odd involution (swapping  $S^+ \leftrightarrow S^-$ ) certifies exactly  $1 + 1 = 2 = |\mathbb{Z}_2|$  invariant scalar kernels (one per orientation), a sheet-even involution certifies *none*. The kernel count  $2 = |\mathbb{Z}_2|$  matches exactly the *two Lagrangian glues* (swapped by single spinor conjugation): the Calderón scalar channel carries precisely the glue ambiguity, resolved by the same sheet choice. *Anti-overclaim control* [E]: the parity theorem is generic over the odd ladder ( $g = 3, 5, 7$ : within-sheet count 0, cross count  $2^{g-1}$ ) — sheet-oddness forces the *half-spinor relation*  $K = (g - 1)/2$ , **not** the value  $g = 5$ ; the  $g$ -selection stays with the independent rank-8/integrality closure. *Consequence*: the QBL residue splits into two named parts [C] (recorded, not claimed): **(a) interface** — “the seam Calderón involution is sheet-odd” (structurally natural: the one-sided collar/double-cover deck is the *same*  $|\mathbb{Z}_2|$  that halves  $c_3 = \frac{1}{2}(4\pi)^{-1}$ , and exchanging cover sides is sheet-odd by construction); **(b) analytic core** — “the two-point kernel certifies only slot-degrees  $\leq 2$ ”. Proving both closes  $\text{QBL} \Rightarrow K=2 \Leftrightarrow g=5$ .

**The Quadratic-Transport Theorem** ([verification/v111\_quadratic\_transport.py]) — **the transport half of the analytic core is a theorem**. Grade seam transport by Clifford degree in the ten field operators on the five-slot Fock space ( $S = \Lambda^\bullet(\mathbb{C}^5)$ ,  $S^+$  the 16-state code; integer Jordan–Wigner model, spanning claims by full-rank certificates mod  $p$ , valid over  $\mathbb{Q}$ ). *Parity* [E]: all linear words are sheet-odd, all 45 quadratic words sheet-even — code-preserving transport has *even* Clifford degree (the same  $\mathbb{Z}_2$  again). *Minimality* [E]: sheet-even words of degree  $\leq 1$  are the scalars alone. *Generation and completeness* [E]: the 45 quadratics ( $= \mathfrak{so}(10)$ , exactly the  $\Lambda^2$  term of the certified tower  $1 + 45 + 210$ ) generate the whole code from the vacuum *and* span the full operator algebra  $\text{End}(S^+)$  ( $\dim = 256$ ) at word length two — every code operation whatsoever is a word in pair transport, and all higher even-degree elements (e.g. the 210 quartics) are redundant. Hence

$$\text{the transport degree is exactly 2}$$

— degree  $\leq 1$  generates nothing, degree 2 generates everything: a theorem, not a hypothesis. *Anti-overclaim control* [E]: the  $g = 3$  world behaves identically (15 quadratics generate  $\text{End}(S^+) = 16$ ) — the theorem selects the *degree*, not the rank; the  $g$ -selection stays with Pascal closure + rank-8. *Residue after v109–v111* [C]: QBL’s remaining content is two *interface* statements — (a) the seam Calderón involution is sheet-odd; (b’) the certified *state* inventory is the  $\leq 2$ -slot tower. Nothing about transport remains hypothetical.

**The Self-Counting Channel** ([verification/v112\_selfcounting\_channel.py]) — **the Pascal closure is an identity of the certified channel**. For odd  $g$ , negation  $w \mapsto -w$  maps the  $S^+$  weights bijectively into the opposite sheet, so the Cartan-neutral kernels of the certified channel are exactly the pairs  $(w, -w)$  — *one per code state*:  $\#\{\text{neutral kernels}\} = \dim S^+ = 2^{g-1}$  exactly [E]. Grading the *same* neutral set by pair-degree  $m$  gives sizes  $\binom{g}{m}$ ,  $m = 0..K$  — for  $g = 5$  the triple  $(1, 5, 10)$  [E]. Two countings of one set:

$$2^{g-1} = \sum_{m \leq K} \binom{g}{m} \text{ is the channel counting itself}$$

(symbolically verified for all odd  $g = 3..13$ ) — the Pascal *closure* (v2/v108) is not an extra requirement but an identity of the certified channel. *Hodge fold* [E]: the code’s minus-sign grading  $\binom{g}{2j}$  equals the folded binomial  $\binom{g}{\min(2j, g-2j)}$  with  $\min(2j, g-2j) \leq K$  — every code sector appears at pair-degree  $\leq K$ . *Anti-overclaim*: the identity holds for every odd  $g$  and carries no  $g$ -selection; the selection stays with the rank-8 closure  $g + N_{\text{fam}} = 8$  and ladder integrality. *Residue re-type* [C] (recorded, not claimed): clause (b’) loses its closure burden — given the sheet-odd kernel, the channel *automatically* contains one neutral kernel per code state at pair-degree  $\leq K$ ; what remains physical is the single input “the seam certifies through one scalar two-point kernel” (for the established free  $c = 8$  seam net, Wick extends the one kernel to the entire even tower by determinants). QBL is thereby the structural consistency frame of the carrier choice; the *selection* is carried by rank-8 + integrality.

**One kernel is the whole net** ([verification/v113\_quasifree\_kernel.py]) — **the QBL input merges with the holomorphy premise**. The remaining input is now anchored in the established Gate-A objects. *Bookkeeping* [E]:  $c(D_5)_1 = \frac{45}{9} = 5 = g_{\text{car}}$ ,  $c(A_3)_1 = \frac{15}{5} = 3 = N_{\text{fam}}$ , and the carrier net  $(D_5)_1 \times (A_3)_1 = SO(10)_1 \times SO(6)_1$  is  $10 + 6 = 16$  *free Majorana fermions* ( $c = \frac{16}{2} = 8$ ); the whole

extension tower carrier  $(\mu=16) \rightarrow SO(16)_1(\mu=4) \rightarrow (E_8)_1(\mu=1)$  carries the *same* sixteen fermions — only the certified glue grows (index  $2 \times 2 = 4 = |\mu_4|$ ). *Wick/Pfaffian* [E]: in the exact ten-Majorana model *all* 210 vacuum four-point and all 210 six-point functions equal the Pfaffian of the single two-point kernel — one kernel determines all correlations. *The kernel is a Calderón involution of rank  $g$*  [E]:  $M = \mathbf{1} + iA$  with  $A^2 = -\mathbf{1}$ , so  $P = M/2$  is a projection with rank  $P = 5 = g_{\text{car}}$  and  $\varepsilon = iA$  an involution; at seam level (sixteen Majoranas) the rank is  $8 = \text{rank } E_8$ :

the central charge is the rank of the one kernel

( $c = 5$  carrier block,  $c = 8$  seam hull); and the polarization fixes the vacuum *uniquely* (joint annihilator kernel one-dimensional). *Premise merge* [C] (*recorded, not claimed*): “single scalar two-point kernel” = “the seam state is quasi-free” = the defining property of the free-fermion net the holomorphy gate already posits — QBL adds *no* assumption beyond Gate A; when the gate closes, the carrier choice closes [E] along free net  $\Rightarrow$  one kernel  $\Rightarrow$  sheet-odd  $\Rightarrow$  self-counting tower  $\Rightarrow$  complete transport  $\Rightarrow$  carrier. Gate typing [C]/[O] unchanged.

### Hardening (P1): the Gauss–Bonnet origin of $c_3 = \frac{1}{8\pi}$

$c_3 = 1/(8\pi)$  is not an arbitrary number: it is the *one-sided Gauss–Bonnet normalisation of the seam*. The seam construction compactifies the oriented two-dimensional normal slice  $N_\Sigma$  to a sphere  $S^2$ , so Gauss–Bonnet gives the seam curvature integral

$$\oint_{S^2} K dA = 2\pi \chi(S^2) = 4\pi \quad (\chi(S^2) = 2),$$

and the boundary datum is *one-sided* (the Calderón/double-cover construction uses a single collar), contributing a factor  $1/|\mathbb{Z}_2| = \frac{1}{2}$ . Hence

$$c_3 = \frac{1}{|\mathbb{Z}_2| \oint_{S^2} K dA} = \frac{1}{2 \cdot 4\pi} = \frac{1}{8\pi}$$

equivalently  $c_3 = \frac{1}{2}(4\pi)^{-1}$ , the one-sided 2D heat-kernel coefficient. This locates both factors precisely:

- the *discrete*  $8 = |\mathbb{Z}_2| \cdot 2 \cdot \chi(S^2) = 2 \cdot 2 \cdot 2$  *coincides with*  $h(D_5) = \text{rank } E_8 = \varphi(30)$  — the geometric and the bootstrap readings of the “8” agree;
- the *continuous*  $\pi$  is the irreducible Gauss–Bonnet / 2D-Gaussian primitive ( $\int e^{-x^2} = \sqrt{\pi}$ ), the single primitive the self-consistency loop also isolates.

*Status*: this *hardens* (P1) — it gives  $c_3$  a named geometric origin and reconciles the “8” with  $E_8$  — but does *not* eliminate  $\pi$ ;  $\pi$  remains the one continuous primitive. [C]

**The one-paragraph version.** There is a boundary (the seam). It yields  $c_3 = 1/(8\pi)$  and a five-slot carrier  $g_{\text{car}} = 5$ , split 3 colour + 2 weak. Two base charges  $y_- = -1/3$  and  $y_+ = 1/2$  generate the whole fermion family; the 16 even-occupation states are one generation, three families give  $3 \times 16 = 48$ . One seed  $u = \varphi_0^{\text{ret}}$  drives four low-energy observables. Finally  $\alpha_\star^{-1} \approx 137$  is the exponential engine: the electroweak scale takes one fifth of it ( $e^{-\alpha_\star^{-1}/5}$ ), the cosmological constant takes it twice ( $e^{-2\alpha_\star^{-1}}$ ), and the Hubble scale is the square root of  $\Lambda$ . A small grammar generates many sectors.

## 2 Headline: the Pascal compiler on five carrier slots

The strongest new statement is that three of the theory’s separate structures are the *same* truncated row of Pascal’s triangle on the five-slot carrier.

**Family = action ladder = transition count (one Pascal row)**

The even exterior algebra of  $E$  ( $\dim E = g_{\text{car}} = 5$ ), the action ladder, and the  $E_8$  root count are one Pascal row:

$$S^+ = \Lambda^0 E \oplus \Lambda^2 E \oplus \Lambda^4 E, \quad (\dim \Lambda^0, \dim \Lambda^2, \dim \Lambda^4) = (1, 10, 5), \quad \dim S^+ = 1 + 10 + 5 = 16,$$

$$A_{\text{EW}} : \mathcal{A}_H : \mathcal{A}_\Lambda = \binom{5}{0} : \binom{5}{1} : \binom{5}{2} = 1 : 5 : 10 = (\dim \Lambda^0 E, \dim \Lambda^4 E, \dim \Lambda^2 E),$$

$$|R(E_8)| = \dim S^+ (\dim S^+ - 1) = 16 \cdot 15 = 240.$$

Family (exterior degree), cosmology (action charge) and  $E_8$  (root transitions) read the same  $\{1, 5, 10\}$ . **[E]**

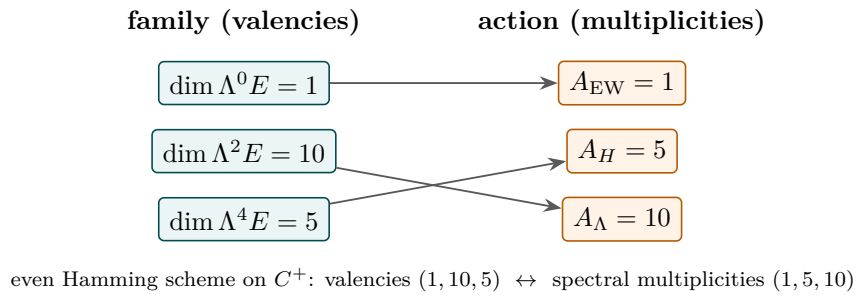


Figure 1: Exterior–Action duality. The Standard-Model family is the valency side of the even Hamming scheme on the five-slot code; the cosmological action ladder is its spectral (multiplicity) side. The  $\Lambda$  (pair) sector  $\Lambda^2 E$  has dimension  $\binom{5}{2} = 10$ , the origin of the decuple exponent.

Natural sector assignment:

$$EW \leftrightarrow \Lambda^0 E \text{ (singlet)}, \quad H \leftrightarrow \Lambda^4 E \cong E^\vee \text{ (dim 5)}, \quad \Lambda \leftrightarrow \Lambda^2 E \text{ (dim 10)}.$$

The cosmological constant lives on the *pair* sector  $\Lambda^2 E$ ,  $\dim = \binom{5}{2} = 10$ ; this is why the decuple bridge has exponent 10 —  $\Lambda$  is a product over all  $\binom{5}{2}$  carrier pairs ( $K_5$  edges), not over the full 16-state family.

## 2.1 $g_{\text{car}} = 5$ as the Pascal closure

A family is exactly the complete Pascal truncation to degree two:

$$\dim S^+ = 2^{g_{\text{car}}-1} = \binom{g_{\text{car}}}{0} + \binom{g_{\text{car}}}{1} + \binom{g_{\text{car}}}{2} \iff g_{\text{car}} = 5 \quad (16 = 1 + 5 + 10), \quad \textbf{[E]}, \quad (2)$$

the unique positive-integer solution of  $f(g) := 2^{g-1} - 1 - \frac{g(g+1)}{2} = 0$ . *Proof of uniqueness.* Direct evaluation gives  $f(1)=-1$ ,  $f(2)=-2$ ,  $f(3)=-3$ ,  $f(4)=-3$ ,  $f(5)=0$ ,  $f(6)=+10$ ; so  $g=5$  is the only root in  $\{1, \dots, 6\}$ . For  $g \geq 6$  the forward difference is  $f(g+1) - f(g) = 2^{g-1} - (g+1) \geq 2^5 - 7 = 25 > 0$  and increasing, so  $f$  is strictly increasing on  $[6, \infty)$  from  $f(6)=10 > 0$ ; hence  $f(g) > 0$  for all  $g \geq 6$  and there is *no* further root. Thus  $g_{\text{car}} = 5$  is the unique solution (no finite-range caveat). [\[verification/v2\\_carrier\\_pascal.py\]](#) This is sharper than  $2g_{\text{car}} = \binom{g_{\text{car}}}{2}$ : the family code (even states,  $2^{g-1}$ ) equals the action code (sum of the first three Pascal entries) only at  $g_{\text{car}} = 5$ .

**Carrier uniqueness theorem:  $g_{\text{car}} = 5$  and the 3+2 split are forced [E]**

The carrier rank and its colour/weak split are not chosen — they are the unique solutions of two integer conditions tied to the  $E_8$  closure and the SM gauge algebra:

$$g_{\text{car}} = 5 \text{ is the unique } g \text{ with } N_{\text{fam}} = \frac{2^{g-1}-1}{g} \in \mathbb{Z}^+ \text{ and } \text{rank } E_8 = g + N_{\text{fam}} = 8 \quad (3)$$

$$(b, s) = (3, 2) \text{ is the unique split of } b + s = g_{\text{car}} = 5, \quad b^2 + s^2 = 13 = |R(A_3)| + 1 \quad (4)$$

The integer-family carriers are  $g \in \{3, 5, 7, \dots\}$  (rank 4, 8, 16), so  $\text{rank } E_8 = 8$  picks  $g_{\text{car}} = 5$  uniquely; and  $b^2 + s^2 = 13$  with  $b+s = 5$  forces (3, 2), reproducing  $\dim \mathfrak{g}_{\text{SM}} = (b^2-1) + (s^2-1) + 1 = 12 = |R(A_3)|$ . The induced hypercharge is anomaly-free ( $\text{Tr } Y = \text{Tr } Y^3 = 0$  over the 16 states). This hardens P2: the *rank* and the *split* are forced; only the seam→carrier interface itself remains a declared input. [\[verification/v14\\_carrier\\_uniqueness.py\]](#)

**The Pascal grammar: one  $\Lambda^\bullet$  mechanism in four places (anti-numerology) [E].** The recurring small integers are *not* coincidences — they are the *same* exterior-algebra (Pascal) mechanism read in four spaces, which is why the binomials of rows 4 and 5 keep reappearing:

| space                    | Pascal reading                                                                                                                            |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| carrier (one generation) | $\dim S^+ = \Lambda^{\text{even}}(5) = \binom{5}{0} + \binom{5}{2} + \binom{5}{4} = 1 + 10 + 5 = 16$                                      |
| scale ladder (cosmology) | $A_{\text{EW}} : \mathcal{A}_H : \mathcal{A}_\Lambda = \binom{5}{0} : \binom{5}{1} : \binom{5}{2} = 1 : 5 : 10$                           |
| flavor $u/d$ readout     | $\ \text{Pl}(K)\ _1 = \binom{4}{0} + \binom{4}{1} + \binom{4}{2} = 1 + 4 + 6 = 11 = 16 - g_{\text{car}}$                                  |
| mass volume              | $\det M_{\text{SM}} \sim (\varphi_0^{\text{ret}})^{6+9+10} = (\varphi_0^{\text{ret}})^{25} = (\varphi_0^{\text{ret}})^{g_{\text{car}}^2}$ |

So the carrier is the even row 5, the scale ladder is the lower row 5, the  $u/d$  leg is the cumulative row 4 up to  $\deg u^c = 2$ , and the mass volume closes on  $g_{\text{car}}^2$ . *One* grammar ( $\Lambda^k$  of the 5-slot carrier and the  $4 = |\mu_4|$  family fundamental), not four lucky hits. [\[verification/v44\\_carrier\\_exterior.py\]](#) [\[verification/v45\\_family\\_exterior.py\]](#) [\[verification/v46\\_grand\\_mass\\_volume.py\]](#)

## 2.2 One master moment generates 40, 41, 240, $\delta_2$

With  $X = 6Y$ , the quadratic hypercharge trace on one family is  $\text{Tr}_{S^+} X^2 = 120 = 5!$ , and it generates the whole integer chain:

$$\sum_{f,j} L_{f,j} = \frac{\text{Tr}_{S^+} X^2}{3} = 40, \quad 10b_1 = \frac{\text{Tr}_{S^+} X^2}{3} + 1 = 41, \quad (5)$$

$$|R(E_8)| = 2 \text{Tr}_{S^+} X^2 = 240, \quad \delta_2 = 4! \text{Tr}_{S^+} X^2 c_3^8.$$

The flavor budget, the  $U(1)$  budget, the  $E_8$  count and the second seam defect all hang on the second hypercharge moment  $5!$ . [E]

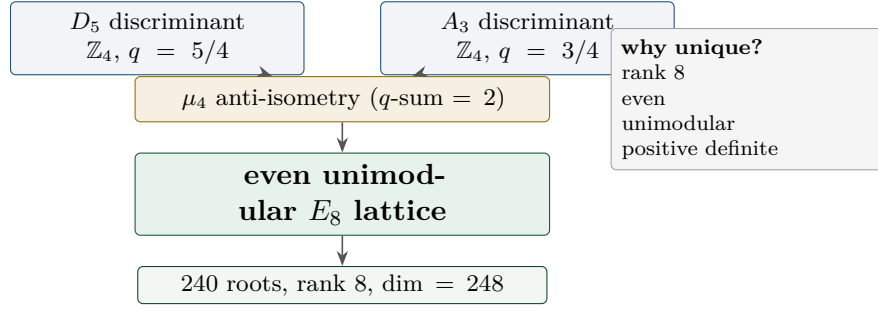
## 2.3 Compact $E_8$ compression and the role of $\nu^c$

$$\begin{aligned} |R(E_8)| &= \dim S^+ \cdot g_{\text{car}} \cdot N_{\text{fam}} = 16 \cdot 5 \cdot 3 = 240, \\ \dim E_8 &= \dim S^+(\mathcal{A}_H + \mathcal{A}_\Lambda) + (g_{\text{car}} + N_{\text{fam}}) = 240 + 8 = 248 \end{aligned} \quad (6)$$

with  $N_{\text{fam}} = (2^{g_{\text{car}}-1} - 1)/g_{\text{car}} = 15/5 = 3$  and  $\text{rank } E_8 = g_{\text{car}} + N_{\text{fam}} = 8$ . [E]. The right-handed neutrino  $\nu^c$  is the Pascal singleton  $\Lambda^0 E = (1, 1)_0$  that closes  $16 = 1 + 5 + 10$ ; removing it breaks the truncation and shifts  $\alpha^{-1}$  by  $2.16 \times 10^{-3}$ .

### 3 The $D_5 \times A_3$ compiler: $E_8$ as a closure, not an input

This section upgrades the  $E_8$  atlas (Section 10) from a counting coincidence to an explicit lattice construction. The whole content is consolidated and verified in the standalone companion note “*The Even Carrier Code and the  $\mathbb{Z}_{30}$  Coxeter Compiler.*” The thesis: the carrier code is the  $D_5$  half-spinor, the family geometry is an  $A_3$ , and  $E_8$  is the standard unimodular glue closure of  $D_5 \oplus A_3$  — with the four punctures  $\mu_4$  as the glue code.



$D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$  — the unique even unimodular rank-8 glue  $[E]$  exact (lattice).

#### The compiler in one line:

five carrier slots  $\Rightarrow D_5$ , four family corners  $\mu_4 \Rightarrow A_3$ ,  $(D_5 \oplus A_3) + \mu_4\text{-glue} \Rightarrow E_8$ .

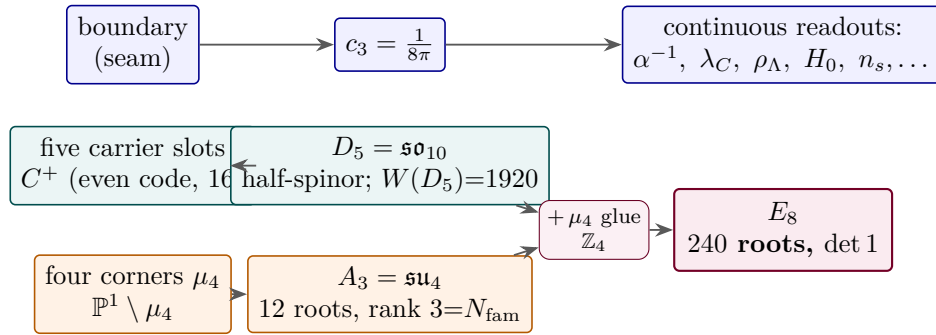
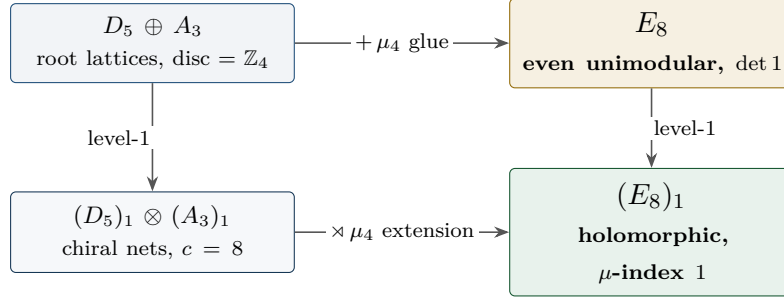


Figure 2: The explanation hierarchy. The boundary fixes the seam seed  $c_3$ ; the five-slot even carrier code is the  $D_5$  half-spinor; the four punctures  $\mu_4$  give  $A_3$ ; their common  $\mathbb{Z}_4$  discriminant ( $\mu_4$ ) glues  $D_5 \oplus A_3$  into the unimodular  $E_8$ .  $E_8$  is the *closure*, not the input;  $c_3$  dresses the continuous observables.

**The glue is *one* operation in two categories.** The same order- $|\mu_4|=4$  discriminant glue closes  $D_5 \oplus A_3$  into  $E_8$  *twice over*: once on *root lattices* (the unimodular pushout  $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ , **v1**) and once on *chiral conformal nets* (the simple-current / anyon-condensation extension  $(D_5)_1 \otimes (A_3)_1 \rtimes \mu_4 \cong (E_8)_1$ , **v92/v281**). These are not two coincidences but the two functorial faces of a single commuting square (**v234/v277/v260**); the boundary realisation on which the strict-TOE residual turns (**SEAM.EQUIV.01**) is exactly the statement that the raw seam realises this square — now *closed modulo cited theorems on its MMST route* **SEAM.EQUIV.MMST.01** (the lattice/ $S_3$  stack pins it at every computable level, Lean **FORM.SEAM.MMST.01**; only the cited MMST/OS continuum-existence theorems stay **[O]**, **v367/v368/v376–v379/v336**; the twistor route **SEAM.EQUIV.TWISTOR.01** and the unconditional parent stay **[O]**). The extension leg of that citation package is itself re-founded at net level (**[verification/v469\_seam\_crossedproduct\_route.py]**): the 128-spinor step is the *local*  $\mathbb{Z}_2$  simple-current crossed product ( $h_s(SO(16)_1) = 16/16 = 1 \in \mathbb{Z}$ , the Longo–Rehren locality criterion; KLM  $\mu = 4/2^2 = 1$ ), on 1995–2001 peer-reviewed subfactor theory, with the lattice-VOA preprint route demoted to an independent second witness, and the realisation input reduced to invariant level (R1'). Premise (i) of that package ( $c_-=8$ ) now also

carries a *ground-state* witness ([[verification/v489\\_seam\\_modular\\_commutator.py](#)]): the Kim–Shehab–Kim modular commutator  $J(A, B, C) = (\pi/3)c_-$ , evaluated on the exact BdG ground state of the collar model, gives  $c_- = 8 = g_{\text{car}} + N_{\text{fam}}$  for the 16-layer carrier (per layer 0.500000 at  $L=48$ ; trivial and orientation-flip controls pass) — independent of both the Chern route (v367) and the KLM count (v378). **SEAM.EQUIV.01** stays [O].



One operation, two categories: the order- $|\mu_4|=4$  glue is the lattice pushout  $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$  and the simple-current / anyon-condensation extension  $(D_5)_1 \otimes (A_3)_1 \rtimes \mu_4 \cong (E_8)_1$ ; the square commutes. [E] exact.

### 3.1 Carrier = $D_5 = \mathfrak{so}_{10}$ (proved)

The 16 even-occupation states  $C^+ = \{x \in \mathbb{F}_2^{g_{\text{car}}} : |x| \text{ even}\}$  are exactly the weights of the  $D_5$  half-spinor  $\frac{1}{2}(\pm 1, \dots, \pm 1)$  (even number of minus signs). The flip-weight-4 relation is the **Clebsch graph**  $\text{SRG}(16, 5, 0, 2)$  and the flip-weight-2 relation is its complement  $\text{SRG}(16, 10, 6, 6)$ ; the *affine* automorphism group of the scheme (translations  $\rtimes$  coordinate permutations) is the  $D_5$  Weyl group,

$$\boxed{\text{Aut}_{\text{aff}}(C^+) = C^+ \rtimes S_5, \quad |\text{Aut}_{\text{aff}}(C^+)| = 2^{g_{\text{car}}-1} g_{\text{car}}! = 1920 = |W(D_5)| = r |R(E_8)|} \quad [\text{E}], \quad (7)$$

(the linear code automorphism group alone is only  $S_5$ ; the order-1920 group is the affine/scheme one) and the Clebsch edge count is the flavor budget  $|E(R_4)| = \frac{16 \cdot 5}{2} = 40 = |R(D_5)| = \sum_{f,j} L_{f,j}$ .

### 3.2 Family = $A_3 = \mathfrak{su}_4$ from $\mathbb{P}^1 \setminus \mu_4$ (proved, with the metric settled)

The carrier geometry fixes  $X_f^\circ \cong \mathbb{P}^1 \setminus \mu_4$ ,  $H_1 = \mathbb{Z}^3$ , rigid  $D_4$  generated by  $z \mapsto iz$ ,  $z \mapsto 1/z$ . The twelve puncture-difference cycles  $\gamma_i - \gamma_j$  carry the residue pairing  $\text{Res}(\omega_i, \gamma_j) = \delta_{ij}$  and form the  $A_3$  root system:

$$\boxed{\#\{\gamma_i - \gamma_j\} = 12 = |R(A_3)| = \dim \mathfrak{g}_{\text{SM}}, \quad \text{rank} = 3 = N_{\text{fam}}, \quad \text{Gram} = \text{Cartan}(A_3), \quad \det = 4} \quad [\text{E}]. \quad (8)$$

The corner rotation  $z \mapsto iz$  acts on  $H^1 = \mathbb{C}^3$  as the  $A_3$  Coxeter element (order  $4 = h(A_3) = |\mu_4|$ , spectrum  $\{i, -1, -i\}$ ,  $\chi = \Phi_4 \Phi_2$ ), an isometry of the Cartan form — the exact analogue of the  $E_8$  Coxeter element reading  $\Phi_{30}$ .

**The  $\mu_4$  clock is the order-4 character of the  $E_8$  cycle, and  $(2, 3, 5)$  is the icosahedron** ([[verification/v223\\_coxeter\\_totative\\_clock.py](#)], [[verification/v219\\_icosahedral\\_mckay.py](#)]). The two “Coxeter” clocks are one object read at two scales. The  $E_8$  exponents are the totatives  $(\mathbb{Z}/30)^\times = \{1, 7, 11, 13, 17, 19, 23, 29\}$ , and the multiplicative order-4 element 7 ( $7^2=19$ ,  $7^4=1 \pmod{30}$ ) generates a literal  $\mu_4$  clock  $\langle 7 \rangle = \{1, 7, 13, 19\}$  that permutes the four conjugate Coxeter 2-planes — the same  $\mu_4$  as the seam corner rotation. And the period  $30 = 2 \cdot 3 \cdot 5$  itself is not a free factorisation: by the McKay correspondence  $E_8$  is the exceptional *top* of the tower of finite  $SU(2)$  subgroups ( $2T \rightarrow \hat{E}_6$ ,  $2O \rightarrow \hat{E}_7$ ,  $2I \rightarrow \hat{E}_8$ ), so its branch data is the *icosahedron* and the three atoms  $(|\mathbb{Z}_2|, N_{\text{fam}}, g_{\text{car}}) = (2, 3, 5)$  are its rotation-axis orders. The binary icosahedral group  $2I$  (order  $120 = |R^+(E_8)|$ ) has nine irreducible-representation degrees  $\{1, 2, 2, 3, 3, 4, 4, 5, 6\}$  equal to the affine- $E_8$  Kac marks, with  $\sum d_i = 30 = h(E_8)$  and  $\sum d_i^2 = 120$ . This is a *backward certificate* of the closed  $E_8$  [E] (it explains the  $(2, 3, 5)$  choice [C]; it does *not* replace the P2 input).



**The metric, settled (former [C]).** The transcendental geometric (Green) metric  $G_{ij} = -\log |p_i - p_j|$  on  $\mu_4$  is only  $D_4$ -symmetric, with a regularisation-independent anisotropy  $-\log 2$  on the root space, so it is *not* proportional to the  $A_3$  Cartan form. But it does not need to be: the  $E_8$ -relevant  $A_3$  is the canonical *residue* pairing (integral,  $S_4$ -symmetric,  $= \text{Cartan}(A_3)$ ), and the geometric metric is a  $D_4$ -equivariant deformation preserving the root system, Weyl group and Coxeter element. So the lattice structure is exact [E]; the analytic metric is a characterised deformation, not an open gap.

### 3.3 The $\mu_4$ glue and the $5/4 + 3/4 = 2$ discriminant theorem (proved)

The two factors have equal discriminant groups and  $E_8$  is unimodular, so the glue index is exactly  $|\mu_4|$ :

$$\text{disc}(D_5) = \text{disc}(A_3) = \mathbb{Z}_4, \quad [E_8 : D_5 \oplus A_3] = \sqrt{\frac{\det D_5 \det A_3}{\det E_8}} = \sqrt{\frac{4 \cdot 4}{1}} = 4 = |\mu_4|. \quad (9)$$

The discriminant-form norms sum to the  $E_8$  root norm, and — strikingly — they are pre-existing TFPT constants:

$$\boxed{\begin{aligned} q(D_5) &= \frac{5}{4} = \frac{\delta_2}{\delta_{\text{top}}^2}, & q(A_3) &= \frac{3}{4} = \xi_{\text{tree}}, \\ q(D_5) + q(A_3) &= 2 = |E_8 \text{ root}|^2 \end{aligned}} \quad [\text{E}] \quad [\text{verification/v1\_e8\_glue.py}] \quad (10)$$

This is literally the  $E_8$  spinor-root split  $\frac{1}{2}(\pm)^8 = \frac{1}{2}(\pm)^5 \oplus \frac{1}{2}(\pm)^3$ ,  $5 \cdot \frac{1}{4} + 3 \cdot \frac{1}{4} = 2$ .

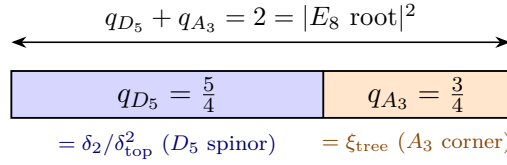


Figure 3: The discriminant-norm glue. The  $E_8$  spinor root  $\frac{1}{2}(\pm)^8$  (norm 2) splits across  $D_5 \times A_3$  as  $5 \cdot \frac{1}{4} + 3 \cdot \frac{1}{4}$ ; the two halves are the pre-existing TFPT constants  $\delta_2/\delta_{\text{top}}^2 = \frac{5}{4}$  and  $\xi_{\text{tree}} = \frac{3}{4}$ .

The standard branching is realised explicitly:

$$\mathfrak{e}_8 = (45, 1) \oplus (1, 15) \oplus (10, 6) \oplus (16, 4) \oplus (\overline{16}, \overline{4}), \quad 248 = 45 + 15 + 60 + 64 + 64, \quad 240 = 40 + 12 + 60 + 64 + 64, \quad (11)$$

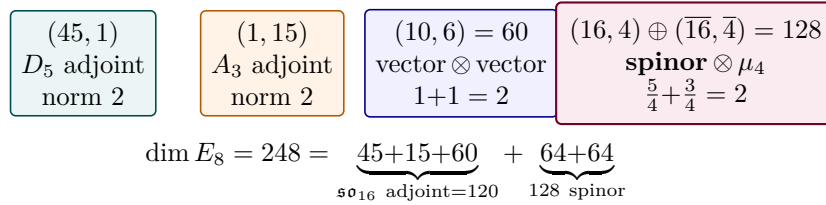


Figure 4: The  $E_8 \supset D_5 \times A_3$  branching as norm-2 blocks. The adjoint half is the hypercharge moment  $\text{Tr}_{S^+} X^2 = 120 = \dim \mathfrak{so}_{16}$ ; the spinor half is two sheets over the four corners  $\mu_4$ .

and assembling  $D_5 \oplus D_3 \subset D_8$  with the even spinor coset produces all 240 roots (norm 2, Cartan  $\det = 1$ , every root an integer combination of one simple-root basis): *the 240-metrisation is no longer blind —  $E_8$  is built.* [E]

**Hecke structure of the glue census is lattice-native (HECKE.GEOM.01).** The  $\mu_4$ -glue class thetas that refine the 240-root branching ( $240 = 52 + 64 + 60 + 64$  over  $\mathbb{Z}/4$ ) now carry a machine-checked Hecke package ([verification/v535\_hecke\_from\_geometry.py]): Kneser  $p$ -neighbours realise



$\#iso\_lines = \sigma_3(p) \# \mathbb{P}^3(\mathbb{F}_p)$  at  $p = 2, 3, 5, 7$ ; the marked neighbour-sum is the affine element  $\nu_p = a \text{Id} + b T_p$  with cusp output  $a_3 = -4$ ,  $a_5 = -2$ ; and the seven census  $q$ -series span exactly the 2-adic oldform hull of  $E_4$  and  $f_8$  (levels  $\{1, 2, 4, 8, 16\}$  and  $\{8, 16\}$ ). Classical scaffolding named classical; weight-4  $\rightarrow \text{GL}(1)$  stays [O]; no RH claim. [E] (in-suite mechanics)

**Eichler trace layer at the  $E_8$  lattice (HECKE.GEOM.EICHLER.01).** The same census channels now carry a machine-checked Eichler/Witt package ([verification/v536\_eichler\_trace\_layer.py]; 23 checks,  $\sim 30$  s; builds on HECKE.GEOM.01 without duplicating Kneser/ $\nu_p$ /oldform checks). (A) The elementary Witt shadow  $\lambda_{\text{Eis}} = \sigma_3(\# \mathbb{P}^3 - \sigma_3) = L - \sigma_3^2$  is closed in  $p$  (identically and for all odd  $p \leq 100$ ). (B) The Eichler identity  $\lambda_{\text{geom}} = \lambda_{\text{Eis}} + a_p^2$  is two-sided at  $p = 3$  (live type-(i)/(ii) split:  $352 = 364 - 12$ ) and at  $p = 5$  via a T36-validated freeze ( $3784 = 3906 - 122$ ); the assembler  $R(p) = \sigma_3 - 1 - c(p^2)/8$  equals  $a_p^2$  with Kneser filter = identity. (C) Closed Type-A/B local densities  $N_A = \min(240(1 + p^3), p^7 + p^4 - p^3 - 1)$  (B empty iff  $p^4 < 241$ ;  $p = 7$  live 82560/743040; injectivity at  $p = 11, 13$ ). (D) Signed extraction  $a_p = -c(p)/8$  from geometric Shell( $p$ ) recovers  $(-4, -2, +24)$  and  $b \in \{24, 124, 368\}$ . (E)  $O(1)$  assembler for all odd  $p \leq 31$  against the eta product  $f_8$ . Classical Eichler/Siegel/Witt named classical; the claim is in-suite two-sidedness and the closed compiler densities. Honest fences: Shell( $p^2$ ) for  $p \geq 7$  out of scope; fibre refinement scoped to  $\{5, 7, 11\}$ ; weight-4  $\rightarrow \text{GL}(1)$  stays [O]; no RH claim; no marker moves. [E] (mechanics) / [O] (weight drop)

**Half-integral bridge (HECKE.GEOM.HALFINT.01).** The weight-4 cusp  $f_8$  now has a machine-checked half-integral companion in the compiler theta-monoid ([verification/v537\_halfintegral\_bridge.py]; 20 checks,  $\sim 90$  s; companion to HECKE.GEOM.01/HECKE.GEOM.EICHLER.01). (A) In the 70-element weight-5/2 theta-monoid (plain and  $q^2$ -scaled  $\vartheta_2, \vartheta_3, \vartheta_4$ ) there is exactly one object with the signed Shimura identity  $\text{Sh}_{t=2, \psi=1}(g) = -8 f_8$  (witness  $g = \vartheta_2(q^2)^2 \vartheta_3(q^2) \vartheta_4 \vartheta_4(q^2)$ , verified coefficientwise through  $n \leq 120$ ; three further monoid elements lift to scales  $+8, -16, +16$ ). (B)  $g$  is an exact  $T(p^2)$ -eigenform of weight 5/2, trivial nebentypus, with eigenvalues  $a_p(f_8)$  for  $p = 3, 5, 7, 11, 13$   $(-4, -2, 24, -44, 22)$ ; trial FE of the 5/2-side around centre 5/4 with  $N = 512$ ,  $\varepsilon = +1$ . (C)  $U_4(g) = 0$  exactly (mass mod 4 supported on  $\{1, 2\}$ ); level  $32 = 4 \cdot 8$  with  $M = 8$  even lies outside Kohnen 1982 — the plus-route is structurally closed (negative result part of the claim). (D) With AFE-based twist  $L$ -values ( $\varepsilon_d = \chi_d(8)$ ;  $\varepsilon = -1 \Rightarrow L(2) = 0$ ), the Waldspurger quotient  $R(d) = b(|d|)^2 / (|d|^{3/2} L(f_8 \times \chi_d, 2))$  is constant on ten fundamental discriminants  $d \equiv 1 \pmod{8}$  ( $R = 23.1873585645\dots$ , relative spread  $\leq 10^{-12}$ , including  $d = 89$ ); the class  $d \equiv 5 \pmod{8}$  vanishes structurally. Classical Shimura/Kohnen/Waldspurger/Baruch–Mao named classical; the claim is the unique signed-scale monoid preimage, the in-suite chain, and the *measured* constant  $R$ . Honest fences: GL(2) centre  $s = 2$ , not the  $\xi$ -line; no RH claim; ZETA.HP.CARRIER untouched; no marker moves. [E] (Shimura/Hecke/Kohnen fence) / [C] (measured  $R$ )

**Compiler relative-trace identity (HECKE.GEOM.RTF.01).** The three preceding packages are machine-checked as three *projections* of one finite relative-trace identity [ $E_8$ /glue orbital counting] = [Newform/Waldspurger spectral sum] ([verification/v538\_relative\_trace\_identity.py]; 18 checks,  $\sim 14$  s; promotion of discovery T53; companion to HECKE.GEOM.01/HECKE.GEOM.EICHLER.01/HECKE.GEOM.HALFINT.01). (R1)  $\text{Tr}_V(\nu_p \circ \pi)$  is computed *both sides independently*: geometric  $a_p = -c(p)/8$  from the glue census (no modular input) equals spectral  $a_p$  from  $\eta(2\tau)^4 \eta(4\tau)^4$  (no geometry) for all  $(p, \pi) \in \{3, 5, 7\} \times \{\text{Id}, \pi_{\text{Eis}}, \pi_{\text{cusp}}, \pi_{E_4}, \pi_{f_8}\}$  (anchors e.g.  $p=7$ : full trace 727680,  $f_8$ -channel 19840). (R2) The Eichler split  $\lambda_{\text{geom}} = \lambda_{\text{Eis}} + a_p^2$  is the RTF orbit decomposition (principal orbit  $\rightarrow \lambda_{\text{Eis}} = L - \sigma_3^2$ ; elliptic  $\rightarrow a_p^2$ ; cross terms cancel; anchors  $\lambda_{\text{Eis}} = 336/3780/19264$ ,  $a_p^2 = 16/4/576$ ). (R3) The period side is lattice counting:  $g$  is the quaternary form  $n = (x^2 + y^2)/2 + 2z^2 + u^2 + 2w^2$  ( $x, y$  odd; sign  $(-1)^{|u|+|w|}$ ), with  $b_{\text{rep}} = b$  through  $n \leq 200$ , and [signed lattice count] $^2 = R \cdot |d|^{3/2} \cdot L(f_8 \times \chi_d, 2)$  at  $R = 23.1873585645$  (rel  $\sim 10^{-12}$ ). Verdict ONE-FORMULA. Classical Jacquet-type RTF / Eichler / Witt / Waldspurger / theta correspondence named classical; NEW is the compiler-native realisation and bilateral independence. Honest fences: the identity is *finite* ( $p \in \{3, 5, 7\}$ , finite  $d$ -windows); the *infinite* RTF (sum over all  $p$  and  $d$  + archimedean term) remains the named open step; no RH claim; not ZETA.HP.CARRIER-relevant; no marker moves. [E] (R1/R2 exact) / [C] (R3 measured  $R$ )

**Weil structure of the compiler family (RTF.GNS.WEIL.01).** The compiler family carries a fully identified Weil structure up to *two explicitly isolated obstructions* ([[verification/v539\\_weil\\_structure\\_family.py](#)]; 25 checks,  $\sim 6$ s; promotion of discovery T55/T61/T62/T63; companion to HECKE.GEOM.RTF.01). (A) The canonical GNS pairing is the direct integral over the Gelfand spectrum of the character subalgebra: metric  $\ell^2(d, b^2/|d|)$  diagonal; fibres  $\sigma(d) = (\chi_d(3), \dots, \chi_d(13))$  partition the live family;  $\Lambda_{\text{fam}}$  constant on each fibre; twist-mixing certificate per fibre  $\leq 5.037 \times 10^{-16} < 10^{-12}$ . (B) The trivial Sato–Tate isotype is algebraically the GL(1) core  $G_0 = (1 + Y)/(1 - Y) = \zeta_p(w - 3)^2/\zeta_p(2w - 6)$ , globally  $\zeta(u)^2/\zeta(2u) = \sum 2^{\omega(n)} n^{-u}$  (Dirichlet coefficients exact through  $n \leq 2000$ ; character expansions  $\Phi_1 = \chi_0 + \chi_2$ ,  $\Phi_2 = \chi_4 - \chi_2$ ,  $\Phi_3 = 2\chi_0 - \chi_4 + \chi_6$ ,  $\Phi_4 = \chi_8 - \chi_6$ ; core series  $c_{k,0} = [1, 0, 2, 0, 2, 0]$ ). (C) The family form satisfies the exact linear relation  $Q_{\text{fam}} = 2Q_\zeta(g) - 2Q_\zeta(g_b) + \text{Arch} + \text{Corr}$  ( $\text{Prime}_F = 2P_\zeta(g) - 2P_\zeta(g_b)$ ) with max rel  $9 \times 10^{-16}$ ;  $Q_{\text{fam}} = Q_F + \text{Corr}$  with rel  $2 \times 10^{-16}$ . **Obstruction 1:** the doubling term enters with a *minus* (family positivity does *not* imply  $Q_\zeta \geq 0$ ). **Obstruction 2:** the Plancherel correction is demonstrably non-automorphic (no polar/arch absorption; no finite Euler factor; identified as  $e^{-\sum p^{-u}}$ -type). Finite-class positivity  $Q_{\text{fam}} \in [4.369, 11.486]$  is measured [C]; dense-class positivity stays open / RH-adjacent and is *not* claimed. Classical Weil 1952 / Gelfand / SU(2)–Chebyshev / Sato–Tate /  $2^\omega$ -series / GNS named classical; NEW is the compiler-native identification and the machine-checked isolation of the two obstructions. Honest fences: *not* “almost RH” — the claim structurally shows why family positivity does not imply RH; ZETA.HP.CARRIER untouched; no marker moves. [E] (A/B/C/obstructions exact) / [C] (finite-class  $Q_{\text{fam}}$ )

**Amplitude route and positive linear carrier (RTF.GNS.AMP.01).** The route out of the square plane behind the two RTF.GNS.WEIL.01 obstructions is now machine-checked ([[verification/v540\\_amplitude\\_linear\\_carrier.py](#)]; 34 checks,  $\sim 3$ s; promotion of discovery T67–T72; companion to HECKE.GEOM.RTF.01/RTF.GNS.WEIL.01). (A) The amplitude Dirac  $D = \begin{pmatrix} 0 & V \\ V^\top & 0 \end{pmatrix}$  with  $V[d, m] = \sqrt{w_d} b(d) \chi_d(m)$ ,  $w_d = |d|^{-1}$ , exists exactly and is Hecke-equivariant:  $D = D^\top$ ,  $D^2 = \text{diag}(VV^\top, K)$  with  $V^\top V = K$  (rel  $2.7 \times 10^{-16}$ ), spectral  $\pm$ -symmetry;  $m$ -side intertwining  $VA_p^\top = \hat{A}_p V$  exact on the  $p$ -free locus ( $p = 3, 5, 7$ ) and  $d$ -side Shimura  $b(dp^2) = b(d)(a_p - \chi_d(p)p)$  integer-exact on 644 pairs. (B) The geometric polarisation  $b = N_+ - N_-$  is exact (parity-split lattice count of the quinary form, plus the  $\Theta$ -monomial identity coefficientwise);  $\Theta = N_+ + N_-$  is a *pure* Siegel–Weil Eisenstein  $T(p^2)$ -eigenform with eigenvalues  $\sigma_3(p) = 1 + p^3$  ( $p \leq 13$  exact; cusp component zero), carrying the Cohen seed identity  $\Theta(d) = -48 L(-1, \chi_d)$  exact-rational on 159 live  $d$  (Cohen 1975:  $H(2, d) = L(-1, \chi_d)$ ; generalised Bernoulli  $B_{2, \chi}$ ). (C) Every coefficient bilinear form inherits the even- $k$  deletion: the Cauchy–Littlewood window lemma with *free* Satake symbols has numerator  $1 - stuv X^2$ , hence  $1 - p^6 X^2$  pair-independently at  $st = uv = p^3$  (five channels checked); simultaneously the deletion *is* the square-class double counting of the towers,  $\text{fam}[1, 0, 2, 0] + 2b[0, 2, 0, 2] + \delta_{k1} = [2, 2, 2, 2]$  (generating-function exact; Möbius square sieve). (D) The positive linear carrier  $\ell^2(d, \mu)$  with  $\mu = 48 |L(-1, \chi_d)| |d|^{-a}$  carries *full* weights  $\lambda_k = 1 + p^{3k} - (\chi p)^k$  (layer  $[1, 1, 1, 1]$ ; no even- $k$  kill) and the plus balance  $Q = Q_\zeta(g_-) + Q_\zeta(g_+)$ ,  $g_\pm = e^{\pm 3u/2} g$  (rel  $2 \times 10^{-15}$ ); the  $b$ /doubling kernel enters *with plus* — the sign of the v539 minus inverted. (E) The completed functional equation  $\Lambda_\Theta(s) = 8^{1-s} \Lambda_{\Theta^\dagger}(5/2 - s)$  holds exactly (Fricke closed via the Jacobi inversions; rel  $\sim 10^{-40}$  modular and in the strip; poles  $5/2$  and  $0$  swapped; centre  $5/4$ , plus-balance locus off-centre by exactly  $1/4$ ); the guaranteed cone and the gap functional are FE-self-dual/-covariant. **Open boundary as content:** the positivity is *Euler-region* positivity (edge  $L$ -values, not the central line); the residual distance to the Weil cone is the FE-covariant gap functional  $\lambda^*$  on  $n \equiv 6 \pmod{8}$  — named, measured, Farkas-certified, and not removable by any finite signed theta library. Classical Cohen 1975 / Siegel–Weil / Jacobi–Fricke / Landen / Hecke split-Mellin / Cauchy–Littlewood / Shintani / Weil 1952 / Farkas named classical; NEW is the compiler-native consolidation. Honest fences: *not* “almost RH”; ZETA.HP.CARRIER untouched; no marker moves. [E] (A–E exact) / [C] (sampled cone facts;  $\lambda^*$  named open)

**Matching lemma and transport ledger package (RTF.GNS.LEDGER.01).** The discovery proof package T78–T85 is now load-bearing as one module in which every core check is *re-*

computed, not cited ([`verification/v541_matching_lemma_ledger.py`]; 33 checks,  $\sim 10$ s; companion to `HECKE.GEOM.RTF.01/RTF.GNS.WEIL.01/RTF.GNS.AMP.01`). (A) *Window proof*. The matching lemma of the hybrid certificate recipe is *proved exact-integer on*  $[4, 10^6]$ : the certificate  $7S(j) < 40A(j)$  (i.e.  $\rho_{\text{design}}F(j) < 1$  with  $\rho = 21/20$ ) holds at *every* atom  $j \leq 10^6$  (0 violations over 939 870 clash atoms; full enumeration, proven no-overflow, big-integer rechecks), with the exact Fraction margin  $X = 22242575693/270725400000 = 0.082159$  at the exact  $\text{argmax } j^* = 554400$  and  $\rho_{\text{crit}} = 1.1440 > 21/20$ ; the maximal-greedy identity makes the certificate *uniform* in the target set  $M$  and the greedy weights. Structure laws at 0 tolerance: the 8-law  $\Theta(4n) = 8\Theta(n)$  ( $n \leq 250000$ ); the seed-tower divisor formula  $\Theta(4^a s f^2) = 8^a \Theta(s) \alpha_s(f)$  and the  $\psi$   $w$ -table (all  $n \leq 50000$ ; Euler factor and  $w$ -recursion sympy-exact); the Cohen seeds  $\Theta(s) = c L(-1, \chi_\Delta)$  with *one* exact-rational constant per residue class  $\{-48, -80, -8, -8\}$  on all squarefree  $s \leq 2000$ ; the bracket  $\Theta \leq 2|\psi| \leq 3\Theta$  on the full window. (B) *Transport ledger*. The ledger  $Q_{\text{Weil}} = Q_{\text{cert}} + \Delta_{\text{arch}} + \Delta_2$  closes *exactly*:  $\Delta_{\text{pole}} \equiv 0$  and  $\Delta_{\text{conv}} \equiv 0$  are *proven identities* (sympy kernel collapse; quadrature  $2.8 \times 10^{-16}$ ; exact scale invariance), the residual is  $4.4 \times 10^{-16}$  on 24 exact-autocorrelation battery rows, and the odd-prime side of the full Weil functional agrees *exactly* with the certified plus combination  $P_\zeta^{\text{odd}}(h) = P_\zeta(g_-) + P_\zeta(g_+)$  (rel  $3.9 \times 10^{-16}$ ): the prime side was never the wall. (C) *Character-exact signed envelope*.  $-\psi = (\chi_{-4} + \frac{1}{4}\chi_8 + \frac{1}{4}\chi_{-8}) \cdot \Theta$  coefficient-exact on all odd  $n \leq 50000$  on both independent builds (character system solved sympy-exact); the signed certificate  $7S_{\text{net}} < 40A$  holds with 0 violations and margin factor  $8.86\times$ ; the confinement is a *set equality* on  $10^6$ :  $\{\text{zero-credit clash atoms}\} = \{\chi_{-4}\text{-coherent odd composites}\}$ . (D) *The archimedean term is internal*. Via Legendre duplication  $(2\pi)^{-s}\Gamma(s) = \frac{1}{2}\Gamma_{\mathbb{R}}(s)\Gamma_{\mathbb{R}}(s+1)$  (sympy-exact +  $1.8 \times 10^{-41}$ ) the ledger kernel obeys  $k_\zeta = K_{\text{fam}} - K_{\text{shift}}$  *pointwise* ( $7.9 \times 10^{-31}$  on 51 points; Gauss digamma anchor at  $t = 0$ ), and  $\Delta_{\text{arch}}(h) = A_{\text{fam}}(h) - A_{\text{shift}}(h)$  on the battery subset at rel  $9.9 \times 10^{-16}$  — I5 becomes the self-consistency of *one* heat family. (E)  *$\lambda$ -equivariant channel*. The CM carrier  $g_\lambda = \sum_{\mathbf{a}} \lambda_1(\mathbf{a}) q^{N_{\mathbf{a}}}$  exists exactly in two independent routes ( $\mathbb{Z}[i]$  lattice sum of  $z^4$  vs Cornacchia reconstruction, 0 mismatches  $n \leq 10^4$ ; CM laws exact;  $\text{supp}(c_1) = \{\mathbb{Z}[i]\text{-norms}\}$  as a set equality with  $c_1 \neq 0$  at every coherent atom;  $v_3^2 \equiv r_2$  on  $10^6$ ); the canonical phase mean  $\mu_1 = c_1/(c_0 d^2) \in [-1, 1]$  is exact, and the  $\lambda$ -window certificate over all  $j \leq 10^6$  gives  $\max 7|S_\lambda|/(40A) = 0.0653$  strictly below the unlifted 0.2365 (factor  $3.6\times$ ; exact Fraction rechecks) — the sign-mixing of the Grossencharacter phases *is* the cancellation channel; coherent targets are reachable only by coherent rescalings (T81 reachability). (F) *Frontier facts* [C]: unlifted coherent-chain crossing  $k^* = 14$  (exact Fractions,  $\log_{10} N^* = 23.1$ ); absolute Robin tail factor 6.16 at  $J = 10^6$  and divergent — the constant route closes nothing beyond the window. **Named limits as content:** (i) *one* classically-shaped open lemma remains — the correlated-cancellation lemma on credit-rich non-coherent tail atoms (uniform form); provably-shaped  $\neq$  formal proof; (ii) I5 in one-family form ( $Q_{\text{cert}} + \Delta_2 + A_{\text{fam}} - A_{\text{shift}} \geq 0$ ) is the single remaining TFPT-specific object — by the closed ledger *equivalent* to Weil positivity  $\iff$  RH (Weil 1952): an equivalence *typing*, not a progress claim. Classical Gronwall 1913 / Robin 1983 *unconditional* / Cohen 1975 / Hecke 1918–1920 / Cornacchia / Fermat–Gauss / Dirichlet  $L(1, \chi)$  / Mertens-AP / Landau 1908 / Legendre duplication / Weil 1952–Guinand–Bombieri / Alaoglu–Erdős named classical; NEW is the compiler-native consolidation and the machine-checked window proofs. Honest fences: no RH-evidence language; *not* “almost RH”; ZETA.HP.CARRIER untouched; no marker moves. [E] (A–E exact) / [C] (frontier facts; two named limits open)

**The margin-chain identities of phase 2 (PRIME.MARGIN.IDENT.01).** The identity/theorem block of the discovery parts T128–T131 is now load-bearing as one *narrow* module ([`verification/v542_margin_chain_identities.py`]; 44 checks,  $\sim 0.2$ s; companion to `RTF.GNS.LEDGER.01`), recomputed on small frame-A windows: 24 seam instances ( $n = 3 \dots 97$ ,  $\rho = 1.25 \dots 4.00$ , every inverted or diagonalised matrix  $\leq 300$ ), 43 graded/fine grid pairs and 400 randomised profiles, each identity a numerical residual against a preregistered tolerance *plus* a mutation control that must fail by  $\geq 10^{-3}$  relative. *Border geometry*.  $\sum_i (1 - g_i) = t_l + t_r$  exactly ( $5.5 \times 10^{-13}$ ), so the  $\kappa$  weights are a probability vector and the flat border vector gives  $\kappa_{\text{flat}} = 1$ ; the (T) four-term identity  $\tau_{\text{dn}} = \|y\|^2 - m_{\text{prot}} + m_{\text{fill}} - V_{\text{bord}}$  holds at  $2.2 \times 10^{-16}$ , with  $\tau$  recomputed independently from the full odd overlap matrix ( $1.1 \times 10^{-13}$ ). *Profile and curvature*.

$p_{N-1} = 2 - p_0 + 2E$  at  $2.3 \times 10^{-16}$ , hence the equivalence  $2E > p_0 \iff p_{N-1} > 2$  with 0 exceptions (both branches realised), and the Abel/Hölder chain  $\kappa \leq 2 - p_0 + \frac{N-2}{N} \sum_j |\Delta p_j - \overline{\Delta p}| + (p_{\max} - p_{N-1})$  holds per instance with 0 violations — while the same chain *without* the curvature term is violated on 88/400 profiles, which is what makes the curvature term load-bearing. *Assembly and compression.* The matrix-free two-scale assembly equals  $J^T Q J$  ( $5.2 \times 10^{-16}$ ), the graded overlap operator equals  $J$ , and the C ea/Strang defect identity  $S_{\text{graded}} - S_{\text{uniform}} = R^T X^{-1} R$  holds as a *matrix* identity on all 43 grids ( $2.1 \times 10^{-14}$ ) with a PSD defect — the compression error is one-sided *by* the identity. *Sandwich, sign, counting.* The secular sandwich  $\varepsilon/(\|u\|^2 + \varepsilon/\mu_1) \leq \lambda_{\min}(Q) \leq \varepsilon/\|u\|^2$  holds on all 48 sections at which  $A > 0$  and  $\varepsilon > 0$  are *checked* (relative width  $\leq 1.4448$ ), with Albert’s sign equivalence verified in both directions;  $S^{-1} > 0$  entrywise (measured 48/48) implies a sign-constant and simple ground vector (Perron–Frobenius), with a control matrix where hypothesis and conclusion fail together; and  $n_{\text{run}} \leq n_{\text{cross}} + 1 \leq s_2 + 2$  together with  $\sum_j |\Delta p_j - \overline{\Delta p}| = \text{TV}(P)$  hold without exception on a non-vacuous battery. **Named limits as content:** every statement is *per instance on small windows* — nothing is uniform in the zone index; the  $\kappa$  law is *not* claimed (T129 broke it), only the chain underneath it; positivity of the pole-free section at depth stays open and carries no floor value here; the Perron step has a *measured* hypothesis; and the defect identity says nothing about the *size* of the defect. Abel / H lder / Schur–Haynsworth 1968 / Albert 1969 / C ea 1964–Strang / Yserentant 1986 / Perron–Frobenius–Ostrowski 1937 named classical; Weil 1952 cited, never used as a criterion. [E] (per-instance identities; nothing uniform, no marker moves)

**The lumped  $M$ -matrix pair (PRIME.MMATRIX.IDENT.01).** The identity block of the discovery part T136 is load-bearing as a second narrow module ([verification/v543\_lumped\_pair\_identities.py]; 35 checks,  $\sim 1$ s; companion to PRIME.MARGIN.IDENT.01) on 20 frame-A seam windows ( $n = 4 \dots 59$ ,  $\rho = 1.25 \dots 4.00$ ,  $h = 25 \dots 299$ , every factorised matrix  $\leq 300$ ) giving 39 matrices with positive off-diagonals, 312 shifted ladder rungs and 8 whitened two-grid rungs. *The pair and the congruence.* With  $\Delta$  the positive off-diagonal part and  $L_\Delta = \text{diag}(\Delta \mathbf{1}) - \Delta$ , the lumped pair  $A_B = A + L_\Delta$  is Stieltjes and row-sum preserving ( $1.6 \times 10^{-15}$ ) with  $A_B \succeq A$  — an *upper* bound on  $\lambda_{\min}(A)$ , never a floor;  $A = A_B^{1/2}(I - W)A_B^{1/2}$  holds as a matrix identity ( $4.4 \times 10^{-15}$ ) with  $\lambda_{\max}(W) = \rho(L_\Delta A_B^{-1})$ , and the equivalence  $\lambda_{\max}(W) < 1 \iff A \succ 0$  is checked in *both* directions, so the Loewner sandwich is circular by itself. *Anchor, ladder, Varga.* With  $x = A_B^{-1} \mathbf{1} \geq 0$  and  $A_B^{-1} \geq 0$  entrywise (Ostrowski 1937, checked),  $\lambda_{\min}(A_B) \geq 1/\max_r x_r$  on all 39 rows (sharpest ratio 0.9070) from one solve and one sign check; lumping commutes with diagonal shifts and  $M$ -matrix inverses are entrywise monotone (Berman–Plemmons 1979), so the shifted ladder is *structurally* empty (0 exceptions of 312); and for the regular splitting  $A_B = D_0 - N_0$  Varga’s  $\rho(J) = \tau/(1 + \tau)$  is an identity ( $4.8 \times 10^{-15}$ ) whose Collatz–Wielandt bracket at the anchor vector gives an a-priori  $\rho(J) \leq 0.9861 < 1$  with no eigenvalue anywhere. *One negative result [X].* The  $k$ -scanned M18 majorant is *valid* at every  $k$  (39 pairs, 0 violations) and still stays above its preregistered bar  $1/2$  (best 3.080) while the measured bad mass is  $\leq 0.0120$ : the loss is the bound, the localisation term alone exceeds the bar, and the bar is not moved. **Named limits as content:** per instance on small windows, nothing uniform in the zone index, in  $D$  or in the gap; the T136  $D$ -exponents are fits and stay in the sandbox; the anchor and Varga rows are implications with hypotheses checked at the instance. Stieltjes / Ostrowski 1937 / Fan 1958 / Berman–Plemmons 1979 / Varga 1962 / Collatz 1942–Wielandt 1950 named classical; Weil 1952 cited, never used as a criterion. [E] (per-instance identities) + [X] (one closed route; no marker moves)

**The long-lag support structure (PRIME.LONGLAG.SUPP.01).** The structure block of the discovery part T137 is load-bearing as the companion module ([verification/v544\_long\_lag\_support.py]; 24 checks,  $\sim 0.2$ s) on the same 20 windows (3–25 prime-power atoms, 20555 positive off-diagonal edges, 455 loaded stripes, 39 bracketed lumped comparisons). *Support and amplitude.* The kernel splits exactly as  $c = c_{\text{arch}} - c_{\text{comb}}$  with  $c_{\text{comb}} \geq 0$ , and the archimedean section alone has *all* off-diagonals  $\leq 0$  on 20/20 windows, so every positive off-diagonal is comb-generated; all 20555 positive edges have their Hankel index  $M - 1 - r - t$  inside the atom-hat index set (per-window fraction exactly 1.000000), so the support is a union of anti-diagonal stripes at prime-power atom positions, each stripe a perfect matching with *exactly* orthogonal edge vectors; and  $\Delta_{rt} \leq c_{\text{comb}}[M - 1 - r - t]$  on all



20555 edges (sharpest ratio 0.99998532), with the Toeplitz lag in place of the Hankel index failing on 20415 of them. *The Green bracket*. For the lumped comparison,  $q_J = 1 - \min_r 1/(d_r x_r) < 1$  is an identity ( $8.2 \times 10^{-16}$ ) needing no measurement, and the Neumann partial sum brackets the Green function entrywise from below *because the discarded terms are nonnegative*, the anchor-weighted tail from above, on 39 rows. *A death certificate* [X]. Gershgorin, every row-sum norm and every Collatz–Wielandt quotient at every *positive* weight are upper bounds for  $\rho(|E|)$ ,  $E = B^{-1}L_\Delta$ ; one rigorous lower bound therefore decides them all at once:  $\rho(|E|) \geq 1.3141 \geq 1$  on 20/20 rows, so no member of the absolute-value envelope family can *ever* certify  $\lambda_{\max}(W) < 1$  — while the *signed* radius  $\rho(E) = \lambda_{\max}(W) \leq 0.99995781$  on the same rows. The absolute value alone destroys the cancellation. **Named limits as content:** per instance on small windows, no decay law and no amplitude law; the support statement is one inclusion; the amplitude bound does *not* bound  $\lambda_{\max}(W)$ , and the structure bound assembled from it does not beat the margin (T137: 0 of 35 windows) and is not promoted; the bracket is for the lumped comparison, not the target. Gershgorin 1931 / Neumann series / Haynsworth 1968 named classical; Demko–Moss–Smith 1984 cited as a model and *not* applied (the comparison is dense); Weil 1952 cited, never used as a criterion. [E] (per-instance structure) + [X] (one family closed; no marker moves)

**The Hardy-core identities (PRIME.HARDY.IDENT.01).** The identity blocks of the discovery parts T140 and T141 are load-bearing as the third companion module (36 checks,  $\sim 1$  s) on 11 frame-A windows ( $n = 8 \dots 139$ ,  $h = 58 \dots 285 \leq 300$ , 193–5102 positive off-diagonal edges,  $D$  spanning a factor 3.15). *Form lifting and finite core*. With  $C = \text{diag}(\sqrt{\Delta})M$  and  $M$  the edge–interval incidence,  $\text{Gram} = CHC^\top$  ( $3.0 \times 10^{-15}$ ), so  $\text{rank}(\text{Gram}) \leq h - 1$ ; the covering kernel in closed geometric form  $K_{rs} = W([r \wedge s, r \vee s])$  equals  $M^\top \Delta M$  ( $1.3 \times 10^{-15}$ ); and  $\rho(W) = \lambda_{\max}(K^{1/2}HK^{1/2})$  is an *exact* eigenvalue, agreeing with  $1 - \lambda_{\min}(A, A_B)$  ( $5.4 \times 10^{-14}$ ) and with  $\lambda_{\max}(\text{Gram})$  ( $7.8 \times 10^{-16}$ ) — three independent routes, no truncation — while  $H = \text{diag}(s) + L_N$  ( $4.7 \times 10^{-16}$ ) reads the sign law as mass plus a long-range Dirichlet form. *The four Hardy identities*.  $\rho(W) = u^\top H u / u^\top K^+ u$  at  $u = K^{1/2}v$  ( $6.6 \times 10^{-13}$ );  $L_N = D^\top B D$  with the signed crossing kernel  $B_{kl} = \sum_{r \leq k \wedge l, k \vee l < s} N_{rs}$  ( $4.9 \times 10^{-16}$ );  $Q = B_+ \mathbf{1}$  ( $1.4 \times 10^{-15}$ ), so the Cauchy–Schwarz step is the Gershgorin diagonal step; and  $DKD^\top = L_\Delta$  on the interior nodes ( $2.7 \times 10^{-14}$ ), whose diagonal is the endpoint edge mass. The crossing kernel of  $|N|$  does *not* reproduce  $L_N$ : the signs are load-bearing. The closed Muckenhoupt quantity  $A_{M0} = \max_k Q_k(\Delta \mathbf{1})_{k+1}$  follows from two routes agreeing to  $8.3 \times 10^{-15}$ ; its *size* is reported, not promoted. The checkerboard split  $\text{Band}_b = D + A_{\text{even}} + A_{\text{odd}}$  is entrywise exact on 9 (window,  $b$ ) pairs, with three Weyl steps dominating. *Two shapes, promoted as valid and not by size*. Every Loewner step is Cholesky-certified; the additive Hardy chain dominates  $\rho(W)$  on 11/11 ( $3.074$ – $10.758\times$ ), its exact Weyl split dominates too ( $1.691$ – $2.503\times$ ) and floors the additive family, and the joint shape  $\Lambda(H, Y) \cdot \Omega(Y)$  dominates on 11/11 for a positive definite Jacobi  $Y$  — the same  $Y$  without its mass term is singular and the certificate refuses. *A negative typing* [X]. Over the  $D$  range the exactly bounded object spreads  $1.36\times$  while every absolute variant spreads  $\geq 54.94\times$ ; the power laws ( $D^{-0.138 \pm 0.094}$  signed against  $D^{-0.776 \pm 0.352}$  and  $D^{-2.748}$ ,  $D^{-2.820}$ ,  $D^{-2.918}$ ) are fits and labelled as fits. **Named limits as content:** per instance on small windows, nothing uniform in the zone index or in  $D$ ; the bound rows promote *validity* and not size — neither shape beats the target — and the finite-core row is an identity, so the zone-uniform discrete Hardy inequality stays open. Abel summation / Cauchy–Schwarz / Weyl 1912 / Gershgorin 1931 / Muckenhoupt 1972 / Bradley 1978 / Opic–Kufner 1990 / Gantmacher–Krein 1950–1960 named classical; Weil 1952 cited, never used as a criterion. Machine-checked: [verification/v545\_hardy\_core\_identities.py]. [E] (per-instance identities) + [X] (one route typed; no marker moves)

**The capacity-chain identities (PRIME.CAPCHAIN.IDENT.01).** The identity spine of the capacity cycle T142–T145 is load-bearing as the fourth companion module (26 checks,  $\sim 0.5$  s) on 12 frame-A windows ( $n = 4 \dots 139$ ,  $h = 26 \dots 285 \leq 300$ ,  $D$  spanning  $7.6 \times 10^{-3}$ – $2.8 \times 10^{-2}$ ; the covering kernel positive definite on 11, where the first two items run). *The capacity decomposition*.  $K^{-1} = D^\top J^{-1}D + xx^\top / \text{cap}$  with  $J = DKD^\top$  ( $3.8 \times 10^{-15}$  entrywise),  $x = K^{-1}\mathbf{1}$ ,  $\text{cap} = \mathbf{1}^\top K^{-1}\mathbf{1}$ , as a matrix identity ( $1.4 \times 10^{-13}$ ); under the congruence the Dirichlet half is an *orthogonal projection* —  $\Omega = 1$  exactly ( $5.1 \times 10^{-12}$ ), where T141 had guessed 20.7–2724 — and its complement is the

capacity rank-one. *The exact capacity-Rayleigh identity.* The assembled quotient for  $1 - \rho(W)$  equals  $v^\top(K^{-1} - H)v/v^\top K^{-1}v$  for every  $v$  ( $9.5 \times 10^{-14}$ ), reproduces  $1 - \rho(W)$  at the exact top eigenvector under the declared 1/gap conditioning, and satisfies the inf property; on the constant vector the whole denominator is the capacity term. *Cholesky nestedness.*  $\text{cap}_E([a, a+j])$  is a prefix sum of squared forward-substitution entries of one triangular solve (3404 interval capacities at  $2.0 \times 10^{-14}$ , exhaustive on the median window); the reversed ordering fails at  $\geq 4.6 \times 10^{-1}$ . *The whitening certificate.*  $\lambda_{\max}(W) \leq \kappa_{\text{up}} \leq$  Gershgorin ceiling (1.2245–1.4236 against  $\leq 2.1213$ ), direction-correct, refusing at the shrunk shift. *The  $M_4$  split and the certified chain.* The mass half of Maz'ya's Markov step is a pointwise theorem,  $\sigma_{\text{tot}} = 0.2197\text{--}0.4425 < 1$  on every window; licence 4 holds with  $|R|$  and not  $R^+$  (witness  $[[1, -\frac{1}{2}], [-\frac{1}{2}, 1]]$  built in); and  $S_1'$  is certified per window on both routes, better route  $c_0 = 2.5154\text{--}4.2265$  against the classical Dirichlet value 4, with Charikar's cited density constant bracketed by exhaustive enumeration. **Named limits as content:** per instance on small windows, nothing uniform in the zone index or in  $D$ ; the  $c_0$  is measured at the minimiser's own layer cake and the a-priori level lemma (T145's L1) stays open — Maz'ya's strong-type half is not used as a theorem anywhere. Maz'ya 1985 / Muckenhoupt 1972 / Fukushima–Ōshima–Takeda 1994 / Miclo 1999 / Charikar 2000 / Goldberg 1984 / Gershgorin 1931 named classical; Weil 1952 cited, never used as a criterion. Machine-checked: `[verification/v546_capacity_chain_identities.py]`. **[E]** (per-instance identities and certified inequalities; no marker moves)

**The level-lemma identities (PRIME.LEVEL.LEMMA.01).** The a-priori core of T146 is load-bearing as the fifth companion module (20 checks,  $\sim 0.4$ s) on 12 frame-A windows ( $n = 4 \dots 139$ ,  $m = 26 \dots 285 \leq 300$ ), 9 random positive definite forms for the form-independent items, and a stress ladder  $m = 64 \dots 288$  — the companion to the capacity-chain module, whose certified chain carried a constant measured at the minimiser's layer cake: here the same constant is a functional of the form alone. *The union identity for nested cakes.*  $\sum_{j,l} c_j c_l |S_j \cup S_l| = 2T\|\psi_t\|_1 - \|\psi_t\|^2$  — form-independent and  $O(m)$ , the reason the cake base is movable; exact (0.0) against the  $K \times K$  double sum and against union sizes computed with no nestedness used (controls  $3.2 \times 10^{-2}$  /  $4.8 \times 10^{-3}$ ). *The counting lemma at a general base.*  $G(b) \leq 2b^2\|v\|_\infty\|v\|_1/\|v\|^2 + \varepsilon$  for any vector at all 6 preregistered bases; the classical dyadic 8 recovered at  $b = 2$  exactly, the base free (gain  $4/b_*^2 = 3.7612$ ); the one-level-down cake breaks the domination on every draw, so the +1 is load-bearing. *The resolvent delocalisation bound.*  $\psi = \lambda R\psi$  ( $9.0 \times 10^{-14}$ , an identity — no Davis–Kahan step anywhere)  $\Rightarrow \|\psi\|_\infty \leq \lambda_{\text{up}} \max_j \|Re_j\|$  with Rayleigh–Ritz  $\lambda_{\text{up}}$  (1.0592–1.2883 of the gap) and residual-paid certified column norms, within 1.061–1.337 of the truth; the lever: the columns sit 2.85–10.17 below the trivial ceiling  $1/\lambda_{\min}$ , giving  $\Gamma = 1.8356\text{--}2.2797$  against the useless  $\sqrt{m} = 5.1\text{--}16.9$ . *The composite level lemma.*  $c_0^{\text{ap}} = 2b^2\Gamma \min(1, \Gamma_1) + \varepsilon = 3.9042\text{--}4.8488$  (slack 1.986–2.418 over the measured constant, under the ceiling  $16 = 4 \times$  Maz'ya's Dirichlet 4), the certified window inequality  $\text{cert\_lam\_max}(R) \leq c_0^{\text{ap}} \Psi_{\text{abs}}$  on 12/12, chain loss 0.0423–0.1272 of the exact gap; the built-in no-go discriminator ( $R = aa^\top + \varepsilon I$ ,  $a_i = i^{-1/2}$ :  $\Gamma$  grows  $3.676 \rightarrow 6.798$ , ratio  $1.85 \geq 1.5$ ,  $c_0^{\text{ap}} = 11.10$  exceeds every window value) fails exactly where it must, the theorem half surviving and the Dirichlet control flat (1.0063). **Named limits as content:** per instance on small windows — a finite list with an explicit maximum; nothing uniform in the zone index or in  $D$ , and explicitly no statement for all  $D$  (the asymptotic delocalisation of the Green columns stays open, typed open); the energy-route  $\sigma_{\text{tot}}$  stays measured. Maz'ya 1985 / Rayleigh 1877–Ritz 1909 / Cauchy–Schwarz / Miclo 1999 / Davis–Kahan 1970 (cited, deliberately not used) / Charikar 2000 / Goldberg 1984 / Gershgorin 1931 named classical; Weil 1952 cited, never used as a criterion. Machine-checked: `[verification/v547_level_lemma_identities.py]`. **[E]** (per-instance identities and certified inequalities; no marker moves)

**The Green/Szegő identities (PRIME.GREEN.SZEGO.IDENT.01).** The identity/certificate core of T147+T148 is load-bearing as the sixth companion module (21 checks,  $\sim 0.4$ s) on 12 frame-A windows ( $n = 4 \dots 139$ ,  $m = 26 \dots 285 \leq 300$ ), 7 random positive definite forms for the form-independent items, and a weight-class model ladder  $m = 64 \dots 288$  — the companion to the level-lemma module: here the two scalars its constant is made of are certified. *The factorisation identity.*  $\Gamma = \sqrt{m} \lambda_{\text{up}} \max_j \|Re_j\| = \sqrt{Q^*} \cdot S_w$  with  $S_w = \lambda_{\text{up}} \|R\|_F$  purely spectral and  $Q^* =$

$m \max_j (R^2)_{jj} / \text{tr}(R^2)$  purely geometric — exact at  $2.2 \times 10^{-16}$ ; the certified factorisation dominates the true  $\Gamma$  everywhere ( $S_w = 1.1186\text{--}1.5777$ ,  $Q^* = 2.0879\text{--}2.8634$ ). *The bottom-block constant.*  $\sum_j (\Pi_B)_{jj} = |B|$  exactly ( $3.3 \times 10^{-16}$ ;  $B$  the preregistered  $\theta = 10$  block,  $|B| = 3\text{--}5$ ), so  $Q_B \geq |B|$  is a theorem; measured  $Q_B/|B| = 1.5091\text{--}1.7980$  inside the preregistered band  $[1, 2.2]$  against the Szegő/Widom sharp reference  $2|B|$  — the coordinate-projector control overshoots by  $\geq 3.9$ . *The second-difference  $\ell^1$  ceiling.*  $|a_k| \leq \min(1, \|s_k\|_\infty \|L_0^P \psi\|_1 / \mu_k^P)$  from the exact Dirichlet eigenpair, with the  $m$ -free closed form  $\|a\|_1 \leq 2\sqrt{\nu} + 1$  ( $\nu = (m+1)^{3/2} \|L_0 \psi\|_1 / (2\sqrt{2})$ ) and the chain  $m\|\psi\|_\infty^2 \leq 2\|a\|_1^2$ ; Dirichlet calibration  $\|a\|_1 = 1$  exact,  $\nu \rightarrow \pi$ . *The layer-cake  $S_w$  certificate.*  $\text{LDL}^T$  inertia counts equal the direct spectral counts on all 96 (window, rung) pairs, and the layer cake certifies  $S_w \leq 1.3288\text{--}1.8220$  (true  $1.1186\text{--}1.5777$ ); dropping the bottom layer breaks it everywhere. *The Liouville transform.*  $m\|\psi\|_\infty^2 \leq 2\kappa_\Lambda \text{ceil}(\hat{\varphi})^2$  with  $\varphi = \Lambda^{-1/2}\psi$  and a-priori  $\kappa_\Lambda = \max \Lambda / \min \Lambda = 1.3868\text{--}2.5805$ , stepwise on every bottom-block mode; the weight-class discriminator at matched  $\kappa_\Lambda$  (mismatch 0.0013): the BV multiplier flat (1.003), the rough one (TV larger by 162.6) grows by 19.9 — the roughness, not the conditioning. **Named limits as content:** per instance on small windows — a finite list with an explicit maximum; nothing uniform in the zone index or in  $D$ , and explicitly *no* statement for all  $D$  (the one open lifting input,  $\nu_L \leq F(\text{form})$   $m$ -free / bounded TV( $\log \Lambda$ ), stays open, typed open); the symbol-level order-2 hypothesis is refuted (T148:  $f(0) < 0$  on every surface window) and enters nowhere as a theorem;  $\sigma_{\text{tot}}$  stays retired as a route. KMS 1953 / Widom 1958–Basor–Ehrhardt 2009–Böttcher–Silbermann (the mechanism’s address, never an authority) / Sylvester 1852–Bunch–Kaufman 1977 / Euler 1735 / Rayleigh 1877–Ritz 1909 / Davis–Kahan 1970 (cited, computed vacuous, not used) / Gershgorin 1931 named classical; Weil 1952 cited, never used as a criterion. Machine-checked: [verification/v548\_green\_szego\_identities.py]. [E] (per-instance identities and certified inequalities; no marker moves)

**The gauge/parity identities (PRIME.GAUGE.PARITY.IDENT.01).** The identity/certificate core of T149+T150 is load-bearing as the seventh companion module (21 checks,  $\sim 0.3$ s) on 12 frame- $A$  windows ( $n = 4 \dots 139$ ,  $m = 26 \dots 285 \leq 300$ ), 48 gauge assemblies (Jacobi, const, moving-geometric-mean, and the arithmetic-free archimedean diagonal), the full-section parity items on the 6 windows with  $M \leq 300$ , and a parity control model at  $m = 2001$  — the companion to the Green/Szegő module: here the gauge *freedom* around that machinery and the parity *structure* underneath it are certified. *The gauge freedom.* For any positive diagonal  $\tilde{\Lambda}$ ,  $\lambda_{\min}(A) \geq \text{cert } \lambda_{\min}(\tilde{E}) \cdot \min_j \tilde{\Lambda}_j$  is a valid certified lower bound (an identity plus one Rayleigh step; family maximum valid);  $\Gamma = \sqrt{Q^*} S_w$  holds gauge-invariantly at  $2.4 \times 10^{-16}$ ; the const gauge has  $\text{TV}(\log \tilde{\Lambda}) = 0$  *exactly* with the certified entrywise sandwich  $\sigma = 1.3868\text{--}2.5805$ ; on the shifted indefinite form the completed-Cholesky floor refuses on 12/12. *The two-regime discriminator.* The flutter smoother removes TV by  $\geq 2.88\times$  while the Liouville-transported  $\nu_{\tilde{L}}$  moves  $\leq 0.0088$  — even the rough re-gauge moves it  $\leq 0.0046$  — while the untransported functional jumps 0.41–9.76 under the same rough gauge: the roughness is in the form, not in the multiplier (T149’s withdrawal of the TV hypothesis, wired per instance). *The explicit zone diagonal.*  $\Lambda_r = c_0 - c_{M-1-2r} + \sum_{s \neq r} \max(c_{|r-s|} - c_{M-1-r-s}, 0)$  at residual  $5.9 \times 10^{-16}$ , the exact split  $c = c^{\text{arch}} + c^{\text{atom}}$  with  $c^{\text{atom}} \leq 0$ , the closed atom budget  $\|c^{\text{atom}}\|_1 \leq 4B\sqrt{N}$  ( $B = 1.038821$  verified on the table); the archimedean section alone is indefinite on 12/12 (the atoms are co-responsible for the positivity — the additive perturbation reading is structurally empty). *The parity compression.*  $U_-^T T_M(c) U_- = T - H$  exactly (cross block  $1.4 \times 10^{-16}$ ): the reflection-odd section is the compression of the full symmetric Toeplitz section onto its antisymmetric parity sector, and the negative inertia of the sign-changing symbol sits entirely in the *even* sector (odd sector zero,  $\text{LDL}^T$ ) —  $f(0) < 0$  explained rather than worked around. *The parity-sine calibration.* The parity sines are exact eigenpairs of the parity Laplacian (corner 3 forced by antisymmetry),  $\nu_k^P = \pi k^2$  on the exact model to  $3.0 \times 10^{-5}$ , and the per-window ladder  $\nu_k^P \leq Ck^2$  holds with certified  $C = 11.70\text{--}17.28$  and the derived per-mode ceiling  $m\|\psi_k\|_\infty^2 \leq 2(2\sqrt{C}k + 1)^2$ . **Named limits as content:** per instance on small windows — a finite list with an explicit maximum; nothing uniform in the zone index or in  $D$ , and explicitly *no* statement for all  $D$  (the one open input, an  $m$ -free  $C$  in the odd-sector ladder for a sign-changing



symbol, stays open, typed open);  $\text{TV}(\log \Lambda)$  stays withdrawn as a hypothesis;  $\sigma_{\text{tot}}$  stays retired as a route. KMS 1953 (parity sector) / Widom 1958–Basor–Ehrhardt 2009–Böttcher–Silbermann (the parity sectors as the algebra’s natural objects; the sign-changing symbol is exactly what Basor–Ehrhardt does not cover) / Sylvester 1852–Bunch–Kaufman 1977 / Chebyshev 1852 (verified on the table) / Abel / Weyl 1912–Bauer–Fike 1960–Davis–Kahan 1970 (cited, computed dead, not used) / Rayleigh 1877–Ritz 1909 / Gershgorin 1931 named classical; Weil 1952 cited, never used as a criterion. Machine-checked: [\[verification/v549\\_gauge\\_parity\\_identities.py\]](#). [E] (per-instance identities and certified inequalities; no marker moves)

**The odd-sector identities (PRIME.ODD.SECTOR.IDENT.01).** The identity/certificate core of T151 is load-bearing as the eighth companion module (19 checks,  $\sim 0.2$ s) on 12 frame-A windows ( $n = 4 \dots 139$ ,  $m = 26 \dots 285 \leq 300$ ), parity control models at  $m = 241$  and  $m = 2001$ , and a no-go size ladder  $m = 64 \dots 256$  — the companion to the gauge/parity module: here the *reroute* off the quadratic ladder is certified — through the pencil against the parity Laplacian and one elementary Sobolev step, to a per-mode bound *linear* in  $k$ . *The grid step-over.* The parity sines live on the shifted grid  $\theta_k = 2\pi k/N$  with  $\mu_k^P = 2 - 2\cos \theta_k$  exactly; the symbol has  $f(0) = -47.6 \dots -6.3 < 0$  with a nonempty negative window  $[0, \theta_c)$ , and  $\theta_c/\theta_1 = 0.371\text{--}0.485$  on every window: the odd grid steps over the negative window — odd-sector positivity is a grid fact. The honest negative beside it, promoted as content: the symbol-only pencil floor is vacuous ( $\rho \geq 1$  on 12/12) and the symbol moves by  $0.45\text{--}3.27$  across one mode spacing — the local model is a *matrix* statement, not a symbol statement. *The certified bottom ladder.*  $\lambda_k(A) \leq S\mu_k^P$  for  $k \leq 8$  by completed  $\text{LDL}^T$  counts (Sylvester 1852),  $S = 1.1019\text{--}6.4725$  — order 2 in  $k$ , with  $f(0) < 0$  absorbed by the Rayleigh floor  $L_P \geq \mu_1^P I$ . *The Sobolev step.*  $\|\psi\|_\infty^2 \leq 2\|\psi\|_2 \|\nabla \psi\|_2$ , licensed by the odd sector’s virtual node  $\psi_{-1} = 0$ ; on the exact parity model the pencil is  $(1, 1)$  (certified  $1.1 \times 10^{-12}$ ) and the per-mode price is exactly  $\pi$  ( $5.1 \times 10^{-8}$ ). *The closed archimedean minimum.*  $\min_r \Lambda_r^{\text{arch}} = c_0^{\text{arch}} - c_{M-1}^{\text{arch}}$  exactly (residual 0.0), attained at  $r = 0$ , from two per-window-checked shape properties of the smooth kernel. *The linear per-mode bound.* With the certified pencil floor  $\kappa = 0.1939\text{--}0.3599$  and  $K_{\text{bot}} = S$ ,  $b_k = 2m\sqrt{K_{\text{bot}}\mu_k^P/\kappa} \leq C_S k$  with  $C_S = 12.081\text{--}26.953 \leq 2\pi\sqrt{K_{\text{bot}}/\kappa}$  — linear where the quadratic ladder grew, nothing additive anywhere; on the T145 no-go the bottom pencil ratio explodes ( $\times 15.8$  over  $m = 64 \dots 256$ ) while the parity model holds 1. **Named limits as content:** per instance on small windows — a finite list with an explicit maximum; nothing uniform in the zone index or in  $D$ , and explicitly *no* statement for all  $D$  (the one open input, the  $m$ -freeness of the bottom pencil ratio  $R = K_{\text{bot}}/\kappa = 3.7362\text{--}18.6607$ , stays open, typed open); the symbol route stays certified vacuous. KMS 1953 / Widom 1958–Basor–Ehrhardt 2009–Böttcher–Silbermann (the symbol question’s address — the computed answer: the pointwise symbol language does not extend to the bottom mode) / Weyl 1912 / Parlett / Hardy–Littlewood–Pólya 1934–Agmon 1965 / Sylvester 1852–Bunch–Kaufman 1977 / Rayleigh 1877–Ritz 1909 / Wilkinson 1968–Higham 2002 / Chebyshev 1852 named classical; Weil 1952 cited, never used as a criterion. Machine-checked: [\[verification/v550\\_odd\\_sector\\_identities.py\]](#). [E] (per-instance identities and certified inequalities; no marker moves)

**The fixed-size Ritz ceiling certificate (PRIME.RITZ.CEIL.01).** The certificate core of T154 is load-bearing as the ninth companion module (16 checks,  $\sim 0.4$ s) on the 12 *deepest* prime-power zones admitting a frame-A window inside the cap ( $n = 29 \dots 139$ ,  $m = 96 \dots 291 \leq 300$  — declared, not tuned: on the shallowest zones  $n \leq 11$  more than one bottom direction leaves the sine block and the fixed-depth certificate is not claimed there), a no-go size ladder  $m = 48 \dots 288$ , parity controls at  $m = 64, 128, 256$  and 3 Dirichlet control windows — the companion to the odd-sector module: here the *fixed-size* replacement for the ceiling step is certified. *The direction lemma.*  $\lambda_k(A) \leq \theta_k$  for every  $k \leq \dim$  — Ritz values are upper bounds for the eigenvalues of the *same index* (Courant–Fischer 1920 / Cauchy 1829), verified against exact spectra with zero one-sided excess — so the ceiling needs *no* residual argument; the reversed reading is refuted on 12/12 (overshoot  $\geq 18.6$ ). *The sixteen-column certificate.*  $\text{span}\{t_1 \dots t_8\} + A^{-1}L_P \text{span}\{t_1 \dots t_8\}$  (dimension 16 fixed, independent of  $m$ ; 8 completed Choleskys of matrices  $\leq 8 \times 8$ ) carries  $K^F = 1.1019\text{--}1.5845$  attaining the inertia-certified  $K_{\text{bot}}$  to  $\leq 9.9 \times 10^{-7}$  against the declared cap  $10^{-4}$  on 12/12 — the 8 size- $m$

LDL<sup>T</sup> counts retired from the ceiling step (controls: the sines alone overshoot 41.6–739; corrupted Green columns break the attainment on 12/12). *The one-direction anatomy*. Median of the seven aligned angles 0.03–0.14°, the largest 83.79–89.70° (measured): the single misaligned direction *is* the collapse price  $\lambda_1(A)/(t\mu_1^P) = 4.408\text{--}7.042$ , recovered in full per window by one Cholesky of  $A - \gamma I$  (size  $m$ , certified, not fixed size). *The no-go discriminator*. On the T145 reconstruction  $c(l) = 1/(1+l)$  the floor survives ( $t = 0.05$ ) but  $K_{\text{bot}}$  explodes  $\times 35.4$  and the ceiling explodes with it  $\times 57.5$  — no manufactured closure; the Dirichlet control (the Hankel half removed) is certified non-positive on 3/3: the reflection half is load-bearing for positivity. **Named limits as content:** per instance on small windows — a finite list with an explicit maximum; nothing uniform in the zone index or in  $D$ , and explicitly *no* statement for all  $D$ ; the attainment is not promoted to a theorem (the no-go overshoots — it is a property of the real section); the two open inputs after T154 (R1', an  $m$ -free floor for  $\lambda_{\min}(B_{HH})$  on the bulk parity block; R2', an  $m$ -free lower bound for the  $L_P$  floor on the Ritz complement, arithmetic-free and worth 91–100% of the collapse price) stay open, typed open. Courant–Fischer 1920–Cauchy 1829 / KMS 1953 / Schur 1917–Haynsworth 1968 / Sylvester 1852–Bunch–Kaufman 1977 / Weyl 1912 / Temple 1928–Kato 1949 (a floor device, not used here) / Kaniel 1966–Paige 1971 (quoted, never a bound) / Björck–Golub 1973 / Wilkinson 1968–Higham 2002 / Chebyshev 1852 named classical; Weil 1952 cited, never used as a criterion. Machine-checked: [verification/v551\_ritz\_ceiling\_certificate.py]. [E] (per-instance theorems and certified inequalities; no marker moves)

**The four fixed-size angle instruments (PRIME.ANGLE.INSTR.01).** The instrument cores of T155 and T157 are load-bearing as the tenth companion module (18 checks,  $\sim 0.4$ s) on the same declared surface as the Ritz module (the 12 deepest in-cap frame-A windows,  $n = 29 \dots 139$ ,  $m = 96 \dots 291 \leq 300$ ; a no-go size ladder  $m = 48 \dots 288$ ; parity and Dirichlet controls at  $m = 64, 128, 256$ ) — none of the four closes an angle, and none is promoted as anything but a per-instance instrument. *The complement-floor certificate (T155)*.  $\min_{v \perp W} v^T L_P v \geq \mu_{13}^P - \lambda_{\max}(M^{1/2}(I - GG^T)M^{1/2})$  with  $G = T_{12}Q_W$  — an identity plus one certified Rayleigh bound, a theorem for every  $m$  and every subspace; never above the exact size- $m$  complement bottom, agreement  $\leq 1.01 \times 10^{-3}$  against the declared cap  $2 \times 10^{-3}$  on this small surface, both closed-form controls to  $2.0 \times 10^{-11}$  (including the *worthless* configuration  $W = \text{span}\{t_2..t_{13}\} \rightarrow \mu_1^P$ : the instrument reports its own worthlessness); a corrupted overlap row moves it by  $\geq 0.549$ . *The sine-block confinement (T157)*. From the certified pencil floor and ladder ceiling alone,  $\|\gamma_H\|^2 \leq \lambda_1/(t\mu_{17}^P) \leq (S/t)/\rho_{17} = 0.0153\text{--}0.0245 < 1$  on 12/12 — the bottom eigenvector lives 97.5–98.5% inside the first 16 parity sines by a per-instance theorem (measured tail  $6.7 \times 10^{-6}\text{--}3.0 \times 10^{-5}$ ); with the pencil ceiling  $\kappa$  instead of the ladder  $S$  the bound is vacuous (17.6–477), and the inflated floor is refuted at  $e_1(A)$  by  $3.3 \times 10^3\text{--}1.0 \times 10^5$ . *The Schur-entry identity (T157)*.  $s = \mu_1^P t_1^T A^{-1} t_1 = (B^{-1})_{11}$  exactly ( $1.1 \times 10^{-11}$ ),  $(B^{-1})_{LL} S_L = I$  to  $5.8 \times 10^{-11}$  with  $S_L$  the fixed-size  $16 \times 16$  Schur complement the chain already forms:  $1/s = 4.0460\text{--}5.8962$  is an  $O(1)$  number carried by *one* entry, while the inversion-free Cauchy–Schwarz ceiling misses by  $250\text{--}7.27 \times 10^3$  (the cancellation is nearly complete); a one-index shift breaks the identity by 1.55–2.15. *The adaptive Lipschitz ceiling (T157)*. Interval bisection with the local Lipschitz constant certifies  $\inf f^{\text{arch}}/(4\sin^2(\theta/2)) \geq 0.25$  on 12/12 windows in 519–1201 evaluations (a uniform grid at the global constant would need up to  $61 \times$  more); fed a target above the true grid minimum it *refuses* on 12/12. *The no-go discriminator*. On  $c(l) = 1/(1+l)$  the floor survives ( $t = 0.05$ ) while the confinement goes vacuous ( $\times 52.0$ ) and  $(S_L)_{11}$  explodes ( $\times 60.0$ ) against the flat 4.05–5.90 real band; parity control exact to  $4.9 \times 10^{-14}$ , the Dirichlet residual hits  $2/\sqrt{N}$  to  $7.4 \times 10^{-5}$ . **Named limits as content:** per instance on small windows — nothing uniform in the zone index or in  $D$ , and explicitly *no* statement for all  $D$ ; nothing about the angles is closed — the two open terms after T157 (an  $m$ -free lower bound on  $\hat{g}_1^2$ , equivalently an  $m$ -free upper bound on the one Schur diagonal entry; an  $m$ -free R1'' domination with a non-shrinking margin) stay open, typed open. KMS 1953 / Schur 1917–Haynsworth 1968 / Courant–Fischer 1920–Cauchy 1829 / Cauchy–Schwarz (refuted as a ceiling, wired as a control) / Kantorovich 1948 / Temple 1928–Kato 1949 (floor devices, not used here) / Sylvester 1852–Bunch–Kaufman 1977 / Wilkinson 1968–Higham 2002 / Chebyshev 1852 / Szegő 1915–Widom 1958 (a named address, never a licence) named classical; Weil 1952 cited, never

used as a criterion. Machine-checked: [verification/v552\_angle\_instruments.py]. [E] (per-instance identities and certified inequalities; no marker moves)

**The exact-form identities (PRIME.EXACT.FORM.IDENT.01).** The theorem cores of T158 and T159 are load-bearing as the eleventh companion module (25 checks,  $\sim 0.2$ s), recomputed on the same declared surface as the two modules above (12 deepest in-cap frame-A windows,  $m = 96 \dots 291 \leq 300$ ; no-go ladders  $m = 48 \dots 288$  on *both* T145 forms; parity and Dirichlet controls at  $m = 64, 128, 256$ ) — the algebra of the sixteen-form itself, and it closes neither open term. *The Thomson dual form and the positive Cholesky ladder (T158).*  $s = (B^{-1})_{11} = \max_x (2x_1 - x^T Bx)$  with the direction *executed* (8 random trials per window never exceed  $s$ ); the ladder  $g_K$  from one Cholesky of the leading  $16 \times 16$  block: terms strictly positive, partial sums monotone,  $1/s \leq 1/g_{16} \leq 1/g_1 = B_{11}$  literally, tight to 1.0642–1.1522; a 5% corruption of one off-diagonal entry makes the Cholesky *refuse* on 12/12. *The two-dimensionality (T158).*  $\text{span}\{t_1, A^{-1}L_{Pt_1}\}$  attains  $s$  exactly ( $9.0 \times 10^{-13}$ ) because  $L_{Pt_1} = \mu_1^P t_1$ ; without the Green column the two-sine span misses by 1.42–4.8, the corrupted column by  $255\text{--}7.7 \times 10^3$ . *The gauge identity and the signature (T159).* The reflection-odd section annihilates constant lag vectors entrywise with *zero* tolerance, hence  $\sum_d w_d = 0$  to  $1.1 \times 10^{-14}$  of  $\sup |w|$ : the form is blind to the lag mass of  $c$  — exactly where the archimedean and atom halves are  $h^2$ -sized and cancel; the Toeplitz-minus-Hankel signature holds to  $1.4 \times 10^{-16}$ ; a one-index Hankel shift breaks the identity loudly, and the T145 form is not even a parity section (signature defect  $7.4 \times 10^{-2}$ ). *The exact lag sum and the two closed scalars (T159).*  $x^T B_{LL} x = \sum_d c_d w_d$  to  $2.7 \times 10^{-16}$  of the absolute scale, with the honest price measured (relative to the cancelled total only  $1.4 \times 10^{-11}$  on this small surface);  $w_0 = \sum_k x_k^2 / \mu_k^P$  and  $2w_0 - w_1 = \|x\|^2$  exact. *The Z-matrix law and the Collatz–Wielandt floor (T159).*  $B_{HH}^{\text{arch}}$  is a symmetric Z-matrix *raw* (no tolerance, 12/12); the closed sign-based floor  $\min_k (B_{HH}^{\text{arch}} \mathbf{1})_k = 0.9093\text{--}1.1284 \geq 0.25$  with the theorem direction verified against the exact  $\lambda_{\min} = 0.9138\text{--}1.1432$ ; the atom block obeys *no* sign law (a coin toss) and the full-block row-sum floor is negative — admissible and vacuous; the checkerboard similarity (an isometry) destroys the raw law: an entrywise fact, not a spectral one. *The no-go discriminator.* The T145 family breaks on three axes of this module’s structure, while on the  $c(l) = 1/(1+l)$  reconstruction the ladder theorem survives and  $1/g_{16}$  explodes ( $1187 \rightarrow 7.11 \times 10^4$ ) — the instrument reports the collapse. **Named limits as content:** per instance on small windows — nothing uniform in the zone index or in  $D$ , and explicitly *no* statement for all  $D$ ; the two open terms after T159 (the one explicit pairing inequality for  $R2''$ , needing correlation instead of sizes; a third device for the atom off-band block for  $R1''$ ) stay open, typed open. Maz’ya 1985 / Miclo 1999 / Schur 1917–Haynsworth 1968 / KMS 1953 / Dirichlet 1829 / Perron 1907–Frobenius 1912 / Collatz 1942–Wielandt 1950 / Gantmacher–Krein 1950 named classical; Weil 1952 cited, never used as a criterion. Machine-checked: [verification/v553\_exact\_form\_identities.py]. [E] (per-instance identities and certified inequalities; no marker moves)

**The sampling/harmonics identities (PRIME.SAMPLING.HARM.01).** The closed theorem cores of T160 and T161 are load-bearing as the twelfth companion module (21 checks,  $\sim 0.5$ s), recomputed on the same declared surface as the module above (12 deepest in-cap frame-A windows,  $m = 96 \dots 291 \leq 300$ ) plus a declared  $\Lambda$ -battery of six log-spaced cut-offs  $X = 10^3 \dots 3.2 \times 10^5$  (finite von-Mangoldt sums only, no matrix anywhere) — what the sixteen-form *pairs against*, and it closes no open term. *The sampling identity (T160).* The atom half of the pairing equals  $-\frac{1}{2} \sum_{n \leq X} (2\Lambda(n)/\sqrt{n}) \hat{w}(\log n)$  ( $\hat{w}$  the linear interpolant of the lag weights; exact to  $5.6 \times 10^{-15}$  of the absolute atom scale on 12/12): identically a finite combination of sums  $\sum_n \Lambda(n) n^{-1/2} \cos(t \log n)$  at 32 explicit frequencies; a one-index grid shift breaks the identity loudly. *The moment laws and the TV theorem (T160).*  $m_0 = 0$ ,  $m_1 = -[S_0^2 + 2 \sum_j P_j^2] \leq 0$  and  $m_2 = -[2S_1 - (M-1)S_0]^2 \leq 0$  (a perfect square), both strictly negative on every window; and U3 as a theorem with *checked* hypotheses:  $\text{TV}(c^{\text{arch}}) = 3.98\text{--}4.90 \leq |c_0| + 2 \sup < 6$ , closed and window-independent — the full atom-including kernel violates the monotone hypothesis on 12/12 (the wired mutation). *The harmonic-frequency theorem and the PNT main term (T161).*  $t_j(2\alpha) = 2\pi j$  *exactly*: the 32 frequencies are the Fourier harmonics of the log-window, and the closed parameter-free prediction  $(\sqrt{X} - 1)/(\frac{1}{4} + t^2)$  (de la Vallée Poussin 1896) matches the finite sums to 0.0007–0.097 of the trivial mass against the declared cap 0.12; a

wrong Mellin pole and a mass-matched uniform fake measure both fail loudly — the agreement is a property of the von-Mangoldt measure. *The head split and the scale-free Bernstein rate (T161).*  $c_d^{\text{arch}} = \Psi_d + D\hat{G}(dD, D)$  with  $\Psi_d = -\frac{1}{2}[(d+1)\log(d+1) - 2d\log d + (d-1)\log(d-1)]$  closed,  $D$ -free and  $m$ -free, exact to  $3.2 \times 10^{-16}$  at every lag;  $\rho^* = (3 + \sqrt{5})/2 = 2.618034$  is the root of  $x^2 - 3x + 1$ , scale-free (interval ratio 5 only), with every window's  $\rho(D) = 2.588\text{--}2.608$  below it. *The fraction bound and the sign refutation (T161).*  $|\text{off-diagonal}| \leq \frac{1}{4}|Q^{\text{arch}}|$  certified per window (0.171–0.212 on 12/12), with the *refuted* sign law wired as the negative control (arch half negative, off-diagonal block positive, 176/240 pairs violate the per-pair form) — nothing single-signed is promoted. **Named limits as content:** per instance on small windows — nothing uniform in the zone index or in  $D$ ; finite  $\Lambda$ -sums allowed and used, *no* RH statement, and the  $\delta$  triage of T161 (required depth against RH strength) is a documentation item, not a module claim; the open terms after T161 (R-A' the prime-free log-moment; R-B' the  $m$ -free  $16 \times 16$  Gram fraction bound; R-C' one split of the pairing with  $\delta_{\text{eff}} < \frac{1}{2}$ ; R-D a fifth R1'' device) stay open, typed open. Dirichlet 1829 / Abel 1826 / Fejér 1915–Schur 1917 / Bernstein 1912 / Chebyshev 1852–Mertens 1874 / de la Vallée Poussin 1896 / Clausen–Lerch / KMS 1953 named classical; Weil 1952 / Bombieri 2000 cited, never used as a criterion. Machine-checked: [verification/v554\_sampling\_harmonics\_identities.py]. [E] (per-instance identities and certified inequalities; no marker moves)

**The Pareto/total-variation identities (PRIME.PARETO.TV.01).** The closed theorem cores of T162 and T163 are load-bearing as the thirteenth companion module (23 checks,  $\sim 0.4$ s), recomputed on the same declared surface as the module above (12 deepest in-cap frame-A windows,  $m = 96 \dots 291 \leq 300$ ) with the one arithmetic input  $\kappa = 0.038821$  *verified* at every jump point of  $\psi(x)/x$  on the von-Mangoldt table — the *price structure* of the pairing, and it closes no open term beyond the negative closure that the floor inequality itself is. *The exchange law (T163).*  $\delta_{\text{bnd}}(x) = \frac{1}{2} + \log(2\kappa g_{16}\text{TV}(x)/P(x))/\log X$  is an identity at all 612 grid points of the declared Fejér knob ( $5.1 \times 10^{-16}$ ), with the crossing criterion  $\delta_{\text{bnd}} < \frac{1}{2} \iff P > 2\kappa g_{16}\text{TV}$  an exact equivalence at every point: price and demand are two coordinates of one object; the wrong constant breaks the law loudly. *The total-variation floor (T163).*  $w_0 = \|a\|^2$  ( $4.9 \times 10^{-16}$ ),  $\text{TV} \geq |w_0|$  by telescoping, and  $x_1 = 1$  forces  $\mu_1^P \text{TV} = 5.00\text{--}28.84 \geq 1$  on all 708 trial vectors built — the closed floor  $1/\mu_1^P = 1/(4\sin^2(\pi/N)) \sim h^2$  — so every sub- $\frac{1}{2}$  point pays  $P > 2\kappa g_{16}/\mu_1^P$ : a flat-price sub- $\frac{1}{2}$  demand is impossible in the parity sector on this surface (closes R-C''' negatively); the inadmissible vector  $0.01x^*$  undercuts the floor and is recognised (the wired negative control). *The chain-derived price axis (T163).*  $\sigma = 1$  is exactly the  $K = 1$  rung of the T158 Cholesky ladder ( $4.1 \times 10^{-16}$ ) and  $\sigma = \infty$  the  $K = 16$  top rung ( $1.2 \times 10^{-12}$ ); the knob is monotone in both coordinates (its curve *is* the Pareto front); a corrupted sixteen-block makes the Cholesky refuse. *The Mellin cell moments and the Abel step (T162).* The  $k$ -th Mellin rung  $\sum_d w_d C_d(-(2k + \frac{1}{2}))$ , closed, agrees with an independent quadrature to  $2.8 \times 10^{-13}$  ( $k = 1 \dots 4$ ); the boundary-free Abel identities hold at levels 1–3 ( $8.3 \times 10^{-17}$ ), the optimal level is one for the closed reason  $32\pi/\alpha = 40.6\text{--}59.4 > 1$ , and a gauge violation breaks by its closed prediction ( $1.2 \times 10^{-15}$ ). *The Lerch/Frullani log-moment (T162).*  $L = \sum_d \Psi_d w_d$  on three routes (Wronskian form  $2.8 \times 10^{-14}$ , integral with the  $d = 1$  peel  $1.8 \times 10^{-15}$ ,  $2\log 2$  to  $2.2 \times 10^{-16}$ ,  $\Psi_1 = -\log 2$  exact); plus the monotone no-go staircase ( $1.00 \rightarrow 3.49 \rightarrow 10.81 \rightarrow 24.98$ , exact null control). **Named limits as content:** per instance on small windows — nothing uniform in the zone index or in  $D$ ; the T163  $h$ -exponents are sandbox fits, *not* consumed; finite  $\Lambda$ -sums allowed and used, *no* RH statement; the open terms after T163 (R-E the successor, both arms prime-free; R-B''' the  $h$ -uniform positivity margin; R-D a fifth R1'' device) stay open, typed open. Fejér 1915 / Mellin 1896 / Abel 1826 / Chebyshev 1852–Rosser–Schoenfeld 1962 / Legendre / Dirichlet 1829 / KMS 1953 / Frullani 1828–Lerch 1887–Clausen 1832 / Schur 1917–Haynsworth 1968 named classical; Weil 1952 / Bombieri 2000 cited, never used as a criterion. Machine-checked: [verification/v555\_pareto\_tv\_identities.py]. [E] (per-instance identities and certified inequalities; no marker moves)

**The gauge/ $P_{\text{pr}}$  identities (PRIME.GAUGE.PPR.01).** The closed theorem cores of T164 and T165 are load-bearing as the fourteenth companion module (23 checks,  $\sim 0.3$ s), recomputed on the same declared surface as the module above (12 deepest in-cap frame-A windows,  $m = 96 \dots 291 \leq 300$ )



with the one arithmetic input  $\kappa = 0.038821$  *verified* at every jump point of  $\psi(x)/x$  on the von-Mangoldt table — the *gauge structure* of the pairing, and it does not close the one remaining quantifier. *The gauge theorem (T164)*.  $x_1 = 1$  fixes only the *scale* of  $a$ ;  $Q$  and  $TV$  are homogeneous of degree two; hence the demand ratio  $Q/TV$  of the same  $a$ -direction is the *same number in every sector* — unchanged to  $2.9 \times 10^{-11}$  on 2520 sector-by-trial-vector combinations (5 weight families  $\times$  7 shifts  $\times$  6 taper widths), with the pure shift scaling the certified value by exactly  $\theta = \mu_1^P/(\mu_1^P + \kappa_s)$ ; the full space is settled strictly worse ( $\mu_0 = 0$  exactly); the wrong transport map breaks loudly ( $2.0 \times 10^2 - 5.4 \times 10^3$ ). *The  $h^2$  conservation law (T164)*. floor  $\times$  transfer  $= 1/(\mu_1^P + \kappa_s) \cdot (1 + \kappa_s/\mu_1^P) = 1/\mu_1^P$  *identically* at every shift ( $2.2 \times 10^{-16}$ ): the  $h^2$  is conserved, never removed — it lives in the ratio, not in the floor; the wrong transfer misses by 0.9995 at the largest shift. *The power-one spend (T164)*. The  $r$ -ceiling  $r \leq U/L$  with  $L = \lambda_1(A)/\mu_1^P$  is linear in the entry ceiling, so  $d \log r / d \log U = +1$  exactly ( $1.1 \times 10^{-14}$ ); the quadratic consumption reads slope 2 and is recognised. *The  $P_{\text{pr}}$  identity and the predicate equivalence (T165)*.  $P_{\text{pr}} = g_{16} \cdot R \cdot (TV/(t_1 v)^2)/\mu_1^P$  exact to  $4.3 \times 10^{-16}$  on all 192 battery vectors, with the predicates  $R > 2\kappa$  and  $P_{\text{pr}} > 2\kappa g_{16} \text{ tvf}/\mu_1^P$  agreeing as booleans everywhere and every crossing vector paying  $P_{\text{pr}} > 2\kappa g_{16}/\mu_1^P = 15.4\text{--}142.8$  above the preregistered tolerance 10 on 12/12 — the successor question R-F closes negatively by an identity plus a floor; the price with  $(t_1 v)$  unsquared breaks the gauge invariance loudly. *The Schur-cascade direction and the exponent closure (T164/T165)*.  $s = \mu_1^P t_1^T A^{-1} t_1 \geq g_{16} \geq g_1$  window by window with  $P_K = \hat{a} s \geq 1$  (Kantorovich 1948), and the exponent closure  $\log(1/g_1) = \log(g_{16}/g_1) + \log(1/g_{16})$  is exact per instance and on the fitted lists ( $1.3 \times 10^{-15}$ ); plus the monotone no-go staircase and exact Dirichlet/parity controls. **Named limits as content:** per instance on small windows — nothing uniform in the zone index or in  $D$ ; the T164/T165  $h$ -exponents and the measured anatomy are sandbox, *not* consumed; finite  $\Lambda$ -sums allowed and used, *no* RH statement; **the one open object after T165 stays open, typed open:**  $\inf_m g_{16}(m) > 0$ , equivalently a lower bound on the sixteen-step Schur-cascade gain  $g_{16}/g_1$  uniform in  $m$  — a quantifier over a certified flat list. Schur 1917–Haynsworth 1968 / KMS 1953 / Abel 1826 / Chebyshev 1852–Rosser–Schoenfeld 1962 / Kantorovich 1948 / Fejér 1915 / Dirichlet 1829 named classical; Weil 1952 / Bombieri 2000 cited, never used as a criterion. Machine-checked: [verification/v556\_gauge\_ppr\_identities.py]. [E] (per-instance identities and certified inequalities; no marker moves)

**The cascade/vector identities (PRIME.CASCADE.VECT.01).** The closed theorem cores of T166 and T167 are load-bearing as the fifteenth companion module (23 checks,  $\sim 0.1$  s), recomputed on the same declared surface as the module above (12 deepest in-cap frame-A windows,  $m = 96 \dots 291 \leq 300$ ) with the one arithmetic input  $\kappa = 0.038821$  *verified* at every jump point of  $\psi(x)/x$  on the von-Mangoldt table — the *anatomy of the cascade* and the *exactness of the closed vector at  $K = 2$* , and it does not close the one remaining scalar. *The cascade identities (T166)*. The prefix property (one Cholesky of the sixteen-block yields all sixteen rungs; 192 comparisons,  $1.1 \times 10^{-12}$ ), the minor form  $1/g_K = \det G_{[1..K]}/(\mu_1^P \det G_{[2..K]})$  on the raw arithmetic Gram block ( $2.7 \times 10^{-12}$ ; Schur 1917–Haynsworth 1968), and the regression form  $g_K/g_1 = 1/(1 - R_K^2)$  ( $8.8 \times 10^{-13}$ ) — the open object *is* a near-collinearity statement; the wrong minor misses by 0.99–1.45. *The diagonal invariance (T166)*. The gain  $g_K/g_1 = B_{11}(Q_K^{-1})_{11}$  is unchanged under  $B \rightarrow DBD$  on 144 random positive diagonals ( $2.7 \times 10^{-12}$ ): the cascade does not see the KMS normalisation — the gain belongs to the arithmetic Gram block alone; a non-diagonal congruence moves it by 2.7–3.8. *The exact  $K = 2$  identity (T167)*. The closed vector  $u = (1, -Q_{21}/Q_{22})$  attains  $1/g_2 = Q_{11}(1 - r_{12}^2)$  exactly ( $1.1 \times 10^{-12}$ ) and  $B_{11}g_2 = 1/(1 - r_{12}^2)$  identically ( $1.2 \times 10^{-12}$ ): at  $K = 2$  the fast-null-vector is free — the third dress collapses onto the second, one scalar built from three closed lag sums; the wrong-sign vector pays  $4r_{12}^2/(1 - r_{12}^2) = 6.7 \times 10^2\text{--}1.5 \times 10^4$ . *The pivot identity and the unification (T167)*.  $u^T Q_K u = 1/g_K + \delta^T Q_K \delta$  for  $\delta_1 = 0$  ( $2.0 \times 10^{-12}$  on 252 evaluations) — every candidate excess is an exact number; the sign-aligned entry attainment  $|x^{*T} E x^*| = \varepsilon S_K$  ( $3.8 \times 10^{-16}$ ) makes the entry threshold exact, and the unification  $(1/g_2)/S_2 = (1 - r_{12}^2)/(1 + 3r_{12}^2)$  ( $1.2 \times 10^{-12}$ ) makes the three dresses *one* inequality; the  $\delta_1 = 0.1$  mutation breaks by exactly the closed cross term  $2\delta_1$ . *The trial theorem and the two must-breaks (T166/T167)*. All 288 random trials with  $u_1 = 1$  bound  $g_K$  from below (min ratio 15.14; Schur 1917); the  $L_P$  no-go gives cascade gain exactly 1



(deviation 0.0); the scramble type-change makes  $B_{11}$  itself lose positivity on a majority of draws on 12/12 windows — at  $K = 2$  the collapse under scrambling is a change of type, not a factor; plus exact Dirichlet/parity controls. **Named limits as content:** per instance on small windows — nothing uniform in the zone index or in  $D$ ; the T166/T167  $h$ -exponents and the measured anatomy are sandbox, *not* consumed; finite  $\Lambda$ -sums allowed and used, *no* RH statement; **the one open object after T167 stays open, typed open:** R1, the single scalar — an  $m$ -free upper bound on  $1 - r_{12}^2 = 1 - Q_{12}^2/(Q_{11}Q_{22}) \leq C h^{-3+\varepsilon}$ , three closed lag sums and nothing else. Schur 1917–Haynsworth 1968 / KMS 1953 / Rayleigh 1877–Courant–Fischer 1920 / Cauchy 1829 / Kato 1949 (cited as the closed-off route, never used) / Chebyshev 1852–Rosser–Schoenfeld 1962 / Dirichlet 1829 named classical; Weil 1952 / Bombieri 2000 cited, never used as a criterion. Machine-checked: `[verification/v557_cascade_vector_identities.py]`. **[E]** (per-instance identities and certified inequalities; no marker moves)

### 3.4 Glue-norm arithmetic, and three readings of 41, 60, 120

The two glue norms generate the carrier integers through sum, difference and product. With  $\dim S^+ = 16$ :

$$\boxed{16(q_D + q_A) = 32 = 2^{g_{\text{car}}} = h(D_5)h(A_3), \quad 16(q_D - q_A) = 8 = r(E_8) = h(D_5), \quad 16 q_D q_A = 15 = \dim S^+ - 1} \quad \text{[E]}, \quad (12)$$

so the glue already contains the full carrier exhaustion  $2^{g_{\text{car}}}$ , the  $E_8$  rank, and the non-trivial transition count  $\mathcal{A}_H + \mathcal{A}_\Lambda = 15$ ; and  $|\mu_4| = (q_D - 1)^{-1} = (1 - q_A)^{-1} = 4$  is the inverse displacement of the two norms from the unit norm.

**Spine Quotient Lemma: the recurring factors are adjacent quotients of  $(2, 3, 4, 5)$**  **[E]**  
(`[verification/v74_compiler_micro_lemmas.py]`)

The integers of the spine are  $(|\mathbb{Z}_2|, N_{\text{fam}}, |\mu_4|, g_{\text{car}}) = (2, 3, 4, 5)$ . Every “mysterious”  $O(1)$  factor in TFPT is an *adjacent quotient* of this chain, not a separate special number:

$$\frac{|\mathbb{Z}_2|}{N_{\text{fam}}} = \frac{2}{3}, \quad \frac{N_{\text{fam}}}{|\mu_4|} = \frac{3}{4} = q(A_3), \quad \frac{|\mu_4|}{g_{\text{car}}} = \frac{4}{5}, \quad \frac{g_{\text{car}}}{|\mu_4|} = \frac{5}{4} = q(D_5), \quad \frac{|\mu_4|}{N_{\text{fam}}} = \frac{4}{3}.$$

These are exactly the factors that recur across the theory:  $\lambda_2 = (\frac{2}{3})^6$  (the transport gap = Page recovery rate, and the Koide target  $Q_\star = \frac{2}{3}$ );  $\varepsilon = \frac{3}{4}\varphi_0 = q(A_3)\varphi_0$  (the solar misalignment);  $\varphi_0 = \frac{4}{3}c_3 + 48c_3^4$  with leading  $\frac{4}{3}c_3 = \frac{1}{6\pi}$ ; and  $q(D_5) + q(A_3) = \frac{5}{4} + \frac{3}{4} = 2$  (the  $E_8$  root norm). So  $(\frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \frac{5}{4}, \frac{4}{3})$  are five projections of one discrete chain — not scattered constants. The spine is moreover *one* finite object: as a tetrahedron on  $T = \{e_3(a), p_0(a), e_1(a), e_2(a)\}$  its edge and face products and its volume  $120 = |R^+(E_8)| = 5!$  reproduce the central integer grammar, with the honest negative control that the graph reading  $240 = |\mu_4| \cdot |E(K_4)| \cdot |E(K_5)|$  breaks at  $K_6$  (`[verification/v91_spine_tetrahedron.py]`).

The carrier adjoint  $\dim D_5 = |R(D_5)| + g_{\text{car}} = 45$  then gives a second reading of the  $U(1)$  integer,

$$\boxed{10b_1 = \dim D_5 - |\mu_4| = 45 - 4 = 41 = \underbrace{40}_{\sum L} + \underbrace{(g_{\text{car}} - |\mu_4|)}_{=N_\Phi=1}} \quad \text{[E]}, \quad (13)$$

i.e. the single Higgs doublet is the carrier rank minus the glue corners,  $5 - 4 = 1$ .

A fourth, metric reading of 41 (`[verification/v222_cm_norm_duality.py]`). On the square seam ( $j=1728$ , complex multiplication by the Gaussian integers  $\mathbb{Z}[i]$ ) the  $U(1)$  index is the Gaussian norm of the carrier-glue vector,

$$10b_1 = N_{\mathbb{Z}[i]}(g_{\text{car}} + i|\mu_4|) = g_{\text{car}}^2 + |\mu_4|^2 = 5^2 + 4^2 = 41,$$

and the hexagonal partner ( $j=0$ , Eisenstein  $\mathbb{Z}[\omega]$ ) reads the same  $(3, 2)$  carrier split as the scalaron exponent  $N_{\mathbb{Z}[\omega]}(3+2\omega) = 3^2 - 3 \cdot 2 + 2^2 = 7$ . So 41 (EM) and 7 (scalaron) are the two exceptional CM

norms of the one carrier, square vs. hexagonal — not two unrelated  $O(1)$  numbers [E]. The start defect 60 and the hypercharge moment 120 are “bilingual”:

$$D_{\text{start}} = \dim D_5 + \dim A_3 = 45 + 15 = 60 = \mathcal{A}_\Lambda \cdot |R^+(A_3)| = 10 \cdot 6,$$

$$\text{Tr}_{S^+} X^2 = 120 = \underbrace{60}_{\text{adjoint}} + \underbrace{60}_{\text{mixed (10,6)}}, \quad (14)$$

and  $\dim E_8 = 248 = \text{Tr}_{S^+} X^2 + 2 \dim S^+ |\mu_4| = 120 + 2 \cdot 16 \cdot 4$ . Finally the hypercharge norm and the EW exponent are  $A_3$ -rational,

$$\boxed{\gamma_{\text{car}} = \frac{\mathcal{A}_\Lambda}{|R(A_3)|} = \frac{10}{12} = \frac{5}{6}, \quad I_1^{\text{EW}} = \frac{|E(K_5)|}{|R(A_3)|} + \frac{1}{|\mu_4| |R(A_3)|} = \frac{10}{12} + \frac{1}{48} = \frac{41}{48}} \quad [\text{E}]. \quad (15)$$

*Transport reduction.* The flavor matrix  $L$  has its mod-6 content fixed by a single combinatorial datum:  $\{0, 1, 3\}$  is the *unique* (up to  $D_6$ ) 3-subset of  $C_6 = |R^+(A_3)|$  with all distinct cyclic distances  $\{1, 2, 3\}$ , and the three sectors are its  $D_6$  orbit ( $u \equiv \{0, 1, 3\}$ ,  $d \equiv 2 - \{0, 1, 3\}$ ,  $e \equiv 2 + \{0, 1, 3\}$ ). The total is fixed by the carrier,  $\sum L = 40 = |R(D_5)|$ , with winding unit  $6 = |R^+(A_3)|$ . Given the residue *set* of a sector, the full ordered packet is reproduced by two universal rules — the lightest generation carries the single +6 winding, and the entries are sorted in decreasing length (monotone hierarchy). The *only* remaining datum is then, per sector, which residue is the first-generation anchor that receives the winding:

$$\boxed{\text{anchors } (a_u, a_d, a_e) = (1, 1, 2) \Rightarrow L_u = (7, 3, 0), L_d = (7, 5, 2), L_e = (8, 5, 3)} \quad [\text{E}] \text{ (given anchors)}. \quad (16)$$

So the open step of a full “Clebsch walk” theorem is sharply isolated: it is exactly these three per-sector anchors. They are *not* free: the three transport cusps are exactly the  $D_5$  dual-root combinations, and each sector is tied to its cusp by the right-handed hypercharge,

$$\boxed{\{1, \frac{2}{3}, \frac{1}{3}\} = \{2y_+, -2y_-, 2(y_+ + y_-)\}, \quad \text{lepton} \leftrightarrow 2y_+, \text{ up} \leftrightarrow -2y_-, \text{ down} \leftrightarrow 2(y_+ + y_-)} \quad [\text{E}]. \quad (17)$$

The canonical-transport-kernel theorem (document 3, Part II) derives the residue packets — and hence the anchors — as the *word-lengths of the family-connection holonomy*  $\text{Hol}_{\mathcal{G}_{D_4}}(\nabla_F^*)$  along the  $D_4$ -invariant geodesic spine of  $\mathbb{P}^1 \setminus \mu_4$  with these cusps as boundary data. The anchors are therefore a genuine  $D_5 \times A_3$  readout (cusps from the  $D_5$  carrier, spine holonomy from the  $A_3$  four-punctured sphere), not an extra parameter. A compact candidate closed form is the nearest-integer cusp map

$$\boxed{a_f = \text{round}(2 \text{ cusp}_f) : (a_u, a_d, a_e) = \text{round}(\frac{4}{3}, \frac{2}{3}, 2) = (1, 1, 2)} \quad [\text{C}], \quad (18)$$

which reproduces the matrix exactly. The remaining open step is now precise and purely geometric: an *independent* closed-form evaluation of the  $D_4$ -spine holonomy word-lengths from the lattice data alone (the same  $\mathbb{P}^1 \setminus \mu_4$  whose residue pairing is  $\text{Cartan}(A_3)$  and whose corner rotation is the  $A_3$  Coxeter element), rather than from the family-connection theorem (document 3).

**Riemann–Hilbert check, and what it does (and does not) fix.** Setting up the rigid  $\mathbb{Z}_4$ -symmetric rank-3  $SU(3)$  local system explicitly — four puncture monodromies  $M_k = R^k C R^{-k}$  with the order-3 cusp class  $C \sim \text{diag}(1, \omega, \omega^2)$  ( $\omega = e^{2\pi i/3}$ , the three cusps as eigenvalue-exponents) and the rotation  $R$  ( $R^4 = \mathbb{K}$ ) — the closure  $M_0 M_1 M_2 M_3 = \mathbb{K}$  has a solution to machine precision, and the unique  $R$  has eigenvalues  $\{i, -1, -i\}$ : the  $A_3$  Coxeter element. This *independently* confirms the rigidity of the family connection and the corner-rotation = Coxeter identity. However, the flat monodromy *alone* does not fix the integer word-lengths  $\{0, 1, 3\}$ : those are *parabolic degrees* (the cusp exponents as parabolic weights, together with the  $\mathbb{Z}_2$  sheet parity), i.e. Hodge/metric data on the spine, not topological monodromy. *Status update (the computation has since been carried out)*: the exact residue normal form  $A_0^*$  with  $\text{spec} = \{0, \frac{1}{3}, \frac{2}{3}\}$  is pinned algebraically

(`[verification/v115_anchor_residue.py]`, `[verification/v116_resonance_uniqueness.py]`), the unitarised holonomy is the exact 24-element  $W(A_3) = S_4$  (`[verification/v117_monodromy_weyl_a3.py]`), and the parabolic weights are the  $\mu_4$ -character degrees of  $H^1(\mathbb{P}^1 \setminus \mu_4)$  — one spectrum in four readouts (`[verification/v137_qplus_cohomology.py]`); the deck itself is *derived* — the integer model selects the sheet-twisted class, no discrete freedom left (`[verification/v141_deck_selection.py]`), and the full Möbius  $D_4$  of the curve matches the integer model parity by parity (`[verification/v146_moebius_d4.py]`). What remains of the old residual is the much smaller selector establishment  $R4'$  (research contracts): derive the *one* lift map of the torsion normal — the pairing values are atom identities (`[verification/v142_frame_integrality.py]`/`[verification/v145_pairing_atoms.py]`). [E]/[C]

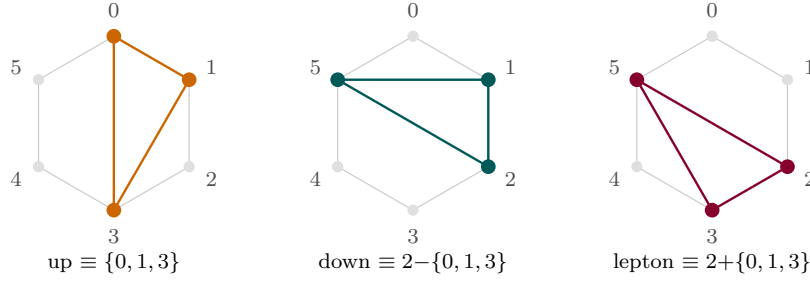


Figure 5: Flavor transport on the hexagon  $C_6 = Z_6 = |R^+(A_3)|$ . The unique distinct-distance triple  $\{0, 1, 3\}$  (cyclic distances  $\{1, 2, 3\}$ ) and its  $D_6$  orbit give the three sectors' residues mod 6; the total  $\sum L = 40 = |R(D_5)|$  and the first-generation  $+6$  winding come from the carrier.

### 3.5 What the compiler outputs for the Standard Model and our reality

Every entry below is an *output* of  $(c_3, g_{\text{car}})$  through the  $D_5 \times A_3$  compiler; none is an independent input. “Reads” names the lattice object the observable comes from.

| SM / cosmology observable                             | Compiler output                                                                                                                                       | Reads              | St.     |
|-------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|---------|
| Number of generations                                 | $N_{\text{fam}} = 3 = \text{rank } A_3 = \dim H^1(\mathbb{P}^1 \setminus \mu_4)$                                                                      | $A_3$              | [E]     |
| Gauge algebra dimension                               | $\dim \mathfrak{g}_{\text{SM}} = 12 =  R(A_3)  = \gcd(6r, 2h)$                                                                                        | $A_3$              | [E]     |
| One generation (16, incl. $\nu^c$ )                   | $S^+ = D_5$ half-spinor $= \Lambda^{0,2,4} E$                                                                                                         | $D_5$              | [E]     |
| Hypercharge spectrum                                  | all $Y$ from $y_{\pm} = \{\frac{1}{2}, -\frac{1}{3}\}$ ; $\text{Tr } X^2 = 120$                                                                       | $D_5$ root data    | [E]     |
| Abelian index $b_1$                                   | $10b_1 = 1 +  \mu_4   E(K_5)  = 1 + 4 \cdot 10 = 41$                                                                                                  | $\mu_4 \times K_5$ | [E]     |
| Fermion family occupancy                              | $\Omega_{\text{adm}} = N_{\text{fam}} \dim S^+ = (N_{\text{fam}} + N_{\Phi})  R(A_3)  = 48$                                                           | $A_3$              | [E]     |
| Fine structure $\alpha_{\star}^{-1} = 137.0359992168$ | root of $F_{U(1)}(\alpha) = 0$ (explicit, unique)                                                                                                     | EM closure         | [E]     |
| Cabibbo angle                                         | $\lambda_C = \sqrt{u(1-u)}$ , $u = \frac{4}{3}c_3 + 48c_3^4$                                                                                          | seed               | [E]     |
| CKM CP phase                                          | $\delta_{\text{CKM}} = \frac{2\pi}{ R^+(A_3) } + N_{\text{fam}} u(1-u) = \frac{\pi}{3} + 3\lambda_C^2$                                                | $A_3^+$ , seed     | [E]/[C] |
| PMNS angles/phase                                     | $\sin^2 \theta_{12} = \frac{1}{3} - \frac{u}{2}$ , $\sin^2 \theta_{13} = e^{-5/6} u$ , $\delta^\nu = \frac{4\pi}{3}$                                  | $A_3^+$ , seed     | [E]/[C] |
| Charged-lepton hierarchy                              | $\det M_\ell \propto u^{h/N_{\text{fam}}} = u^{10}$ ; $(5, 3, 2)$                                                                                     | decuple            | [E]/[C] |
| Mass-hierarchy transport                              | $\sum L = 40 =  R(D_5) $ ; entries on $C_6 =  R^+(A_3) $                                                                                              | $D_5 \times A_3$   | [E]     |
| Strong CP                                             | $\theta_{\text{eff}} = 0$ (determinant erasure, given RP; document 3, Part III)                                                                       | —                  | [E]/[C] |
| Weak CP                                               | holonomy: $\delta_{\text{CKM}} - \frac{\pi}{3} = N_{\text{fam}} u(1-u)$                                                                               | $A_3^+$            | [C]     |
| Gravity tree normaliser                               | $\xi_{\text{tree}} = \frac{\dim \mathfrak{g}_{\text{SM}}}{\dim S^+} = \frac{3}{4} = q(A_3)$                                                           | $A_3$ glue         | [E]     |
| Cosmological constant                                 | $\rho_\Lambda / \bar{M}_{\text{Pl}}^4 = \frac{3}{4\pi^2} e^{2\delta_{\text{ph}}} [v_{\text{geo}} / (5\beta_{\text{rad}}^2 \bar{M}_{\text{Pl}})]^{10}$ | decuple            | [E]     |
| Hubble anchor                                         | $H_0 \sqrt{\Omega_\Lambda} = \bar{M}_{\text{Pl}} e^{-\alpha_{\star}^{-1}} / (2\pi)$                                                                   | action             | [E]     |
| Baryon density                                        | $\omega_b = \lambda_C / \mathcal{A}_\Lambda = \lambda_C / 10$                                                                                         | decuple            | [E]/[C] |
| Inflation (scalaron)                                  | $M_{\text{scal}}^2 / \bar{M}_{\text{Pl}}^2 = c_3^{\Omega_{\text{adm}} - 10b_1} = c_3^7$                                                               | deficit $r - 1$    | [E]     |
| Inflation amplitude                                   | $A_s = N_{\star}^2 c_3^7 / 24\pi^2 \approx 2.2 \times 10^{-9}$                                                                                        | deficit            | [C]/[O] |

Marker key: [E] exact / machine-proven · [C] conditional (holds under named hypotheses) · [O] open / axiom · [X] falsifiable kill test. Per-claim fine type in `verification/status_ledger.csv`.

*Reading.* The discrete sector — generations, gauge dimension, family content, hypercharge,  $b_1$ , occupancy — is now *group-theoretic output* of  $D_5 \times A_3$ . The continuous sector —  $\alpha^{-1}$ ,  $\lambda_C$ , the mixings,  $\rho_\Lambda$ ,  $H_0$ , inflation — is the seam seed  $c_3$  and the action exponentials read on the *same* carrier graphs ( $K_5$  pairs, the  $C_6$  hexagon  $= |R^+(A_3)|$ , the decuple  $\mathcal{A}_\Lambda = \binom{g_{\text{car}}}{2} = 10$ ). What used to be a list of separate hits is, structurally, one machine:  $D_5$  carries the family and the flavor budget,  $A_3$  carries the generations and the gauge dimension,  $\mu_4$  glues them (and reappears as the 41 chain, the discriminant norms, and the CP phase step  $2\pi/6$ ), and  $c_3$  dresses everything.

## 4 Grammar I — the carrier grammar (3 + 2)

Machine checks for this section: [verification/v2\_carrier\_pascal.py]

From  $g_{\text{car}} = 5$  and the split  $(b, s) = (3, 2)$  the carrier data follow:

$$\begin{aligned} \gamma_{\text{car}} &= \frac{g_{\text{car}}}{g_{\text{car}} + 1} = \frac{5}{6}, & N_{\text{fam}} &= \frac{g_{\text{car}} + 1}{2} = 3, & \dim S^+ &= 2^{g_{\text{car}} - 1} = 16, \\ \Omega_{\text{adm}} &= N_{\text{fam}} 2^{g_{\text{car}} - 1} = 48, & b_1 &= \frac{g_{\text{car}} 2^{g_{\text{car}} - 2} + 1}{10} = \frac{41}{10}. \end{aligned} \tag{19}$$

### 4.1 Dual-root dictionary (everything from two charges)

**Two carrier roots generate the family and the transport cusps [E].** With  $y_+ = \frac{1}{2}$  and  $y_- = -\frac{1}{3}$  (the determinant-normalised carrier eigenvalues), the full one-family hypercharge spectrum

and the flavor-transport cusps are sums of the two roots:

$$\begin{aligned} Y(Q_L) &= y_+ + y_- = \frac{1}{6}, & Y(u^c) &= 2y_- = -\frac{2}{3}, & Y(d^c) &= -y_- = \frac{1}{3}, \\ Y(L_L) &= -y_+ = -\frac{1}{2}, & Y(e^c) &= 2y_+ = 1, & Y(\nu^c) &= 0, \\ \{1, \frac{2}{3}, \frac{1}{3}\} &= \{2y_+, -2y_-, 2(y_+ + y_-)\}, & \gamma_{\text{car}} &= \text{Tr}_E Y^2 = 3y_-^2 + 2y_+^2 = \frac{5}{6}, \\ \varphi_{\text{base}} &= \frac{1 - \gamma_{\text{car}}}{\pi} = \frac{1}{6\pi}. \end{aligned}$$

All residuals are exactly zero. The Standard-Model charge table is not assumed; it is built by addition from  $\{1/2, -1/3\}$ .

#### The $X = 6Y$ quadratic and the hypercharge Pascal-sign duality [E]

Clearing denominators with  $X = 6Y$  turns the two base charges into the integer roots  $x_+ = 3$ ,  $x_- = -2$  of

$$\boxed{x^2 - x - 6 = 0} : \quad x_+ + x_- = 1 = N_\Phi, \quad x_+ - x_- = 5 = g_{\text{car}},$$

$$-x_+x_- = 6 = |R^+(A_3)|, \quad \sqrt{\Delta} = 5,$$

so the Higgs singleton, the carrier rank and the  $A_3$  hexagon all fall out of one quadratic. Counting the signs of  $X$  over a generation ( $X = 1^{\times 6}, 2^{\times 3}, 6^{\times 1}, (-4)^{\times 3}, (-3)^{\times 2}, 0^{\times 1}$ ) gives the *Pascal triple* again:

$$\boxed{(\#X=0, \#X<0, \#X>0) = (1, 5, 10) = \binom{5}{0}, \binom{5}{1}, \binom{5}{2}},$$

$$10-5=g_{\text{car}}, \quad 5-1=|\mu_4|, \quad 10-1=\text{tr } R,$$

with  $\nu^c \leftrightarrow \binom{5}{0}$  the neutral singleton — the right-handed neutrino is the Pascal zero with a job. The moment ladder adds  $\text{Tr } X^2 = 120 = 5!$  and  $\text{Tr } X^4 / \text{Tr } X^2 = 2280/120 = 19$  (an  $E_8$  exponent; audit). [E]

## 4.2 Carrier arithmetic (proved identities)

The discrete coefficients are not isolated numbers; they are exact carrier traces:

$$10b_1 = 41 = g_{\text{car}}^2 + (g_{\text{car}} - 1)^2 = \Omega_{\text{adm}} \gamma_{\text{car}} + 1 = 1 + h_{E_8} \frac{\dim S^+}{\dim \mathfrak{g}_{\text{SM}}} = 1 + 30 \cdot \frac{16}{12} = 40 + 1, \quad [\text{E}]$$

$$I_1^{\text{EW}} = \frac{41}{48} = \text{Var}_{S^+}(Y) \cdot b_1 = \frac{5}{24} \cdot \frac{41}{10} = \gamma_{\text{car}} + \frac{1}{\Omega_{\text{adm}}}, \quad [\text{E}]$$

$$\text{Tr}_{S^+}(X^2) = 120 = 5! \quad (X = 6Y), \quad \delta_2 = \frac{5}{4}\delta_{\text{top}}^2 = 2880 c_3^8 = 4! \cdot 5! c_3^8 = 4! \text{Tr}_{S^+}(X^2) c_3^8, \quad [\text{E}]$$

$$\frac{5}{4} = \underbrace{\frac{3}{2} \cdot \frac{5}{6}}_{B\gamma} = \underbrace{\frac{D_{\text{start}}}{\Omega_{\text{adm}}}}_{60/48} = \underbrace{\frac{g_{\text{car}}}{g_{\text{car}}-1}}_{5/4} = \frac{\delta_2}{\delta_{\text{top}}^2}. \quad [\text{E}]$$

So:  $b_1$  is a Pythagorean carrier identity ( $5^2 + 4^2$ ); equivalently the anchor-product form  $10b_1 = p_1p_3 + (g_{\text{car}} - p_1) = 4 \cdot 10 + (5 - 4) = 41$  reads “all four glue corners times all ten carrier pairs, plus the one leftover Higgs slot”; the electroweak exponent  $41/48$  is the hypercharge variance times the  $U(1)$  budget; the second seam defect  $2880 c_3^8$  is a product of factorial traces  $4! \cdot 5!$ ; and the universal compression  $5/4 = g_{\text{car}}/(g_{\text{car}} - 1)$  recurs whenever the five-slot carrier projects onto four effective directions. As a minor curiosity, the  $SU(5)$ -tree weak mixing angle rewrites as  $\sin^2 \theta_W^{\text{tree}} = \frac{3}{8} = N_{\text{fam}}/(g_{\text{car}} + N_{\text{fam}}) = N_{\text{fam}}/\text{rank}(E_8)$  (the value is the standard one; the rewrite is a coincidence to note, not a claim). [C]



## 5 Grammar II — the seed grammar (one decoder $u$ )

Machine checks for this section: [verification/v106\_review\_validation.py]

The seed is, exactly, a pure function of  $c_3$ :

$$u = \varphi_0^{\text{ret}} = \frac{4}{3} c_3 + \Omega_{\text{adm}} c_3^4 = \underbrace{\frac{1}{6\pi}}_{\varphi_{\text{base}}} + \underbrace{\frac{3}{256\pi^4}}_{\delta_{\text{top}}} = 0.0531719521768 \quad [\text{E}]. \quad (24)$$

Its two coefficients are themselves carrier data:  $\frac{4}{3} = \dim S^+ / \dim \mathfrak{g}_{\text{SM}} = 16/12 = 8/(g_{\text{car}} + 1)$  (so  $\varphi_{\text{base}} = 1/((g_{\text{car}} + 1)\pi)$ ) and  $48 = \Omega_{\text{adm}}$ . The seed is thus a pure function of  $c_3$  and the carrier counts.

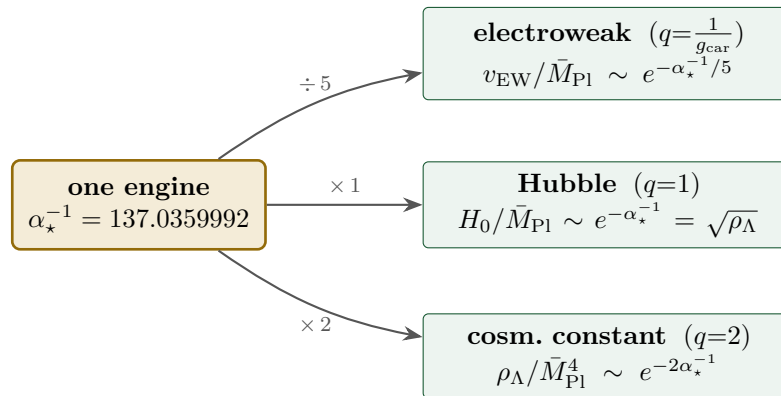
**One seed, four readouts (and their inverses) [E].**

$$\begin{aligned} \lambda_C^2 &= u(1-u), \quad \beta_{\text{rad}} = \frac{u}{4\pi}, \quad \Omega_b = \left(1 - \frac{1}{4\pi}\right)u, \quad \sin^2 \theta_{13} = e^{-5/6} u, \\ u &= \frac{1 - \sqrt{1 - 4\lambda_C^2}}{2} = 4\pi \beta_{\text{rad}} = \frac{\Omega_b}{1 - 1/(4\pi)} = e^{5/6} \sin^2 \theta_{13} \\ &= \sum_{n \geq 0} C_n \lambda_C^{2n+2} = \lambda_C^2 + \lambda_C^4 + 2\lambda_C^6 + 5\lambda_C^8 + \dots \end{aligned}$$

One knob  $u$  turns four dials simultaneously: Cabibbo mixing (the Bernoulli width  $\sqrt{u(1-u)}$ ), cosmic birefringence, the baryon fraction, and the reactor angle. Turning one forces the others; there is no free per-observable tuning. The exponent  $5/6$  is the carrier trace  $\gamma_{\text{car}} = \text{Tr}_E Y^2$ , *not* the Euler–Mascheroni constant  $0.5772\dots$  (audit [O]).

## 6 Grammar III — the action grammar (exponentials of $\alpha_\star^{-1}$ )

**The relation in plain words.** The fine-structure constant is not just “a number near 137”: it is the *one exponential engine* behind three wildly separated scales. Take the same  $\alpha_\star^{-1} = 137.036$  and divide the carrier into it:  $\frac{1}{5}$  of it sets the *electroweak* scale,  $1 \times$  it sets the *Hubble* scale,  $2 \times$  it sets the *cosmological constant*. The three action charges are the Pascal row  $1 : 5 : 10$  of the five-slot carrier, and the Hubble rung is just the square root of the vacuum energy. So one constant fixes the electroweak hierarchy, dark energy and the expansion rate at once.



action charges  $1 : g_{\text{car}} : 2g_{\text{car}} = 1 : 5 : 10 = \binom{5}{0} : \binom{5}{1} : \binom{5}{2}$  (Pascal row of  $K_5$ )

Read every dimensionless ratio through its *action* and ask for the action charge:

$$x = R_x e^{-A_x}, \quad A_x = q_x \alpha_\star^{-1} + \Delta_x, \quad q_x \in \left\{ \frac{1}{g_{\text{car}}}, 1, 2 \right\}. \quad (25)$$

The robust ladder and its residues are

| sector                                            | action $A_x$                                              | charge $q_x$       | residue $R_x$                        | value of $-\ln x$ |
|---------------------------------------------------|-----------------------------------------------------------|--------------------|--------------------------------------|-------------------|
| electroweak $v_{\text{geo}}/\bar{M}_{\text{Pl}}$  | $(\alpha_\star^{-1} + \delta_{\text{ph}})/g_{\text{car}}$ | $1/g_{\text{car}}$ | $g_{\text{car}}\beta_{\text{rad}}^2$ | 36.85             |
| Hubble $H_0/\bar{M}_{\text{Pl}}$                  | $\alpha_\star^{-1}$                                       | 1                  | $1/(2\pi\sqrt{\Omega_\Lambda})$      | 138.68            |
| cosm. const. $\rho_\Lambda/\bar{M}_{\text{Pl}}^4$ | $2\alpha_\star^{-1}$                                      | 2                  | $3/(4\pi^2)$                         | 276.65            |

$$A_{\text{EW}} = \frac{\alpha_\star^{-1} + \delta_{\text{ph}}}{g_{\text{car}}} = 27.5259, \quad \mathcal{A}_\Lambda = 2\alpha_\star^{-1} = 274.072, \quad \boxed{A_{\text{EW}}:\mathcal{A}_H:\mathcal{A}_\Lambda = 1:5:10}. \quad (26)$$

The leading ratio is exact:  $\mathcal{A}_\Lambda/(\alpha_\star^{-1}/g_{\text{car}}) = 2g_{\text{car}} = 10$  (with the transport correction  $\mathcal{A}_\Lambda/A_{\text{EW}} = 9.957$ ). The claim is that the *action charges*, not the raw logarithms, form the ladder.

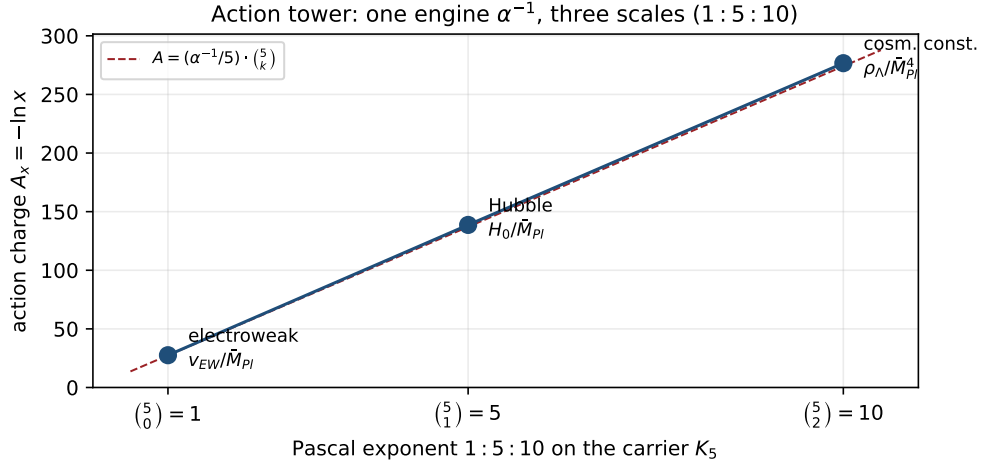


Figure 6: Action tower (data plot, `verification/make_figures.py`). The three action charges  $A_x = -\ln x$  for the electroweak, Hubble and cosmological-constant ratios fall on the line  $A = (\alpha_\star^{-1}/5)\binom{5}{k}$ : one engine  $\alpha_\star^{-1}$  generates all three scales with the Pascal exponents 1 : 5 : 10.

### 6.1 The EM closure: $\alpha_\star^{-1}$ as an explicit fixed point (made explicit here)

The engine  $\alpha_\star^{-1}$  is not an input — it is the unique root of an explicit closure equation (the exact electromagnetic closure theorem). With  $c_3 = 1/(8\pi)$ ,  $\varphi_{\text{base}} = 1/(6\pi)$ ,  $q(\alpha) = \delta_{\text{top}}e^{-2\alpha}$  and the seam response  $\varphi_{\text{seam}}(\alpha) = \varphi_{\text{base}} + q(\alpha)(1 - q(\alpha))^{-5/4}$ ,

$$\boxed{F_{U(1)}(\alpha) = \alpha^3 - 2c_3^3\alpha^2 - \frac{4}{5}c_3^6\left(\sum_{f,j} L_{f,j} + N_\Phi\right) \log \frac{1}{\varphi_{\text{seam}}(\alpha)} = 0} \quad [\mathbf{E}]/[\mathbf{C}], \quad (27)$$

where the coefficient is  $\frac{4}{5}(\sum L + N_\Phi)c_3^6 = \frac{4}{5} \cdot 41 c_3^6 = 8b_1c_3^6$ , so the fine-structure constant reads the *flavor transport budget*  $\sum L = 40$  and the  $U(1)$  index  $b_1 = 41/10$ .

**Lemma 1** (Existence and uniqueness of the EM fixed point).  $F_{U(1)}$  has *exactly one* positive root, and it is simple.

*Proof.* Write  $F_{U(1)}(\alpha) = h(\alpha) - Cg(\alpha)$  with  $h(\alpha) = \alpha^3 - 2c_3^3\alpha^2$ ,  $C = \frac{4}{5}c_3^6M > 0$  ( $M = 41$ ), and  $g(\alpha) = -\log \varphi_{\text{seam}}(\alpha)$ . Since  $q(\alpha) = \delta_{\text{top}}e^{-2\alpha} \in (0, 1)$  is strictly decreasing,  $\varphi_{\text{seam}} = \varphi_{\text{base}} + q(1 - q)^{-5/4}$  is  $C^1$  and decreasing, so  $g$  is  $C^1$ , bounded, and slowly increasing with  $g'(\alpha) = O(q)$  tiny; numerically  $g \in [2.9342, 2.9343]$  on the admissible window. The cubic  $h$  has its interior critical point at  $\alpha_c = \frac{4}{3}c_3^3 = 8.399 \times 10^{-5}$ , with  $h(0) = 0$  and  $h < 0$  on  $(0, \alpha_c)$ . *Monotonicity (corrected endpoint).* Since  $F'_{U(1)}(\alpha) = 3\alpha^2 - 4c_3^3\alpha - Cg'(\alpha)$  and at  $\alpha_c$  the first two terms *cancel* ( $3\alpha_c = 4c_3^3$ ), one has  $F'_{U(1)}(\alpha_c) = -Cg'(\alpha_c) \approx -5.89 \times 10^{-10} < 0$ : the derivative is *not* yet positive at  $\alpha_c$ . Its first zero lies slightly above, at  $\alpha_0 = 8.6264 \times 10^{-5}$  (where  $3\alpha_0^2 - 4c_3^3\alpha_0 = Cg'(\alpha_0)$ ), with  $F'_{U(1)} < 0$  on  $(\alpha_c, \alpha_0)$

and  $F'_{U(1)} > 0$  on  $(\alpha_0, \infty)$ . Now  $F_{U(1)}(0) = -C g(0) < 0$ , and  $F_{U(1)}$  decreases on  $(0, \alpha_0]$  (both  $h$  and  $-Cg$  decrease on  $(0, \alpha_c]$ , then  $F'_{U(1)} < 0$  on  $[\alpha_c, \alpha_0]$ ), so  $F_{U(1)}(\alpha_0) \approx -3.82 \times 10^{-7} < 0$ ; thereafter  $F_{U(1)}$  is strictly increasing to  $+\infty$ . Hence  $F_{U(1)} < 0$  on  $(0, \alpha_0]$  and strictly increasing on  $[\alpha_0, \infty)$ , so it crosses zero *exactly once*. The crossing has  $F'_{U(1)}(\alpha_*) = 1.58 \times 10^{-4} > 0$ , so the root is simple. A sign scan on  $(0, 0.05)$  confirms a single change. *Rigorous version*: an **interval-arithmetic** enclosure (mpmath.iv) brackets the root in  $[0.00729735256220970, 0.00729735256221000]$  with  $F_{U(1)} < 0$  on the lower and  $> 0$  on the upper endpoint (definite signs), and a 200-cell interval partition of  $(\alpha_c, 0.05)$  contains exactly *one* indeterminate (root-bearing) band — a verified uniqueness, not merely a sampled one. [verification/v3\_em\_alpha.py]  $\square$

Solving:

$$\boxed{\alpha_* = 0.00729735256220985\dots, \quad \alpha_*^{-1} = 137.0359992168407\dots} \quad [\mathbf{E}] \quad (28)$$

[verification/v3\_em\_alpha.py] versus CODATA-2022  $\alpha_*^{-1} = 137.035999177(21)$  — a relative deviation  $2.9 \times 10^{-10}$  ( $\approx 1.9\sigma$  on the tiny experimental error; the older 2018 value 137.035999084 gave  $9.7 \times 10^{-10}$ , so the agreement *improved* in relative terms). The exponent  $-5/4$  in  $\varphi_{\text{seam}}$  is the carrier discriminant norm  $q(D_5)$ , and  $\delta_{\text{top}} = 48c_3^4$ , so the closure is built only from  $c_3$ , the carrier, and the flavor budget. This discharges the former open audit point: the EM closure is a stated, existence-/uniqueness-proven, numerically reproduced equation, not a fit.

**Why *this* function: the boundary  $U(1)$  Ward reading (Theorem C) [E]/[C].**  $F_{U(1)}$  is not a chosen ansatz but the reduced Ward identity of the boundary  $U(1)$  partition function  $Z_\partial = Z_{\text{Maxwell}} Z_{\text{Calderón}} Z_{\text{transport}}$ , with three terms whose coefficients are compiler atoms, not fits:

$$\underbrace{\alpha^3}_{\text{Maxwell moment}} \quad \underbrace{-2c_3^3\alpha^2}_{\text{Calderón sheet } (2=|\mathbb{Z}_2|)} \quad \underbrace{-8b_1c_3^6 \log \frac{1}{\varphi_{\text{seam}}}}_{C_6 \text{ transport det.}} = 0, \quad 8b_1 = \frac{4}{5}M = \frac{4}{5}(\sum L + N_\Phi) = \frac{164}{5}.$$

The Calderón factor  $2 = |\mathbb{Z}_2|$  (sheet), the transport power  $c_3^6$  is the  $C_6$  hexagon, and the seam exponent  $-\frac{5}{4} = -q(D_5)$  is the carrier discriminant norm. **Typing**: the three-term decomposition and the coefficient identities ( $8b_1 = \frac{4}{5}M$ ,  $-\frac{5}{4} = q(D_5)$ ) are exact [E]; the *physical origin* as a boundary Ward closure (the lemmas  $\partial_\tau \log Z_{\text{Maxwell}} = \alpha^3$ ,  $\partial_\tau \log Z_{\text{Calderón}} = -2c_3^3\alpha^2$ ,  $\partial_\tau \log \det_\zeta(1 - \mathcal{T}_{C_6, U(1)}) = -8b_1c_3^6 \log \frac{1}{\varphi_{\text{seam}}}$ ) is a conjectured [C] interpretation, not yet machine-proven. [verification/v48\_em\_ward.py]

**The determinant-line (Quillen) form.** ([verification/v341\_alpha\_quillen.py]) The three terms assemble into a single *stationarity* statement for the boundary  $U(1)$  determinant line,

$$\underbrace{[\alpha^3 - 2c_3^3\alpha^2]}_{\delta \log \det \Delta_{U(1)}} = \underbrace{[8b_1c_3^6]}_{\text{ctr.}} \underbrace{[\log \frac{1}{\varphi_{\text{seam}}}]_{\text{seam resp.}}}$$

so  $F_{U(1)} = 0$  is  $\delta \log \det \Delta_{U(1)} + \text{counterterm} + \text{seam response} = 0$ . Every factor is a *named* object, not a knob: the counterterm coefficient is  $8b_1c_3^6 = (\text{rank } E_8) b_1 c_3^6$  (index  $\times$  heat-kernel),  $c_3 = 1/(8\pi)$  is the Gauss–Bonnet boundary coefficient, the exponent  $-\frac{5}{4} = -q(D_5)$  is one carrier discriminant norm and the seam base  $\varphi_{\text{base}} = 1/(6\pi) = \frac{4}{3}c_3$  carries the *other*,  $c_3/\varphi_{\text{base}} = \frac{3}{4} = q(A_3)$  — both glue forms appear, and the split is verified at the root. **Typing (and: no new gate)**. The determinant-line assembly and all coefficient identities are exact [E];  $\alpha_*^{-1}$  itself stays closed [E] (unique root + ablation). This adds *no* open item — it *sharpens* the pre-existing [O]/physical-origin obligation EM.WARD.01 (the “why *this* function” residual, already conjectural) by naming every coefficient. The one still-open part *is* that residual, now *named* as its own tracked target ALPHA.QUILLEN.EXACT.01 ([verification/v382\_alpha\_quillen\_exact.py]): the from-first-principles AQFT/ $\zeta$  proof that  $\delta_\tau(\log \det_\zeta \Delta_{U(1)} + 8b_1c_3^6 \log \varphi_{\text{seam}}) = 0$  — i.e. that  $F_{U(1)}$  is the *exact* Quillen functional of the seam  $U(1)$ , the variational principle *derived*, not assembled. It is a *face* of SEAM.EQUIV.01 (v336) — the seam  $U(1)$  determinant line lives on the same holomorphic net, so once the net is  $(E_8)_1$  its  $U(1)$  sub-determinant is fixed — hence no second independent gate, just the sharpest remaining EM-Ward obligation, now citable on its own. *Honest progress towards it*

(`[verification/v391_alpha_quillen_progress.py]`), without closing it: a solvable model  $\det_\zeta(-\partial_\tau^2 + m^2)$  on a circle  $= (2\sinh(\beta m/2))^2$  has the closed  $m$ -variation  $\beta \coth(\beta m/2) = 2/m + (\beta^2/6)m - (\beta^4/360)m^3 + \dots$ , a heat-kernel/Laurent series — so a  $\zeta$ -determinant's variation is *structured* heat-kernel data, and the counterterm  $8b_1c_3^6$  (a Gauss–Bonnet heat-kernel power) is the right *kind* of object. This *reduces* ALPHA.QUILLEN.EXACT.01 to the named sub-target “the seam  $U(1)$  Seeley–DeWitt coefficient  $= 8b_1c_3^6$ ”, but the target itself stays [O] (an external math proof, v384); the value  $\alpha_\star^{-1}$  stays [E]. A *second honest step* (`[verification/v433_alpha_quillen_heatkernel.py]`) bridges this reduction to the heat-kernel origin below, without closing it: v391's circle reached only the low orders  $a_0, a_1$ , whereas a solvable  $4D$  model — the flat  $\Delta = -\partial^2 + m^2$  on  $T^4$ ,  $\text{Tr } e^{-t\Delta} = V/(4\pi)^2 t^{-2} e^{-tm^2}$  — has  $a_4 = Vm^4/(2(4\pi)^2) \neq 0$ , reaching the  $t^0$  (conformal-anomaly) order the circle could not, the very order in which the quadratic term is grounded below (v342, Gilkey  $a_4$ ). So a  $\zeta$ -determinant's variation is heat-kernel data *at the  $a_4$  order*, and the measure itself is  $c_3$ -built,  $(4\pi)^{-2} = (2c_3)^2 = 4c_3^2$ ; the *actual* seam  $a_4 = 8b_1c_3^6$  and the cubic  $\alpha^3$  Chern moment stay [O] (v384), the seam  $F$ -normalisation [C] (retyped by the fifth step below, v470, as the embedding index  $k_Y=5/3$ ).

**The heat-kernel origin of the terms** (`[verification/v342_em_ward_heatkernel.py]`). The determinant-line form is more than bookkeeping: its terms are textbook heat-kernel coefficients. [E]  $c_3 = 1/(8\pi)$  is the boundary Gauss–Bonnet coefficient ( $\oint_{S^2} K = 2\pi\chi = 4\pi$ ,  $c_3 = 1/(\mathbb{Z}_2 \oint K)$ , the same  $\chi=2$  as the marks, v216). [E] the  $\alpha^2$  (Calderón) term is the Gilkey gauge-curvature coefficient: the  $U(1)$  Laplacian's Seeley–DeWitt  $a_4$  contains  $30\Omega_{ij}\Omega^{ij}/360 = \frac{1}{12}\Omega^2$  (Gilkey, *Invariance Theory*, Thm 4.8.16), and with  $\Omega = i\alpha F$  this forces an  $-(\alpha^2/12)F^2$  contribution to  $\log \det \Delta_{U(1)}$  — so the Calderón  $-2c_3^3\alpha^2$  is the gauge-curvature (Quillen-curvature) piece, not an ansatz. [E] the  $c_3$ -orders  $\{0, 3, 6\}$  are an arithmetic ladder (step 3 = the three channels;  $6 = |R^+(A_3)|$ , the  $C_6$  hexagon), and the  $\pi$ -power test confirms it:  $2c_3^3 = 1/(256\pi^3)$  carries  $\pi^3$ , i.e. *three* boundary  $c_3$ -insertions, not a single bulk coefficient. **Typing:** the boundary structure and the gauge-curvature origin of the quadratic term are heat-kernel-forced [E]; the exact coefficient value is [C] (the seam  $F$ -normalisation — retyped as the embedding index  $k_Y=5/3$  by the fifth step below, v470); the cubic  $\alpha^3$  Maxwell moment (a topological/Chern level) and the full “this is the seam  $U(1)$   $\zeta$ -determinant” claim stay [O] — exactly the EM.WARD.01 residual, unchanged. So the quadratic/curvature half of the EM-Ward origin is now derived; the cubic-moment half remains the honest open part. A *third honest step* (`[verification/v434_alpha_quillen_betafunction.py]`) settles those residuals and shows they are not three independent problems. The counterterm's matter factor is heat-kernel-forced too: by the  $\beta = a_4$  theorem the one-loop  $U(1)_Y$   $\beta$  is the  $a_4$  coefficient, and the carrier hypercharge content gives  $\sum \frac{2}{3}Y_f^2 + \frac{1}{3}Y_s^2 = 41/6$  (SM)  $\Rightarrow b_1 = 41/10$  (GUT, v159/v246), so  $8b_1c_3^6 = b_1(\text{rank } E_8)c_3^6$  is an  $a_4$  content coefficient times a boundary geometry — the same *kind* of object as the Calderón  $a_4$ . The exact value then turns only on the seam  $F$ -normalisation, so the “actual seam  $a_4$ ” residual is the  $F$ -normalisation [C] (one obligation, not two); and the cubic factors  $a^3 - 2c_3^3a^2 = a^2(a - 2c_3^3)$ , exhibiting  $a^3$  as the first moment of the Chern density  $a^2$  — a level whose from-first-principles origin stays [O]. The EM-Ward residual is thus *one* [C] + *one* [O], not three. A *fourth honest step* (`[verification/v435_alpha_quillen_chernlevel.py]`) sharpens that single remaining [O] — residual (2), the cubic  $\alpha^3$  Chern level — without closing it. A  $\pi$ -power / metric-independence test partitions the closure: the three coefficients carry  $\pi$ -powers exactly  $\{0, 3, 6\}$  ( $k_0 = 1$ ;  $-2c_3^3 = -1/(256\pi^3)$ ;  $-8b_1c_3^6 = -41/(327680\pi^6)$ ), so the Maxwell  $a^3$  is the *unique*  $\pi^0/c_3^0$  (metric-independent) term. A heat-kernel boundary coefficient always carries the  $(4\pi)^{-d/2}$  measure; a metric-independent term carries none — the signature of a *topological* term. So the test confirms, now from a test rather than a claim, that  $a^3$  is a level and not a Seeley–DeWitt curvature integral (the other two terms are heat-kernel boundary data). [E] *Conditional* on the Chern–Simons/ $\eta$  reading of the Maxwell moment (the EM1 interpretation, v48), large-gauge invariance quantises the seam  $U(1)$  level to  $\mathbb{Z}$ , so  $k_0 = 1$  is the *unit* (minimal) level — an integer, not a tunable real. [C] What stays [O] is the EM1 lemma itself — the from-first-principles Chern–Simons-boundary proof that  $\partial_\tau \log Z_{\text{Maxwell}} = a^3$  — external math (type A, v384). A *fifth honest step* (`[verification/v470_alpha_inflow_level.py]`) upgrades both leftovers, still without closing the target. [C] The level is computed, not minimal: under the same EM1 reading, the

$a^3$  coefficient  $k_0 = 1$  equals the FHS bulk Chern invariant  $|C| = 1$  of the very p+ip collar that realises S3 of SEAM.EQUIV.01 (v367/v461; control  $C(M=3) = 0$ ) — “why exactly 1” is answered by a computation plus the cited inflow identification (TKNN / Avron–Seiler–Simon quantisation; Callan–Harvey inflow; the APS/Witten  $\eta$ =CS reading of  $\delta \log \det$ ; Quillen/Bismut–Freed determinant-line integrality), replacing the minimality assumption of the fourth step. [E] *The  $F$ -normalisation is the embedding index:*  $k_Y = \text{tr}_5(Y^2)/\text{tr}(T_3^2) = (5/6)/(1/2) = 5/3$  (Ginsparg 1987), so  $(3/5) \times \frac{41}{6} = \frac{41}{10} = b_1$  exactly — the “GUT 3/5” is the inverse embedding index, and once the seam net is  $(E_8)_1$  at level 1 the  $U(1)_Y$  normalisation  $\langle J(z)J(w) \rangle = k/(z-w)^2$  is fixed by it (level-1 current-algebra rigidity, no continuous dial): the seam  $F$ -normalisation [C] has zero independent content — a face of SEAM.EQUIV.01, per the v382 typing. [C] *One phase, two responses:* the collar phase carries  $c_- = 8$  (gravitational, feeding SEAM.EQUIV.01/S3, v456) and  $C = 1$  ( $U(1)$ , feeding EM1) — so discharging the one realisation input feeds both named targets at once. *A sixth honest step* ([verification/v472\_quillen\_detline\_moduli.py]) exhibits that bridge lemma at the finite level. ALPHA.QUILLEN.EXACT.01 names the level as the curvature of the determinant line over the  $U(1)$  moduli (Quillen/Bismut–Freed curvature, Dai–Freed section) — and the fifth step computed the Chern invariant only over the Bloch Brillouin zone, the translation-invariant shortcut. The sixth step computes the Quillen-shaped object itself: the same collar Hamiltonian in real space with twisted boundary conditions — the twist torus  $(\theta_x, \theta_y)$  is the moduli space of flat  $U(1)$  connections — and the Fukui–Hatsugai–Suzuki curvature of the determinant line of the occupied frame over that torus integrates to  $2\pi \cdot 1$  exactly ( $10^{-9}$ ), size-independently ( $L = 4, 6$ ), with clean controls (trivial collar  $\rightarrow 0$ , orientation flip  $\rightarrow -1$ ), the twist-moduli integer equal to the Bloch-BZ integer for all three phases (Niu–Thouless–Wu), and the Fermi gap open over the whole torus (the Dai–Freed section has no zero). So “ $\delta \log \det(\text{seam model}) = \text{the inflow response}$ ” holds verbatim at the finite/model level, where  $\log \det$  of the occupied frame replaces  $\log \det_\zeta$ . [E] (computation) / [C] (the EM1 reading). What stays [O] is the continuum identification “ $\delta \log \det_\zeta(\text{seam}) = \text{the inflow response}$ ” on the abstract seam (the SEAM.EQUIV.01 face); ALPHA.QUILLEN.EXACT.01 stays [O], and  $\alpha_\star^{-1}$  stays [E] regardless. *A seventh step unifies this target with the  $\varphi_0^{\text{ret}}$ -puncture target* ([verification/v484\_seam\_contact\_unit.py]). The  $\{0, 3, 6\}$   $c_3$ -ladder above and the puncture’s per-mark weight  $c_3$  (v483) are one rule — the seam contact unit — and it is not a new unknown: “ $c_3$  per insertion” is the KMS seam unit  $2\pi = 1/(4c_3)$  with  $\frac{1}{4} = 1/|\mu_4|$  (v239), i.e. one bare boundary propagator ( $\frac{1}{2\pi} = 4c_3$ ) orbit-averaged over the four marks. [E] derived on the seam circle for the finite cycle sector (the bare Green function takes integer multiples of  $c_3 \ln 2$  at the  $\mu_4$  separations; the log-det contact expansion is exactly graded in  $c_3$  per insertion), with the grammar already suite-wide (the  $\Lambda$  prefactor  $\frac{3}{4\pi^2} = 48c_3^2$ , v60, carries the same  $\Omega_{\text{adm}} = 48$  at two insertions). [C]/[O] Consequence: ALPHA.QUILLEN.EXACT.01 and HYP.PHIO.PUNCTURE.01 merge into one remaining analytic step — the diagonal zeta-renormalisation at the marks plus the state-multiplicity matching ( $41 = 10b_1$  here,  $48 = \Omega_{\text{adm}}$  there). *An eighth step settles that remaining step at the computable level* ([verification/v485\_contact\_diagonal\_closed.py]). [E] The renormalised coincident-point Green function on a circle of circumference  $\ell$  is  $G_{\text{reg}}(0; \ell) = \frac{1}{\pi} \ln(\ell/2\pi)$  exactly — it vanishes precisely at  $\ell = 2\pi = 1/(4c_3)$ , the KMS seam circumference (v239): the seam scale is the unique zero-self-energy scheme point, and the log is the pure scale channel (matching the fact that  $F_{U(1)}$  carries exactly one log term while the  $\varphi_0^{\text{ret}}$  split is log-free). [E] With the diagonal zero the mark contact calculus resums in closed form (the BFK/gluing route, v151):  $\det(I - uC) = (1 - 4u)(1 + 2u)^2$  with  $u = \varepsilon c_3 \ln 2$  — all orders, the linear term absent because  $\text{Tr } C = 0$ . [E] And the two multiplicities are one state set under two response weights: the same  $48 = 3 \times 16$  admissible states counted flat give the  $\varphi_0^{\text{ret}}$  puncture count  $48 = \Omega_{\text{adm}}$ , and  $Y^2/\text{loop}/\text{GUT}$ -weighted give the alpha budget  $41 = 10b_1 = 40 + 1$  ( $\text{Tr}_{16} Y^2 = \frac{10}{3}$  exact,  $\nu^c = 0$ ; Ginsparg, v470). [C]/[O] Net: every finite/computable piece of the merged target is proven; the single remaining [O] is the identification of the abstract seam  $\zeta$ -det with this glued mark determinant — exactly the v382 typing (a face of SEAM.EQUIV.01), now substantiated computationally rather than asserted.

The full ablation — which inputs are load-bearing and which only set the last digit — is in the Ablation appendix below.



*Inversion test (this round).* Running the closure backwards — inserting the *measured*  $\alpha$  and solving for the integer budget  $M = \sum L + N_\Phi$  — returns  $M = 41.0000001\dots$ , i.e. the external measurement reconstructs the internal discrete budget  $41 = 40 + 1$  to  $10^{-7}$ . This is stronger than “we hit  $\alpha$ ”: the equation reads the discrete 41 out of nature. [E]

## 6.2 The decuple bridge (new, exact headline)

Eliminating  $\alpha_\star^{-1} = g_{\text{car}} A_{\text{EW}} - \delta_{\text{ph}}$  between the electroweak and the cosmological-constant lines gives a direct relation between the two scales:

**Decuple bridge:  $\Lambda$  is the tenth power of the stripped EW scale [E]**

$$\frac{\rho_\Lambda}{\bar{M}_{\text{Pl}}^4} = \frac{3}{4\pi^2} e^{2\delta_{\text{ph}}} \left[ \frac{v_{\text{geo}}}{5\beta_{\text{rad}}^2 \bar{M}_{\text{Pl}}} \right]^{10} \quad (29)$$

where the exponent  $10 = 2g_{\text{car}}$  and  $5 = g_{\text{car}}$ . Verified to a relative residual  $\sim 10^{-159}$ .

*Proof.* From  $v_{\text{geo}}/\bar{M}_{\text{Pl}} = g_{\text{car}}\beta_{\text{rad}}^2 e^{-A_{\text{EW}}}$ ,  $e^{-A_{\text{EW}}} = (v_{\text{geo}}/\bar{M}_{\text{Pl}})/(g_{\text{car}}\beta_{\text{rad}}^2)$ . From  $\rho_\Lambda/\bar{M}_{\text{Pl}}^4 = \frac{3}{4\pi^2} e^{-2\alpha_\star^{-1}}$  and  $\alpha_\star^{-1} = g_{\text{car}} A_{\text{EW}} - \delta_{\text{ph}}$ ,  $e^{-2\alpha_\star^{-1}} = e^{2\delta_{\text{ph}}} e^{-2g_{\text{car}} A_{\text{EW}}} = e^{2\delta_{\text{ph}}} (e^{-A_{\text{EW}}})^{2g_{\text{car}}}$ . Substituting and using  $2g_{\text{car}} = 10$ ,  $g_{\text{car}} = 5$  gives the boxed form.  $\square$

In words: the electroweak scale is not directly the  $\Lambda$  scale, but once the electroweak ratio is stripped of its small  $\beta_{\text{rad}}$  prefactor,  $\Lambda$  is its tenth power, and the “ten” is  $2g_{\text{car}}$ , not a fitted exponent. This is the direct algebraic compression of the action ladder.

## 6.3 The full power tower: $x^1, x^5, x^{10}$ on the carrier graph $K_5$

The decuple bridge is one rung of a tower. Writing  $x := e^{-A_{\text{EW}}} = v_{\text{geo}}/(5\beta_{\text{rad}}^2 \bar{M}_{\text{Pl}})$  for the stripped electroweak unit, all three cosmological scales are powers of the *same*  $x$  with binomial exponents (verified exactly):

$$\frac{v_{\text{geo}}}{5\beta_{\text{rad}}^2 \bar{M}_{\text{Pl}}} = x^{\binom{5}{0}} = x, \quad \frac{H_0}{\bar{M}_{\text{Pl}}} = x^{\binom{5}{1}} R_H = x^5 R_H, \quad \frac{\rho_\Lambda}{\bar{M}_{\text{Pl}}^4} = x^{\binom{5}{2}} R_\Lambda = x^{10} R_\Lambda \quad (30)$$

with residues  $R_H = e^{\delta_{\text{ph}}}/(2\pi\sqrt{\Omega_\Lambda})$  and  $R_\Lambda = \frac{3}{4\pi^2} e^{2\delta_{\text{ph}}}$ . [E]. The exponents are the complete-graph counts of the five-slot carrier  $K_5$ :

$$H \leftrightarrow \text{vertices of } K_5 \text{ } (\# = \binom{5}{1} = 5), \quad \Lambda \leftrightarrow \text{edges of } K_5 \text{ } (\# = \binom{5}{2} = 10).$$

So the cosmological constant is the product of one stripped-electroweak unit over every carrier *pair* ( $K_5$  edge product), and the Hubble scale the product over every carrier *slot* ( $K_5$  vertex product). This is the cleanest single-variable statement of the action ladder.

## 6.4 Hubble is the square root of $\Lambda$ (corollary, exact)

On the comparison layer  $\rho_\Lambda/\bar{M}_{\text{Pl}}^4 = 3\Omega_\Lambda(H_0/\bar{M}_{\text{Pl}})^2$ ; equating to the TFPT line gives

$$\frac{H_0}{\bar{M}_{\text{Pl}}} = \frac{e^{-\alpha_\star^{-1}}}{2\pi\sqrt{\Omega_\Lambda}}, \quad \mathcal{A}_H = -\ln \frac{H_0}{\bar{M}_{\text{Pl}}} = \alpha_\star^{-1} + \ln(2\pi\sqrt{\Omega_\Lambda}) = 138.68. \quad (31)$$

So the Hubble rung is *not* an independent third hit: it is the square root of the vacuum energy plus a geometric cosmology factor. The genuine TFPT content is the single  $e^{-\alpha_\star^{-1}}$ . (The numerical coincidence  $\mathcal{A}_H - \alpha_\star^{-1} \approx \pi^2/6$  is just  $\ln(2\pi\sqrt{\Omega_\Lambda})$  and is  $\Omega_\Lambda$ -dependent; not a claim. [O])

## 6.5 The $\sim 123$ orders of the cosmological constant

$$\log_{10} \frac{M_{\text{Pl}}^4}{\Lambda_{\text{IR}}} = \underbrace{\frac{2\alpha_{\star}^{-1}}{\ln 10}}_{119.028} + \underbrace{\log_{10} \frac{1}{\delta_{\text{top}}}}_{3.920} = 122.948 \approx 123. \quad (32)$$

The “119” is the action  $2\alpha_{\star}^{-1}$ ; the “ $\approx 4$ ” is the seam defect  $\delta_{\text{top}} = \Omega_{\text{adm}} c_3^4$ .

### High-precision cosmological constant: a $\sim 10^{-13}$ prediction [E]

Because  $\alpha_{\star}^{-1}$  is a *derived* number (the EM fixed point, known to 13 digits) and the residue  $3/(4\pi^2)$  is exact, the dimensionless vacuum energy is predicted essentially exactly:

$$\frac{\rho_{\Lambda}}{\bar{M}_{\text{Pl}}^4} = \frac{3}{4\pi^2} e^{-2\alpha_{\star}^{-1}} = 7.12533 \times 10^{-121} \quad [\text{E}] \text{ [verification/v7_gravity_cosmo.py]}, \quad (33)$$

i.e. a dark-energy scale  $\rho_{\Lambda}^{1/4} = (\frac{3}{4\pi^2})^{1/4} \bar{M}_{\text{Pl}} e^{-\alpha_{\star}^{-1}/2} = 2.23747 \text{ meV}$  (reduced Planck mass  $\bar{M}_{\text{Pl}} = 2.435323 \times 10^{18} \text{ GeV}$ ). The *dimensionless* ratio carries no observational input — its uncertainty is that of  $\alpha_{\star}^{-1}$ ,  $\sim 10^{-13}$  relative; the absolute meV value inherits only the Planck-mass uncertainty ( $\sim 2 \times 10^{-5}$ ). Both are *far* sharper than the measured vacuum energy (Planck  $\rho_{\Lambda}^{1/4} \simeq 2.24 \text{ meV}$ , known to  $\sim 1\%$  and entangled with the  $H_0$  tension). So TFPT pins the cosmological constant several orders of magnitude more precisely than it is currently measured — a genuine, falsifiable forward prediction: any future sub-percent  $\rho_{\Lambda}$  must land on 2.2375 meV.

## 6.6 The ladder is binomial; $2g_{\text{car}} = \binom{g_{\text{car}}}{2}$ singles out $g_{\text{car}} = 5$

In units of the electroweak action  $\alpha_{\star}^{-1}/g_{\text{car}}$ , the three rungs are the first three binomial coefficients of the five-slot carrier:

$$\frac{A_{\text{EW}}}{\alpha_{\star}^{-1}/g_{\text{car}}} : \frac{\mathcal{A}_H}{\alpha_{\star}^{-1}/g_{\text{car}}} : \frac{\mathcal{A}_{\Lambda}}{\alpha_{\star}^{-1}/g_{\text{car}}} = 1 : g_{\text{car}} : 2g_{\text{car}} = \binom{g_{\text{car}}}{0} : \binom{g_{\text{car}}}{1} : \binom{g_{\text{car}}}{2} = 1 : 5 : 10. \quad (34)$$

The first two are automatic ( $\binom{g_{\text{car}}}{0} = 1$ ,  $\binom{g_{\text{car}}}{1} = g_{\text{car}}$ ). The non-trivial content is the cosmological-constant charge:

$$2g_{\text{car}} = \binom{g_{\text{car}}}{2} \iff g_{\text{car}} = 5 \quad [\text{E}] \text{ [verification/v2_carrier_pascal.py]}, \quad (35)$$

since  $\binom{g_{\text{car}}}{2} = \frac{g_{\text{car}}(g_{\text{car}}-1)}{2} = 2g_{\text{car}} \iff g_{\text{car}} = 5$ . So the “double overlap” charge  $2g_{\text{car}}$  equals the number of carrier-slot *pairs*  $\binom{g_{\text{car}}}{2}$  *only* at  $g_{\text{car}} = 5$ : the cosmological constant is suppressed by one action unit per carrier pair, and this combinatorial reading is exclusive to the physical rank. It is an independent overdetermination of  $g_{\text{car}} = 5$  from the action grammar (cf. the landscape section).

## 6.7 Discriminant action theorem: the ladder reads the hypercharge discriminant

The  $g_{\text{car}}$  in the action ladder is not merely a rank: it is the square root of the discriminant of the carrier (hypercharge) polynomial. For a general split  $b + s = g$  the carrier polynomial is  $bsY^2 + (s-b)Y - 1 = 0$  with discriminant

$$\Delta_Y = (s-b)^2 + 4bs = (b+s)^2 = g_{\text{car}}^2, \quad \text{so for } 6Y^2 - Y - 1 = 0 : \Delta_Y = 25, \sqrt{\Delta_Y} = 5 = g_{\text{car}}. \quad (36)$$

Hence the action ladder is, exactly,

$$A_{\text{EW}} = \frac{\alpha_{\star}^{-1}}{\sqrt{\Delta_Y}}, \quad \mathcal{A}_H = \sqrt{\Delta_Y} A_{\text{EW}}, \quad \mathcal{A}_{\Lambda} = \binom{\sqrt{\Delta_Y}}{2} A_{\text{EW}} \quad [\text{E}]. \quad (37)$$

The electroweak hierarchy therefore reads the *discriminant of the hypercharge polynomial*, tying the Standard-Model hypercharge derivation directly to the EW-hierarchy and  $\Lambda$  problems.

## 6.8 A two-dimensional action lattice

All scales sit on the integer lattice spanned by two generators — the gauge unit  $\alpha_\star^{-1}/g_{\text{car}}$  and the flavor unit  $L_0 = \ln(1/u)$  — as  $S_x = a_x(\alpha_\star^{-1}/g_{\text{car}}) + b_x L_0 + (\text{residue})$ :

| scale                                | $a_x$               | $b_x$ | axis   |
|--------------------------------------|---------------------|-------|--------|
| $v_{\text{geo}}/\bar{M}_{\text{Pl}}$ | $1 = \binom{5}{0}$  | 0     | gauge  |
| $H_0/\bar{M}_{\text{Pl}}$            | $5 = \binom{5}{1}$  | 0     | gauge  |
| $\rho_\Lambda/\bar{M}_{\text{Pl}}^4$ | $10 = \binom{5}{2}$ | 0     | gauge  |
| $m_e/\bar{M}_{\text{Pl}}$            | 1                   | 5     | flavor |
| $m_\mu/\bar{M}_{\text{Pl}}$          | 1                   | 3     | flavor |
| $m_\tau/\bar{M}_{\text{Pl}}$         | 1                   | 2     | flavor |

The cosmological/gauge sector runs along  $b = 0$  with binomial charges; the charged-fermion sector runs along  $a = 1$  with the integer flavor ladder  $n_f = (5, 3, 2)$ . This is the compact “periodic table” of TFPT scales: two integer labels per scale. **[E]** (charges) / **[C]** (residues).

A third (seam/gravity) generator completes the lattice:  $L_c = \ln(1/c_3) = \ln(8\pi)$ . The scalaron lives purely on it,  $M_{\text{scal}}/\bar{M}_{\text{Pl}} = c_3^{7/2} = e^{-\frac{7}{2}L_c}$ , and the seam defect is  $\delta_{\text{top}} = e^{-(4L_c - \ln 48)}$ . So the three log-generators are  $\{\alpha_\star^{-1}$  (hierarchy),  $L_0 = \ln(1/u)$  (flavor),  $L_c = \ln(1/c_3)$  (seam/gravity) $\}$ . **[E]**

## 7 The Einstein normaliser as a corrected tree value

*Machine checks for this section:* `[verification/v1_e8_glue.py]`

The gravitational normaliser  $\xi = c_3/u$  has a transparent exact form. Writing  $u = \frac{1}{6\pi} [1 + \frac{9}{128\pi^3}]$ ,

$$\xi = \frac{c_3}{u} = \frac{3}{4} \cdot \frac{1}{1 + \frac{9}{128\pi^3}} = 0.748303 \quad \textbf{[E]}. \quad (38)$$

So  $\xi_{\text{tree}} = \frac{3}{4} = \dim \mathfrak{g}_{\text{SM}} / \dim S^+ = 12/16$  is the exact leading value (the inverse of the  $\frac{4}{3} = \dim S^+ / \dim \mathfrak{g}_{\text{SM}}$  family/gauge compression that sits in the seed  $u_{\text{base}} = \frac{4}{3}c_3$ , so  $\xi_{\text{tree}} u_{\text{base}} = c_3$ ), and the deviation is precisely the small topological seed correction  $9/(128\pi^3)$ , i.e. 0.226%. Gravity does not pick up a mysterious factor: first comes  $3/4$ , then the seam defect shifts it minimally.

## 8 $\Lambda$ -anchored metrology closure, and two anchors

*Machine checks for this section:* `[verification/v60_lambda_metrology_branch.py]`

The absolute SI scale is a calibration, not a compiler output. The dimensionless vacuum quotient lets one anchor it to the observed geometric curvature without circularity.

**Theorem 1** ( $\Lambda$ -anchored metrology closure). With  $\lambda_\Lambda := \rho_\Lambda/\bar{M}_{\text{Pl}}^4 = \frac{3}{4\pi^2} e^{-2\alpha_\star^{-1}}$  and the observed geometric curvature  $\Lambda_{\text{geom}}^{\text{obs}} > 0$ ,

$$\bar{M}_{\text{Pl}}^2 = \frac{\Lambda_{\text{geom}}^{\text{obs}}}{\lambda_\Lambda}, \quad G_N^{\text{TFPT}} = \frac{c^3}{8\pi\hbar} \frac{\lambda_\Lambda}{\Lambda_{\text{geom}}^{\text{obs}}} = \frac{3c^3}{32\pi^3\hbar \Lambda_{\text{geom}}^{\text{obs}}} e^{-2\alpha_\star^{-1}}.$$

*Proof.*  $\bar{M}_{\text{Pl}}^2 = 1/(8\pi G_N)$  and  $\Lambda_{\text{geom}} = \rho_\Lambda/\bar{M}_{\text{Pl}}^2$ ; the branch sets  $\rho_\Lambda = \lambda_\Lambda \bar{M}_{\text{Pl}}^4$ , so  $\Lambda_{\text{geom}} = \lambda_\Lambda \bar{M}_{\text{Pl}}^2$ , giving  $\bar{M}_{\text{Pl}}^2$  and then  $G_N = \lambda_\Lambda/(8\pi\Lambda_{\text{geom}}^{\text{obs}})$ ; restoring  $c^3/\hbar$  closes it.  $\square$

**Non-circularity ([O], essential).** The anchor must be the curvature  $\Lambda_{\text{geom}}^{\text{obs}}$  (units  $\text{m}^{-2}$ ), *not* the SI vacuum energy density  $\rho_\Lambda^{\text{obs}} = \Lambda_{\text{geom}} c^4/(8\pi G_N)$ , which already contains  $G_N$ . The curvature follows from  $\Lambda_{\text{geom}}^{\text{obs}} = 3\Omega_\Lambda H_0^2/c^2$ .

**Two independent anchors (crosscheck with teeth).** The metrology can be pinned in two ways that should agree:

1. *Electroweak anchor*: the dimensionless  $G_N v_{\text{phys}}^2 = 4.06717 \times 10^{-34}$  fixes  $G_N$  once  $v_{\text{phys}}$  is calibrated non-gravitationally.
2.  *$\Lambda$ -curvature anchor* (this note):  $G_N = \frac{3c^3}{32\pi^3 \hbar \Lambda_{\text{geom}}^{\text{obs}}} e^{-2\alpha_\star^{-1}}$ .

With the example anchor  $H_0 = 67.36 \text{ km s}^{-1} \text{ Mpc}^{-1}$ ,  $\Omega_\Lambda = 0.6847$  ( $\Lambda_{\text{geom}}^{\text{obs}} = 1.08914 \times 10^{-52} \text{ m}^{-2}$ ) the  $\Lambda$  anchor gives

$$G_N^{\text{TFPT}} = 6.65071 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}, \quad G_N^{\text{TFPT}} / G_N^{\text{CODATA}} - 1 = -3.5 \times 10^{-3}. \quad (39)$$

Closure algebra [E]; SI anchor and numerical match [C] (the cosmological anchor is model dependent; the Hubble tension applies). EW crosscheck: the  $\Lambda$ -pinned  $\bar{M}_{\text{Pl}} = 2.43964 \times 10^{18} \text{ GeV}$  gives  $v_{\text{geo}} = 242.6 \text{ GeV}$ , needing  $Z_{\text{EW}} = (246.22/242.61)^2 = 1.030$ .

**Prefactor branch ([O]).** This note uses  $\Lambda_{\text{IR}}/M_{\text{Pl}}^4 = \delta_{\text{top}} e^{-2\alpha_\star^{-1}}$ . A competing prefactor  $2c_3 e^{-2\alpha_\star^{-1}}$  would differ by  $2c_3/\delta_{\text{top}} = 661.5$  and misscale  $G_N$  by the same factor; the branch must be versioned.

**Mandatory  $\Lambda$ -layer typing ([O]).** Three distinct  $\Lambda$  objects must never be conflated:

$$\underbrace{\frac{\rho_\Lambda}{M_{\text{Pl}}^4} = \frac{3}{4\pi^2} e^{-2\alpha_\star^{-1}}}_{\text{vacuum density quotient (dimensionless)}} \neq \underbrace{\frac{\Lambda_{\text{IR}}}{M_{\text{Pl}}^4} = \delta_{\text{top}} e^{-2\alpha_\star^{-1}}}_{\text{seam-determinant / IR scale (dimensionless)}} \neq \underbrace{\Lambda_{\text{geom}}^{\text{obs}} = \frac{3\Omega_\Lambda H_0^2}{c^2}}_{\text{observed curvature [m}^{-2}\text{]}}.$$

Only  $\Lambda_{\text{geom}}^{\text{obs}}$  carries a dimension and is the non-circular anchor. Keeping the three typed protects the decuple bridge and the metrology closure from a spurious prefactor objection.

**Measurement-near (physical) decuple bridge.** Since  $v_{\text{phys}} = v_{\text{geo}} \sqrt{Z_{\text{EW}}}$  (charged-current residue, not a fit), the decuple bridge in the observable vev reads

$$\frac{\rho_\Lambda}{M_{\text{Pl}}^4} = \frac{3}{4\pi^2} e^{2\delta_{\text{ph}}} \left[ \frac{v_{\text{phys}}}{5\beta_{\text{rad}}^2 \sqrt{Z_{\text{EW}}} \bar{M}_{\text{Pl}}} \right]^{10}. \quad (40)$$

This is the same identity, written in the low-energy charged-current scale rather than the geometric source scale; both forms should appear side by side. [E]/[C]

## 9 The landscape: $g_{\text{car}} = 5$ is overdetermined

*Machine checks for this section:* [verification/v6\_bootstrap.py] [verification/v14\_carrier\_uniqueness.py]

Varying the carrier rank and rebuilding every constant from the same machinery ( $c_3$  fixed), the EM-closure value  $\alpha^{-1}(g_{\text{car}})$  is monotone and meets the observed value only at  $g_{\text{car}} = 5$ :

| $g_{\text{car}}$ | $N_{\text{fam}}$ | $\dim S^+$ | $\Omega_{\text{adm}}$ | $10b_1$   | $\alpha^{-1}(g_{\text{car}})$ | verdict                                             |
|------------------|------------------|------------|-----------------------|-----------|-------------------------------|-----------------------------------------------------|
| 3                | 2                | 4          | 8                     | 7         | 258.15                        | wrong $\alpha$ , wrong family count                 |
| 4                | 2.5              | 8          | 20                    | 17        | 187.31                        | non-integer families                                |
| <b>5</b>         | <b>3</b>         | <b>16</b>  | <b>48</b>             | <b>41</b> | <b>137.036</b>                | <b>SM, 3 families, observed <math>\alpha</math></b> |
| 6                | 3.5              | 32         | 112                   | 97        | 101.30                        | non-integer families                                |
| 7                | 4                | 64         | 256                   | 225       | 75.61                         | wrong $\alpha$                                      |

Several distinct requirements converge on  $g_{\text{car}} = 5$ : integer families ( $g_{\text{car}}$  odd); three families  $(g_{\text{car}}+1)/2 = 3$ ; the observed  $\alpha^{-1}$  on the monotone curve; the SM structure  $(b, s) = (3, 2)$ ,  $\dim S^+ = 16$ ; and  $\Omega_{\text{adm}} = 48$ ,  $10b_1 = 41$ . A clean algebraic shadow:  $\frac{g(g+1)}{2} = 2^{g-1} - 1$  has  $g = 5$  as its non-trivial solution ([C]).

## 10 The $E_8$ carrier atlas — stated unambiguously

Machine checks for this section: [\[verification/v5\\_e8\\_cascade.py\]](#)

In the series  $E_8$  is a *downstream scale grammar*, not a primitive cause: the boundary kernel fixes  $\alpha$ , the carrier rank, families and admissibility without  $E_8$  upstream. Three levels.

### 10.1 Level 1 — original $E_8$ data of the series ([E])

$$|R(E_8)| = 248 - 8 = 240 = 5 \cdot 48 = 5 \Omega_{\text{adm}} = \text{lcm}(\Omega_{\text{adm}}, D_{\text{start}}) = \text{lcm}(48, 60), \quad (41)$$

$$D_{\text{start}} = g_{\text{car}} \cdot \dim \mathfrak{g}_{\text{SM}} = 5 \cdot 12 = 60, \quad \gcd(\Omega_{\text{adm}}, D_{\text{start}}) = 12 = \dim \mathfrak{g}_{\text{SM}}, \quad (42)$$

$$\frac{|R(E_8)|}{D_{\text{start}}} = \frac{240}{60} = 4 = \frac{\Omega_{\text{adm}}}{\dim \mathfrak{g}_{\text{SM}}}, \quad 2880 = D_{\text{start}} \Omega_{\text{adm}} = |R(E_8)| \cdot \dim \mathfrak{g}_{\text{SM}}, \quad (43)$$

$$\kappa_{E_8} = \frac{\gamma_{\text{car}}}{\ln(248/60)} = 0.58723, \quad X(n, r, I_1) = \frac{\bar{M}_{\text{Pl}}}{8} \sin^2 \theta_{13} \left( \frac{60 - 2n}{60} \right)^{\kappa_{E_8}} e^{-(8-r)/64} e^{-12\pi I_1}. \quad (44)$$

**Coxeter/gauge typing of the lcm–gcd pair.** Writing  $D_{\text{start}} = 60 = 2h_{E_8}$  and  $\Omega_{\text{adm}} = 48 = 6 \text{rank}(E_8)$ , the relations become  $\gcd(2h_{E_8}, 6r_{E_8}) = \dim \mathfrak{g}_{\text{SM}} = 12$  and  $\text{lcm}(2h_{E_8}, 6r_{E_8}) = |R(E_8)| = 240$ : the local SM algebra is the gcd of the Coxeter double-period and the  $\mathbb{Z}_6$  rank-occupancy, and the  $E_8$  root count is their lcm. Likewise the EW exponent reads  $I_1^{\text{EW}} = \frac{41}{48} = \frac{h_{E_8}}{36} + \frac{1}{6r_{E_8}} = \frac{5}{6} + \frac{1}{48}$  (Coxeter principal part plus a rank singlet). [E] (arithmetic) / [C] (typing).

### 10.2 Level 2 — new exact counting identities ([E])

$$|R(E_8)| = 240 = 16 \cdot 15 = \dim S^+ (\dim S^+ - 1), \quad (45)$$

$$\dim E_8 = 248 = 8 \cdot 31 = (g_{\text{car}} + N_{\text{fam}})(2^{g_{\text{car}}} - 1), \quad (46)$$

$$|R(E_8)| = 8(31 - 1) = \text{rank}(E_8)(2^{g_{\text{car}}} - 2), \quad \text{rank}(E_8) = 8 = g_{\text{car}} + N_{\text{fam}} = 5 + 3, \quad (47)$$

$$10b_1 = 41 = \frac{|R(E_8)|}{6} + 1, \quad I_1^{\text{EW}} = \frac{41}{48} = \gamma_{\text{car}} + \frac{1}{\Omega_{\text{adm}}}, \quad (48)$$

with  $31 = 2^{g_{\text{car}}} - 1$  the number of non-empty occupation words of the  $[5, 4, 2]$  carrier code.

### 10.3 Level 2.5 — the even-flip transition atlas (a falsifiable program)

Identify  $S^+$  with the even-weight binary code on five bits. A transition  $x \rightarrow y \neq x$  has flip  $F = x \oplus y$ , even and nonempty, so  $|F| \in \{2, 4\}$ ; per state there are  $\binom{5}{2} + \binom{5}{4} = 10 + 5 = 15$  flips, giving

$$240 = 16 \cdot (10 + 5) = \underbrace{160}_{|F|=2} + \underbrace{80}_{|F|=4}, \quad (49)$$

and the 120 undirected flips match the 120  $E_8$  root *pairs* ([E], counting). This makes the atlas conjecture *testable*: a genuine root map must send each flip to a norm-2 vector in  $\mathbb{R}^8$  with inner products in  $\{-2, -1, 0, 1, 2\}$  reproducing the  $E_8$  profile (per root: 1 self, 56 at +1, 126 orthogonal, 56 at -1, 1 anti).

**Honest obstruction ([O]).** The flip atlas splits  $160 + 80$  by flip size; the standard  $E_8$  ( $D_8$ ) split is  $112 + 128$ . Since  $160 + 80 \neq 112 + 128$ , no flip-size-preserving map is an  $E_8$  root system; a metrization, if it exists, must mix flip sizes. Until a norm-2,  $E_8$ -profile embedding is exhibited,  $E_8$  stays a *counting* atlas, not a derived root origin.

**Sharper two-stage audit (Coxeter before Weyl).** A full root-system isomorphism is hard; a much earlier, cheaper test is the Coxeter spectrum. *Stage A*: find a natural order-30 operator  $C_{\text{TFPT}}$  on the even-flip atlas (or its 8-dimensional primitive-character space) with characteristic



polynomial  $\chi_{C_{\text{TFPT}}}(t) = \Phi_{30}(t)$ . *Stage B (only if A holds)*: the norm-2,  $E_8$ -profile, Weyl-closure metrization. Stage A can fail or carry far sooner than B, so it is the right first falsifier.

**Stage A is closed on the primitive-character space** (`[verification/v531_coxeter_tensor_stage_a.py]`).

The explicit integer operator

$$T_{30} = \text{Comp}(\Phi_5) \otimes \text{Comp}(\Phi_6)$$

(the  $4 \times 4$  carrier clock of `v418` tensor the  $2 \times 2$  family/CP hexagon) has *exact* order 30 ( $T_{30}^{30} = I$ ,  $T_{30}^k \neq I$  for  $1 \leq k < 30$ ) and  $\chi_{T_{30}} = \Phi_{30} = t^8 + t^7 - t^5 - t^4 - t^3 + t + 1$  exactly: by CRT ( $\gcd(5, 6) = 1$ ) its eigenvalues are the products (primitive 5th)  $\times$  (primitive 6th) = exactly the eight primitive 30th roots — the carrier clock times the family clock *is* the Coxeter clock, as one integer operator. Its traces are the Ramanujan sums,  $\text{tr}(T_{30}^k) = c_{30}(k) = c_5(k) c_6(k)$  for all  $k$ , with the signed divisor table  $(-1, 1, 2, 4, -2, -4, -8, 8)$  at  $k = (1, 2, 3, 5, 6, 10, 15, 30)$  and the atom readout  $(|\text{tr } T^3|, |\text{tr } T^5|, |\text{tr } T^{15}|) = (2, 4, 8) = (|\mathbb{Z}_2|, |\mu_4|, \text{rank } E_8)$  — eight integer traces type the clock *without* diagonalisation, a cheap kill test for any flip-operator candidate (honest scope:  $\Phi_{15}$  passes the unsigned triple since  $\Phi_{30}(t) = \Phi_{15}(-t)$ ; the *signed* table is the discriminator, and  $\Phi_{16}, \Phi_{20}, \Phi_{24}$  die on the triple already). **[E]** *What stays open ([O])*: a canonical intertwiner from the concrete even-flip atlas onto this primitive space (`COX.FLIP.INTERTWINER`), and Stage B.

**Audit only — non-load-bearing speculation: the dim 496 heterotic count [O]**. *This box feeds no derivation; it is an arithmetic coincidence kept for completeness only, under the no-free-pattern rule.* The even half  $S^+$  and odd half  $S^-$  of the exterior algebra each have dimension  $2^{g_{\text{car}}-1} = 16$  and each carry  $16 \cdot 15 = 240$  transitions. Together the full carrier ( $2^{g_{\text{car}}} = 32$  states) gives

$$\dim S^+(2^{g_{\text{car}}} - 1) = 16 \cdot 31 = 496 = \dim(E_8 \times E_8) = \dim SO(32) = \binom{32}{2} \quad (\text{count}),$$

i.e. the two halves read as two  $E_8$  root atlases, all  $\binom{32}{2} = 496$  pairs as  $SO(32)$  — the two anomaly-free heterotic gauge dimensions. The *dimension* equality is exact; that the carrier algebra *is* either gauge group is unproven and not used anywhere. **[O]** (audit-only). The same fingerprint recurs at checksum level: the dual products of the  $E_8$  invariant degrees sum to  $744 = 3 \cdot 248$  (the  $j$ -function constant), and the identical construction for  $D_{16}$  gives  $1488 = 3 \cdot 496 = 2 \cdot 744$  (`[verification/v532_e8_degree_modular_checksum.py]`) — the checksum measures the common heterotic rank-16/Coxeter-30 structure, *not*  $E_8$  uniqueness. **[E]** (audit-only).

## 10.4 Level 2.7 — the carrier Coxeter theorem (the strongest new bridge)

Use the universal root-system identity  $|R| = \text{rank} \cdot h$  (Coxeter number  $h$ ). With the carrier values  $\text{rank}(E_8) = g_{\text{car}} + N_{\text{fam}} = 8$  and  $|R(E_8)| = \dim S^+(\dim S^+ - 1) = 240$ ,

$$h_{E_8} = \frac{|R(E_8)|}{\text{rank}(E_8)} = \frac{240}{8} = 30 = N_{\text{fam}} \binom{g_{\text{car}}}{2} = N_{\text{fam}} A_{\Lambda} = 2g_{\text{car}} N_{\text{fam}} = 2^{g_{\text{car}}} - 2 \quad \text{[E]}. \quad (50)$$

The Coxeter number of  $E_8$  is exactly three families times the cosmological pair action: each family contributes one  $\Lambda$ -decuple ( $A_{\Lambda} = \binom{5}{2} = 10$ ) to the Coxeter period. Equivalently the action ladder is a Coxeter ladder,

$$A_H = \frac{h_{E_8}}{2N_{\text{fam}}} = 5, \quad A_{\Lambda} = \frac{h_{E_8}}{N_{\text{fam}}} = 10, \quad (51)$$

so Hubble reads half a Coxeter period per family and  $\Lambda$  a full Coxeter period per family. **[C]**

**Rank–Coxeter coupling.** Equivalently  $\dim S^+ = 2 \text{rank}(E_8) = 16$  and  $h_{E_8} = 2(\dim S^+ - 1) = 2 \cdot 15 = 30$ , so  $\dim E_8 = (g_{\text{car}} + N_{\text{fam}})(h + 1) = 8 \cdot 31$ . The two counting forms  $16 \cdot 15 = 8 \cdot 30 = 240$  coincide because  $\dim S^+ = 2r$ . Dropping  $\nu^c$  sends  $\dim S^+ \rightarrow 15$ , which breaks *both*  $\dim S^+ = 2r$  and  $\dim S^+(\dim S^+ - 1) = 240$ :  $\nu^c$  is the state that closes the rank–Coxeter coupling, not an optional singlet. **[E]/[C]**

### 10.5 Level 2.8 — positive-root moment: $|R^+(E_8)| = \text{Tr}_{S^+} X^2 = 5!$

The number of positive  $E_8$  roots equals the master hypercharge moment:

$$\boxed{|R^+(E_8)| = 120 = \text{Tr}_{S^+} X^2 = 5!} \quad [\mathbf{E}], \quad (52)$$

so the second hypercharge moment of one family *is* the positive-root half-system. The whole integer chain then reads off  $|R^+|$ :

$$\sum_{f,j} L_{f,j} = \frac{|R^+(E_8)|}{3} = 40, \quad 10b_1 = 1 + \frac{|R^+(E_8)|}{3} = 41, \quad \delta_2 = 4! |R^+(E_8)| c_3^8. \quad (53)$$

The  $\mathbb{Z}_6$  slice is then  $|R(E_8)|/6 = \frac{|R^+|}{3} = 40 = \Omega_{\text{adm}}\gamma$ : one  $\mathbb{Z}_6$  phase slice of the  $E_8$  root atlas is exactly the full flavor transport budget.  $[\mathbf{E}]$  (arithmetic) /  $[\mathbf{C}]$  (interpretation).

### 10.6 Level 2.9 — the 2·3·5 Coxeter cyclotomic compiler (meta-theorem)

The three hard carrier atoms — weak rank 2, color rank / family count 3, carrier rank  $g_{\text{car}} = 5$  — generate the  $E_8$  data *cyclotomically*, without taking  $E_8$  as input. Set

$$\boxed{h := 2 \cdot 3 \cdot g_{\text{car}} = 30, \quad r := \varphi(h) = \varphi(30) = 8 = g_{\text{car}} + N_{\text{fam}},} \quad (54)$$

where  $h$  is the Coxeter number and  $r = \varphi(2 \cdot 3 \cdot 5) = 8$  the rank. Then the standard  $|R| = rh$ ,  $\dim = r(h+1)$  give

$$|R(E_8)| = rh = 8 \cdot 30 = 240, \quad \dim E_8 = r(h+1) = 8 \cdot 31 = 248, \quad h+1 = 2^{g_{\text{car}}} - 1 = 31. \quad (55)$$

Moreover the  $E_8$  exponents are exactly the primitive residues modulo  $h = 30$ :

$$\text{Exp}(E_8) = (\mathbb{Z}/30\mathbb{Z})^\times = \{1, 7, 11, 13, 17, 19, 23, 29\}, \quad \sum_{m \in (\mathbb{Z}/30\mathbb{Z})^\times} m = 120 = \text{Tr}_{S^+} X^2 = |R^+(E_8)|, \quad (56)$$

so the second hypercharge moment equals the sum of the  $E_8$  exponents and the positive-root count. The  $\mathbb{Z}_6 \times \mathbb{Z}_5 \simeq \mathbb{Z}_{30}$  tact has  $\varphi(30) = 8$  primitive characters, whose phases  $e^{2\pi i m/30}$  are the Coxeter eigenphases of  $E_8$ ; its characteristic polynomial is the 30th cyclotomic polynomial

$$\chi_C(t) = \Phi_{30}(t) = t^8 + t^7 - t^5 - t^4 - t^3 + t + 1. \quad (57)$$

**Reading:**  $E_8$  here is not primarily a root count but the *Coxeter cyclotomic readout of the 2, 3, 5 carrier*.  $[\mathbf{E}]$  (arithmetic) /  $[\mathbf{C}]$  (interpretation).

### 10.7 Level 3 — conjectural geometric readings ( $[\mathbf{C}]$ )

**Conjecture 1** (Carrier transition atlas).  $R(E_8) \cong \{(a, b) \in S^+ \times S^+ : a \neq b\}$  as a structure-bearing transition set, since  $16 \cdot 15 = 240$ . *Open*: a structure-preserving map (roots, lengths, Weyl data), not only the count.

**Conjecture 2** (Least common refinement / rank-8 over the exhausted carrier).  $240 = \text{lcm}(\Omega_{\text{adm}}, D_{\text{start}})$  reads  $E_8$  as the least common refinement of the occupancy lattice and the  $\mathbb{Z}_6$  phase lattice;  $248 = (g_{\text{car}} + N_{\text{fam}})(2^{g_{\text{car}}} - 1)$  reads it as a rank-8 structure over the 31 non-empty carrier words.

**Two routes to  $E_8$  — keep their status separate.** Levels 1–2 are exact arithmetic. There are now *two* distinct routes to  $E_8$ , and they must not be conflated:

- **Glue route (closed,  $[\mathbf{E}]$ ):**  $E_8 = (D_5 \oplus A_3) + \mu_4\text{-glue}$  is built explicitly in Section 3 (disc( $D_5$ ) = disc( $A_3$ ) =  $\mathbb{Z}_4$ , glue index 4, 240 roots of norm 2, simple-root Gram det 1). This closes the former “blind 240-metrisation”.

- **Direct flip-atlas route (open, [O], optional bonus):** the even-flip transition set  $240 = 160 + 80$  is *not* a flip-size-preserving  $E_8$  map, since  $160 + 80 \neq 112 + 128$ . A direct structure-preserving metrisation of the flip graph remains open — a nice-to-have, not load-bearing, because the glue route already delivers  $E_8$ .

So  $E_8$  is closed via the glue; only the direct flip atlas is open.

## 11 $\mathbb{Z}_6$ monodromy, flavor and CP readouts

Machine checks for this section: `[verification/v88_cp_phase_audit.py]`

A single  $\mathbb{Z}_6$  motif recurs in the quotient  $G_{\text{phys}} = (SU(3) \times SU(2) \times U(1)_Y) / \mathbb{Z}_6$ , the transport cusps  $\{1, (2/3)^6, (1/3)^6\}$ ,  $\delta_{\text{ph}} = z_\star^{1/6}$ , and the CP phases

$$\delta_{\text{CKM}}^{(0)} = \frac{\pi}{3}, \quad \delta_{\text{CKM}} = \frac{\pi}{3} + 3\lambda_C^2 = 1.19823 \text{ rad}, \quad \delta_{\text{CP}}^{\text{PMNS}} = \frac{4\pi}{3} = 240^\circ. \quad (58)$$

From the documented inputs  $s_{12} = \lambda_C$ ,  $s_{23} = u/(1 + \lambda_C)$ ,  $s_{13} = \lambda_C^3/3$  one gets the new derived readouts  $|V_{td}| = 0.00908$ ,  $|V_{ts}| = 0.04264$ ,  $J_{\text{CKM}} = 3.33 \times 10^{-5}$ . On the neutrino side, with  $\sin^2 \theta_{13} = e^{-5/6}u = 0.02311$ ,  $\sin^2 \theta_{23} = 0.4557$ ,  $\delta_{\text{CP}} = 4\pi/3$  and the solar candidate  $\sin^2 \theta_{12} = \frac{1}{3} - \frac{u}{2} = 0.30675$ , a near-maximal  $|J_{\text{CP}}^\nu| \approx 0.030$  (sharp readout for DUNE/Hyper-Kamiokande). [C]

### 11.1 CP as triple seed variance; a closed observable quadrilateral

The Cabibbo angle is the Bernoulli width of the seed. The *exact* CKM phase is the family transport holonomy (document 3),  $\delta_{\text{CKM}} = \arg \text{Tr} \Pi_\Delta(\nabla_F^*)$ ; on the small-area branch this has the compact readout form (with an  $O(\lambda_C^4)$  remainder)

$$\lambda_C = \sqrt{u(1-u)}, \quad \boxed{\delta_{\text{CKM}} - \frac{\pi}{3} = N_{\text{fam}} u(1-u) = 3\lambda_C^2} \quad [\text{E}]/[\text{C}] \text{ (readout; exact = holonomy).} \quad (59)$$

The coefficient 3 is the family number  $N_{\text{fam}}$ : weak CP is the  $\mathbb{Z}_6$  base angle  $\pi/3$  plus  $N_{\text{fam}}$  times the seed (Bernoulli) variance. The  $\mathbb{Z}_6$  base angle now has an  $A_3$  root: since  $\mathbb{Z}_6 = |R^+(A_3)| = 6$  (the positive-root hexagon of the family algebra), the CP phases are integer steps of the  $A_3$  holonomy quantum  $2\pi/|R^+(A_3)|$ ,

$$\boxed{\delta_{\text{CKM}} = \frac{2\pi}{|R^+(A_3)|} + N_{\text{fam}} u(1-u), \quad \delta_{\text{CP}}^{\text{PMNS}} = 4 \cdot \frac{2\pi}{|R^+(A_3)|} = \frac{4\pi}{3}} \quad [\text{E}]/[\text{C}]. \quad (60)$$

So weak CP = ( $A_3$  positive-root phase step) + (family count  $\times$  seed variance), while strong CP is the *determinant* sector, erased. The solar angle reads as a carrier root plus a half-seed,  $\sin^2 \theta_{12} = \frac{1}{3} - \frac{u}{2} = -y_- - \frac{u}{2}$  (since  $-y_- = \frac{1}{3}$ ), so PMNS = carrier root + seed +  $A_3$  holonomy phase. [C] Strong CP is killed by the determinant selector ( $\arg \det M_u = \arg \det M_d = 0$ ,  $\bar{\theta} = 0$ ), while weak CP survives as the seed-deformed  $\mathbb{Z}_6$  holonomy: CP is not forbidden, only *determinant* CP is. [C] — the strong-CP closure lives on the admissible QFT branch and is *conditional* on the analytic hypotheses of the QFT closure (document 3, Part III: reflection positivity, temperedness, clustering, mass gap, OS reconstruction); its status is therefore weaker than the discrete lattice identities, even though the semantics “determinant CP forbidden, holonomy CP allowed” is robust. Since  $u = \Omega_b + \beta_{\text{rad}}$  exactly, eliminating the seed closes the four UV-shadow observables into one cycle:

$$\lambda_C^2 = (\Omega_b + \beta_{\text{rad}})(1 - \Omega_b - \beta_{\text{rad}}), \quad \sin^2 \theta_{13} = e^{-5/6}(\Omega_b + \beta_{\text{rad}}), \quad \Omega_b = \left(1 - \frac{1}{4\pi}\right) e^{5/6} \sin^2 \theta_{13}. \quad (61)$$

One measured member of  $\{\lambda_C, \theta_{13}, \Omega_b, \beta_{\text{rad}}\}$  fixes the other three. [E] (the four rows remain operationally distinct readout classes).

## 12 Mass hierarchy as a hexagon with one extra turn

Machine checks for this section: `[verification/v18_quark_yukawa.py]` `[verification/v20_lepton_c_derivation.py]`

The transport length packets are

$$L_u = (7, 3, 0), \quad L_d = (7, 5, 2), \quad L_e = (8, 5, 3), \quad \sum_{f,j} L_{f,j} = 40 = \Omega_{\text{adm}} \gamma_{\text{car}} = 10b_1 - 1. \quad (62)$$

Modulo 6 they lie on related triples of the  $C_6$  hexagon; the first generation carries one extra full turn ( $u, d : 1 + 6; e : 2 + 6$ ), costing  $\lambda_Y^6 = (u(1-u))^3 \approx 1.28 \times 10^{-4}$  and explaining why the first generation is so light — “one turn more”, not a free fit. **[C]** *Flagged proximity ([O], not exact)*: the full-turn cost is close to the topological seam defect,  $\lambda_C^6 / \delta_{\text{top}} = 1.061$  ( $\sim 6\%$ ), hinting that one flavor winding  $\approx$  one seam-defect unit  $\delta_{\text{top}} = \Omega_{\text{adm}} c_3^4$  — a structural smell, not a theorem. Writing the length matrix  $L = R + W$  with  $R = L \bmod 6$  the residue shadow and  $W$  the full-winding part,

$$L = \begin{pmatrix} 7 & 3 & 0 \\ 7 & 5 & 2 \\ 8 & 5 & 3 \end{pmatrix} = \underbrace{\begin{pmatrix} 1 & 3 & 0 \\ 1 & 5 & 2 \\ 2 & 5 & 3 \end{pmatrix}}_R + 6 \underbrace{\begin{pmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}}_W, \quad \sum R = 22, \quad \sum W = 18 = 3 \cdot 6, \quad \sum L = 40,$$

and the column sums are  $(22, 13, 5)$  with  $S_1 = 22 = \sum R$  and  $S_2 + S_3 = 18 = \sum W$ . So the first generation is light because it stores the *entire* residue budget plus the three full  $C_6$  windings; generations 2 and 3 are the winding complement. **[E]** (arithmetic).

### 12.1 Lepton flavor decouple: the charged-lepton determinant carries $u^{10}$

The source masses on the  $v_{\text{geo}}$  branch are  $(\hat{m}_e, \hat{m}_\mu, \hat{m}_\tau) = \frac{v_{\text{geo}}\pi}{\sqrt{2}} (\frac{16}{7}u^5, \frac{4}{3}u^3, \frac{7}{6}u^2)$ . Their product is a pure decouple in the seed:

$$\boxed{\frac{\hat{m}_e \hat{m}_\mu \hat{m}_\tau}{(v_{\text{geo}}\pi/\sqrt{2})^3} = \left(\frac{16}{7} \cdot \frac{4}{3} \cdot \frac{7}{6}\right) u^{5+3+2} = \frac{32}{9} u^{10}} \quad \text{[E]}, \quad (63)$$

with  $5 + 3 + 2 = g_{\text{car}} + b + s = 10 = \binom{5}{2}$ . So the charged-lepton Yukawa powers  $(n_e, n_\mu, n_\tau) = (5, 3, 2)$  are the three carrier numbers  $(g_{\text{car}}, b, s)$ , and the lepton determinant carries the same decouple ( $u^{10}$ ) that the cosmological constant carries in the gauge sector ( $A_\Lambda = 10$ ):  $\Lambda$  and the charged-lepton determinant are two realisations of the same degree- $\binom{5}{2}$  sector.

**Sum–product reading.** The lepton powers  $(5, 3, 2)$  encode *both* decouple numbers at once: their sum is the  $\Lambda$  action, their product is the  $E_8$  Coxeter number,

$$5 + 3 + 2 = 10 = A_\Lambda, \quad 5 \cdot 3 \cdot 2 = 30 = h_{E_8} = 2 \cdot 3 \cdot 5, \quad (64)$$

and  $(5, 3, 2)$  is the prime factorisation of  $h_{E_8} = 30$ . So  $\det M_\ell$ ,  $\rho_\Lambda$  and the  $E_8$  Coxeter period read the same carrier core additively ( $= 10$ ) and multiplicatively ( $= 30$ ). **[E]/[C]**

### 12.2 Why the top Yukawa is $\approx 1$ : the $L = 0$ anchor (Froggatt–Nielsen ladder)

The transport length  $L_f$  is a Froggatt–Nielsen-type charge with expansion parameter  $\lambda_C$ :

$$\boxed{y_f \simeq \lambda_C^{L_f} \times \Lambda_f \quad (\Lambda_f = \mathcal{O}(1)), \quad y_t \simeq 1 \text{ because } L_t = 0} \quad \text{[E]/[C]}. \quad (65)$$

The top sits at  $L_t = 0$  (the *unwound* carrier direction), so  $y_t = \Lambda_t \simeq 1$  ( $\sqrt{2} m_t / v = 0.991$ ); every other charged fermion is suppressed by integer powers of  $\lambda_C = 0.2244$ . Numerically (with  $\mathcal{O}(1)$  sector prefactors  $\Lambda_f \in [0.43, 1.07]$ ):

| $f$                   | $t$  | $b$  | $\tau$ | $c$  | $\mu$ | $s$  | $u, d$                 | $e$  |
|-----------------------|------|------|--------|------|-------|------|------------------------|------|
| $L_f$                 | 0    | 2    | 3      | 3    | 5     | 5    | 7                      | 8    |
| $y_f/\lambda_C^{L_f}$ | 0.99 | 0.48 | 0.90   | 0.65 | 1.07  | 0.94 | $\sim 0.4\text{--}0.9$ | 0.46 |

So the entire charged-fermion hierarchy is one  $\lambda_C$ -ladder anchored by the top at  $L = 0$ ; the lightness of the first generation ( $L = 7, 8$ ) and the heaviness of the top ( $L = 0$ ) are the two ends of the same integer winding spectrum. This is the standard Froggatt–Nielsen mechanism with  $\lambda_C$  as the flavon ratio and  $L_f$  as the FN charge — a clean bridge between TFPT transport and an established flavor mechanism.

### 13 Inflation interface

*Machine checks for this section:* [\[verification/v7\\_gravity\\_cosmo.py\]](#) [\[verification/v86\\_nstar\\_reheating.py\]](#) [\[verification/v494\\_cosmo\\_running\\_mu\\_record.py\]](#)

The documented scalaron is Starobinsky  $R^2$ ; with the pivot  $N_\star = 57.1$ :

$$n_s = 1 - \frac{2}{N_\star} = 0.96497, \quad r = \frac{12}{N_\star^2} = 0.00368, \quad (66)$$

$$\frac{M_{\text{scal}}}{\bar{M}_{\text{Pl}}} = \sqrt{8\pi} c_3^4 = c_3^{7/2} = (8\pi)^{-7/2} \Rightarrow M_{\text{scal}} \approx 3.06 \times 10^{13} \text{ GeV}.$$

$n_s$  matches Planck;  $r = 0.0037$  is the falsifiable target for LiteBIRD/CMB-S4.

**Amplitude (status-clean).** The scalaron mass is a pure seam power,

$$\boxed{\frac{M_{\text{scal}}^2}{\bar{M}_{\text{Pl}}^2} = c_3^{r-1} = c_3^{\Omega_{\text{adm}} - 10b_1} = c_3^7} \quad [\mathbf{E}], \quad (67)$$

an *exact* identity (the exponent  $7 = 48 - 41$  is the post-flavor, post-Higgs occupancy deficit). The cosmology interface reduces the geometric gravity action to an FRW  $R^2$ /scalaron branch in which  $M_{\text{scal}}$  is the Starobinsky mass  $M$ ; the identification is therefore derived in that sector, not posited. With the standard large- $N$  amplitude  $A_s = \frac{N_\star^2}{24\pi^2} (M/\bar{M}_{\text{Pl}})^2$  this gives

$$A_s = \frac{N_\star^2 c_3^7}{24\pi^2} = \frac{N_\star^2 \delta_2}{8640\pi} = 2.17 \times 10^{-9} \quad [\mathbf{C}]/[\mathbf{O}], \quad (68)$$

matching Planck ( $\simeq 2.1 \times 10^{-9}$ );  $\delta_2 = 2880 c_3^8$  is the second seam defect. The seam-power form is exact [\[E\]](#); the amplitude is then determined *within* the  $R^2$  sector, the only residual being the geometric-gravity (Hodge) closure on which that sector rests [\[C\]](#). *Convention note:*  $A_s \propto N_\star^2$ , so the quoted  $2.17 \times 10^{-9}$  uses the pivot  $N_\star = 57.1$ ; the alternative seed pivot  $N_\star = 3/u - 1 = 55.4$  gives  $2.05 \times 10^{-9}$ . A number should only be quoted with  $N_\star$  fixed.

**Running (consistency relation).** The same attractor also fixes the *running* of the tilt parameter-free:  $\alpha_s = \text{d}n_s/\text{d}\ln k = -2/N_\star^2 = -r/6$  and  $\beta_s = -4/N_\star^3$  at leading order; the exact slow roll on the Starobinsky potential gives  $\alpha_s = -7.09 \times 10^{-4}$ ,  $\beta_s = -2.70 \times 10^{-5}$  at the reheating point  $N_\star = 51.4$  (leading-order band  $[-8.0, -5.6] \times 10^{-4}$  over  $N_\star \in [50, 60]$ ). Current data are consistent (Planck 2018  $-0.0045 \pm 0.0067$ :  $+0.57\sigma$ ; P-ACT-LB  $+0.0062 \pm 0.0052$ :  $-1.33\sigma$ ), and since plateau potentials give  $\alpha_s < 0$ , any robust *positive* running would kill the branch. [\[C\]](#) [\[verification/v494\\_cosmo\\_running\\_mu\\_record.py\]](#)

### 14 Further simple bridges

*Machine checks for this section:* [\[verification/v74\\_compiler\\_micro\\_lemmas.py\]](#)

- **Dark-energy equation of state  $w = -1$  (exact).** The TFPT  $\Lambda$  is a genuine vacuum constant  $\rho_\Lambda = \frac{3}{4\pi^2} e^{-2\alpha_\star^{-1}} \bar{M}_{\text{Pl}}^4$ , not a rolling field, so  $w = -1$  identically (no quintessence). A robust measurement  $w \neq -1$  would falsify the  $\Lambda$ -branch. [\[E\]](#) (structural).



- **Effective neutrino number**  $N_{\text{eff}} \simeq 3$ . Three families ( $N_{\text{fam}} = 3$ ) give three light neutrinos and the standard  $N_{\text{eff}} \simeq 3.04$ . [E].
- **Top Yukawa**  $\simeq 1$  as the  $L = 0$  anchor (above): a bridge to the Froggatt–Nielsen flavor mechanism.
- *Flagged (not clean)*:  $m_H/v = 0.509$  is near  $\frac{1}{2}$  (1.7%);  $m_W/m_Z = 0.881$  matches the running  $\sqrt{1 - \sin^2 \theta_W(M_Z)}$  (not the tree  $\sqrt{5/8} = 0.79$ ). No clean closed bridge for the Higgs/ $W/Z$  masses yet. [O]

## 15 Closures that the architecture rests on

Machine checks for this section: [verification/v36\_spectral\_action\_g2.py] [verification/v4\_flavor\_matrix.py]

Five results close or sharpen the discrete core. They are *proved in this document set* — the  $\varphi_0^{\text{ret}}$ -ladder, the H2 realisation and the strong-CP/dynamics gap in document 2 (tfpt\_2\_standard\_model), the  $E_6 \times A_2$  shadow in document 3 (tfpt\_3\_e8\_audit\_bootstrap) — and are collected here because the architecture above depends on them.

1. **The  $\varphi_0^{\text{ret}}$ -ladder (whole spectrum, one formula).** Every fermion mass is  $\hat{m}_{f,j} = \frac{v_{\text{geo}}}{\sqrt{2}} \lambda_Y^{L_{f,j}} \Lambda_{f,j}$  with  $\lambda_Y = \sqrt{\varphi_0^{\text{ret}}(1 - \varphi_0^{\text{ret}})}$ , and the closed evaluation gives the universal form  $\lambda_Y^L \Lambda = \pi c(\varphi_0^{\text{ret}})^k$  — the nine  $\Lambda$  are hexagonal-resolvent outputs (not free numbers); the *charged-lepton* ones are exact, the *quark* ones are reduced to the  $(U_{\text{wall}})$  selector. The  $\varphi_0^{\text{ret}}$ -order ladder is  $t(0) > \{b, c, \tau\}(2) > \{s, \mu\}(3) > \{u, d\}(4) > e(5)$ . [E] (leptons exact, quark ratios closed v49/v71) / [O] (absolute quark scale)
2. **H2 reduced via the Mehta–Seshadri framework.** Mehta–Seshadri supplies the equivalence (stable parabolic  $\leftrightarrow$  irreducible unitary); the TFPT-specific step — “stable = transport-minimal” and the geodesic  $C_6$  word length — is conditional on the unitarity (U) of  $\nabla_F^*$  (document 3) and the parabolic-degree  $\rightarrow C_6$  dictionary. H2 is the sharp realisation statement: the  $D_4$ -equivariant stable structure on  $\mathcal{O}(-2) \oplus \mathcal{O}(-1)^2$  realises the branch with  $\det R = h(D_5) = 8$ , principal 2-minors (2, 3, 5). [E] (given (U))
3. **Strong CP / dynamics via the explicit gap.**  $\theta_{\text{eff}} = 0$  needs only  $\gamma_5$ -Hermiticity + polar mass + sheet involution + reflection positivity (no mass gap). The full dynamics closes on the physical sector because the  $C_6$  transport transfer operator has the *explicit* spectrum  $\{(k/3)^6\}$ , giving the mass gap  $\Delta = 6 \log \frac{3}{2} = 2.43$  in closed form  $\Rightarrow$  OS reconstruction  $\Rightarrow$  measure on the admissible sector. [E] (phys. sector)
4.  **$E_6 \times A_2$  flavor shadow.** The residue matrix sits inside an  $E_8$  secondary branching:  $248 = \|R\|_F^2 + \det R + 2(1^\top R a) N_{\text{fam}} = 78 + 8 + 2 \cdot 27 \cdot 3$ , with  $\|R\|_F^2 = 78 = \dim E_6$ ,  $\det R = 8 = \dim A_2$ . The seven  $E_8$  slices form an audit raster (atlas note). [O]
5. **SM completeness.** The discrete structure and dimensionless core are closed; the dimensionful EW/QCD masses ( $m_W, m_Z, m_H, \sin^2 \theta_W, \alpha_s$ ) are schema-level via  $\mathcal{R}_{\text{SM}}$ , with  $m_H$  resting on the closed UV quartic  $\lambda_\Phi = 1/(16\pi^2)$  (near-criticality). The *exact* solar angle  $\theta_{12}$  (leading  $1/N_{\text{fam}} = \frac{1}{3}$ ) is *conditionally derived*, pending the seam-misalignment lemma — the sharpest remaining SM tension point. [C]

## 16 Status summary and open audit points

Machine checks for this section: [verification/v22\_open\_gates\_audit.py]

| Statement                                                                             | Status | Note                        |
|---------------------------------------------------------------------------------------|--------|-----------------------------|
| Dual-root dictionary ( $Y$ , cusps, $\gamma_{\text{car}}$ , $\varphi_{\text{base}}$ ) | [E]    | exact, from $\{1/2, -1/3\}$ |

|                                                                                                                                                                                 |             |                                                                                                |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|------------------------------------------------------------------------------------------------|
| $10b_1 = 5^2 + 4^2 = \Omega_{\text{adm}}\gamma + 1$                                                                                                                             | [E]         | Pythagorean carrier identity                                                                   |
| $I_1^{\text{EW}} = 41/48 = \text{Var}(Y) b_1$                                                                                                                                   | [E]         | exact                                                                                          |
| $\delta_2 = 4! \cdot 5! c_3^8, \text{Tr } X^2 = 5!$                                                                                                                             | [E]         | factorial traces                                                                               |
| $5/4 = g_{\text{car}}/(g_{\text{car}} - 1)$ universal compression                                                                                                               | [E]         | exact                                                                                          |
| $u = \frac{4}{3}c_3 + 48c_3^4$ ; one-decoder identities                                                                                                                         | [E]         | exact                                                                                          |
| $\xi = \frac{3}{4}/(1 + 9/128\pi^3)$                                                                                                                                            | [E]         | corrected tree value                                                                           |
| Ladder $A_{\text{EW}} : \mathcal{A}_H : \mathcal{A}_\Lambda = 1 : g_{\text{car}} : 2g_{\text{car}}$                                                                             | [E]/[C]     | Hubble rung is a corollary                                                                     |
| Ladder $= \binom{g_{\text{car}}}{0} : \binom{g_{\text{car}}}{1} : \binom{g_{\text{car}}}{2}$ ; $2g_{\text{car}} = \binom{g_{\text{car}}}{2} \Leftrightarrow g_{\text{car}} = 5$ | [E]         | new overdetermination of $g_{\text{car}} = 5$                                                  |
| 2D action lattice $S_x = a(\alpha_\star^{-1}/g_{\text{car}}) + bL_0$                                                                                                            | [E]/[C]     | two integer labels per scale                                                                   |
| <b>Decuple bridge</b> $\rho_\Lambda/\bar{M}_{\text{Pl}}^4 = \frac{3}{4\pi^2}e^{2\delta_{\text{ph}}}[v/(5\beta_{\text{rad}}^2\bar{M}_{\text{Pl}})]^{10}$                         | [E]         | exact, $10 = 2g_{\text{car}} = \dim \Lambda^2 E$                                               |
| Family = ladder = Pascal row $\{1, 5, 10\}$ ; $16 = 1 + 5 + 10$                                                                                                                 | [E]         | central new identification                                                                     |
| $g_{\text{car}} = 5$ via $2^{g_{\text{car}}-1} = \binom{g_{\text{car}}}{0} + \binom{g_{\text{car}}}{1} + \binom{g_{\text{car}}}{2}$                                             | [E]         | Pascal closure (unique)                                                                        |
| $\text{Tr}_{S^+} X^2 = 5!$ generates 40, 41, 240, $\delta_2$                                                                                                                    | [E]         | master moment                                                                                  |
| $ R(E_8)  = 16 \cdot 5 \cdot 3$ ; $\dim E_8 = 240 + 8$                                                                                                                          | [E]         | $E_8$ compression                                                                              |
| Even-flip atlas $240 = 160 + 80$                                                                                                                                                | [E]/[O]     | metrization open ( $\neq 112 + 128$ );<br>Coxeter Stage A closed on the character space (v531) |
| $\delta_{\text{CKM}} - \pi/3 = 3u(1 - u)$                                                                                                                                       | [E]         | CP = triple seed variance                                                                      |
| Observable quadrilateral ( $u = \Omega_b + \beta_{\text{rad}}$ )                                                                                                                | [E]         | closed UV-shadow cycle                                                                         |
| Power tower $v=x^1, H=x^5 R_H, \Lambda=x^{10} R_\Lambda$                                                                                                                        | [E]         | $K_5$ vertices/edges                                                                           |
| $496 = 16 \cdot 31 = \dim(E_8 \times E_8) = \dim SO(32)$                                                                                                                        | [E]/[C]     | heterotic-dimension count                                                                      |
| $A_{\text{EW}} = \alpha_\star^{-1}/\sqrt{\Delta_Y}, \Delta_Y = (b+s)^2 = 25$                                                                                                    | [E]         | discriminant action theorem                                                                    |
| $\hat{m}_e \hat{m}_\mu \hat{m}_\tau \propto \frac{32}{9}u^{10}$ ; $(5, 3, 2) = (g_{\text{car}}, b, s)$                                                                          | [E]         | lepton flavor decuple                                                                          |
| $M_{\text{scal}}/\bar{M}_{\text{Pl}} = c_3^{7/2}$                                                                                                                               | [E]         | scalaron seam power                                                                            |
| $\delta_{\text{CKM}} - \pi/3 = N_{\text{fam}} u(1 - u)$                                                                                                                         | [E]         | weak CP = $N_{\text{fam}} \times$ seed variance                                                |
| $L = R + W$ : $\sum R = 22, \sum W = 18$                                                                                                                                        | [E]         | first generation = residue store                                                               |
| $ R(E_8) /6 = \sum L = 40$ ( $\mathbb{Z}_6$ slice)                                                                                                                              | [E]         | one $\mathbb{Z}_6$ slice = flavor budget                                                       |
| $h_{E_8} =  R /\text{rank} = 30 = N_{\text{fam}} \binom{5}{2} = N_{\text{fam}} A_\Lambda$                                                                                       | [E]/[C]     | carrier Coxeter theorem                                                                        |
| $ R^+(E_8)  = \text{Tr}_{S^+} X^2 = 120 = 5!$                                                                                                                                   | [E]         | positive-root moment                                                                           |
| common decuple $\binom{5}{2} = A_\Lambda = u^{5+3+2} = h_{E_8}/N_{\text{fam}} = 10$                                                                                             | [E]/[C]     | one carrier pair-decuple                                                                       |
| $\lambda_C^6/\delta_{\text{top}} = 1.061$ (winding $\approx$ seam defect)                                                                                                       | [O]         | flagged proximity (not exact)                                                                  |
| $y_f \simeq \lambda_C^{L_f}$ , top $L_t = 0 \Rightarrow y_t \simeq 1$                                                                                                           | [E]/[C]     | Froggatt–Nielsen ladder                                                                        |
| $w = -1$ (true $\Lambda$ ); $N_{\text{eff}} \simeq 3$                                                                                                                           | [E]         | dark energy / neutrino species                                                                 |
| 3-generator lattice $\{\alpha_\star^{-1}, L_0, L_c\}$ ; $M_{\text{scal}} = e^{-\frac{7}{2}L_c}$                                                                                 | [E]         | seam axis $L_c = \ln(1/c_3)$                                                                   |
| $\dim S^+ = 2 \text{rank}(E_8), h_{E_8} = 2(\dim S^+ - 1) = 2^g - 2$                                                                                                            | [E]         | rank–Coxeter coupling ( $\nu^c$ closes it)                                                     |
| $\gcd(2h, 6r) = \dim \mathfrak{g}_{\text{SM}} = 12, \text{lcm} = 240$                                                                                                           | [E]/[C]     | SM algebra = gcd; $E_8$ count = lcm                                                            |
| $I_1^{\text{EW}} = h_{E_8}/36 + 1/(6r) = 41/48$                                                                                                                                 | [E]/[C]     | Coxeter part + rank singlet                                                                    |
| $M_{\text{scal}}^2/\bar{M}_{\text{Pl}}^2 = c_3^{r-1} = c_3^7$ exact; $A_s = N_\star^2 c_3^7/24\pi^2$                                                                            | [E]/[C]/[O] | seam power exact; $A_s$ conditional (Starobinsky)                                              |
| $h_{E_8} = 2 \cdot 3 \cdot 5 = 30, r = \varphi(30) = 8$                                                                                                                         | [E]/[C]     | 2, 3, 5 Coxeter cyclotomic compiler                                                            |
| $\text{Exp}(E_8) = (\mathbb{Z}/30)^\times, \sum = 120, \chi_C = \Phi_{30}$                                                                                                      | [E]/[C]     | exponents = primitive residues                                                                 |
| lepton $(5, 3, 2)$ : $\sum = 10 = A_\Lambda, \prod = 30 = h_{E_8}$                                                                                                              | [E]/[C]     | sum=action, product=Coxeter                                                                    |
| Hubble $H_0/\bar{M}_{\text{Pl}} = e^{-\alpha_\star^{-1}}/(2\pi\sqrt{\Omega_\Lambda})$                                                                                           | [E]         | exact corollary                                                                                |
| $\Lambda$ -anchored closure $G_N = \frac{3c_3^3}{32\pi^3 \hbar \Lambda_{\text{geom}}} e^{-2\alpha_\star^{-1}}$                                                                  | [E]/[C]     | $G_N$ to $-0.35\%$                                                                             |
| $g_{\text{car}} = 5$ overdetermined                                                                                                                                             | [C]         | monotone $\alpha^{-1}(g_{\text{car}})$                                                         |
| $ R(E_8)  = 240 = 16 \cdot 15, 248 = 8 \cdot 31$                                                                                                                                | [E]         | counting identities                                                                            |
| $E_8 \leftrightarrow$ carrier-transition map                                                                                                                                    | [C]         | needs structure map                                                                            |
| $\delta_{\text{CP}} = 4\pi/3, \theta_{12} = \frac{1}{3} - \frac{u}{2}$                                                                                                          | [C]         | PMNS readouts                                                                                  |
| $ V_{td} ,  V_{ts} , J_{\text{CKM}}, J_{\text{CP}}^\nu; n_s, r$                                                                                                                 | [C]         | derived readouts                                                                               |

**Open audit points:** (1) *discharged* — the EM closure  $F_{U(1)}(\alpha) = 0$  is now stated explicitly (Grammar III), with existence/uniqueness and root  $\alpha_\star^{-1} = 137.0359992168$  ( $2.9 \times 10^{-10}$ ,  $1.9\sigma$  vs CODATA-2022); (2) keep  $\gamma_{\text{car}} = 5/6$  vs  $\gamma_E$  distinct; (3) type  $M_{\text{Pl}}$  vs  $\bar{M}_{\text{Pl}}$  and “pure” ( $\mathcal{A}_\Lambda = 274.07$ ) vs “total” ( $-\ln(\delta_{\text{top}} e^{-2\alpha_\star^{-1}}) = 283.10$ ,  $-\ln(\rho_\Lambda/\bar{M}_{\text{Pl}}^4) = 276.65$ ) actions consistently; (4) version the  $\Lambda$  prefactor branch; (5) the scalaron-mass  $\rightarrow$  Starobinsky- $M$  identification is the *only* condition for  $A_s = N_\star^2 c_3^7/(24\pi^2)$  (the seam power  $M_{\text{scal}}^2/\bar{M}_{\text{Pl}}^2 = c_3^7$  itself is exact); (6) the  $E_8$  *lattice* is now built explicitly as  $(D_5 \oplus A_3) + \mu_4$ -glue (Section 3), so the former “blind 240-metrisation” is closed. The flavor status (companion parabolic-weight note) now splits cleanly: the *anchor multiset*  $\{1, 1, 2\}$  is derived (Schur–Horn + stability) [E]; the *sector assignment*  $(1, 1, 2)$  follows from the  $D_5$  cusp ordering [E]/[C]; the *first-generation* +6 *winding* is the hierarchy convention [C]; the six *non-anchor residues* are then forced (residue sets =  $D_6$  orbit of  $\{0, 1, 3\}$ , minus anchor, monotone) [E]. The remaining geometric input at the time of writing was (H2), the specific  $D_6$  cusp labelling on  $\mathbb{P}^1 \setminus \mu_4$  — sharply characterised by the residue-matrix spectral selector ( $\det R = h(D_5)$ , principal minors  $(2, 3, 5)$ ), with a naive cusp-position shortcut explicitly ruled out [C]. The explicit Riemann–Hilbert construction confirms the rigid  $SU(3)$  system and its  $A_3$  Coxeter rotation. *Status update:* the H2 dictionary has since been made *exact* (hexagon = sign-twisted family spectrum, addresses pinned, margins forced, v118–v122), the deck is *derived* (v141), and the residue is the  $R4'$  selector establishment — two line pairings (v139/v142); what stays geometric is only the parabolic realisation premise (GATE.QGEO, research contracts).

**Honest negatives (reported to keep the discipline visible).** The TFPT lepton ladder gives a Koide ratio  $Q = (\sum m_\ell)/(\sum \sqrt{m_\ell})^2 = 0.6645$ , close to but *not* the empirical  $2/3 = 0.66667$  ( $-0.33\%$ ); TFPT is Koide-*like* but does not reproduce the precise relation. The baryon asymmetry  $\eta_B \approx 6 \times 10^{-10}$  and the proton/electron ratio  $m_p/m_e = 1836$  do *not* fall on a clean action charge or  $L_0$  ladder; they remain outside the present grammar. These are stated so the strong items are not read as a uniform success.

## 17 Consolidated picture

**The three nested grammars.**

**Carrier:**  $3 + 2 \Rightarrow \{y_-, y_+\} = \{-\frac{1}{3}, \frac{1}{2}\} \Rightarrow Y$ ,  $\dim S^+ = 16$ ,  $N_{\text{fam}} = 3$ ,  $\Omega_{\text{adm}} = 48$ ,  $b_1 = \frac{41}{10}$ ,

**Seed:**  $u = \frac{4}{3}c_3 + 48c_3^4 \Rightarrow \{\lambda_C, \beta_{\text{rad}}, \Omega_b, \sin^2 \theta_{13}\}$ ,

**Action:**  $\alpha_\star^{-1} \Rightarrow A_{\text{EW}} = \frac{\alpha_\star^{-1}}{g_{\text{car}}}$ ,  $\mathcal{A}_\Lambda = 2\alpha_\star^{-1}$ ,  $\mathcal{A}_H = \alpha_\star^{-1} \Rightarrow \frac{\rho_\Lambda}{\bar{M}_{\text{Pl}}^4} = \frac{3}{4\pi^2} e^{2\delta_{\text{ph}}} \left[ \frac{v_{\text{geo}}}{5\beta_{\text{rad}}^2 \bar{M}_{\text{Pl}}} \right]^{10}$ .

A small grammar generates many sectors: the Standard-Model packet, flavor, the electroweak hierarchy, the cosmological constant, the metrology, and the inflation/CMB interface. The strongest new statements are exact: the decouple bridge ties  $\Lambda$  to the tenth power of the stripped electroweak scale with  $10 = 2g_{\text{car}}$ ; the Hubble scale is the square root of  $\Lambda$ ; and the Einstein normaliser is the corrected tree value  $\frac{3}{4}/(1 + 9/128\pi^3)$ . None of this is claimed as a proof of a final theory; it is offered as a tightening of the existing architecture for your assessment.

**The compiler in one line:** TFPT = even carrier code + seed decoder + 2·3·5 Coxeter compiler. From the three atoms  $(2, 3, g_{\text{car}}=5)$ :

$$h_{E_8} = 2 \cdot 3 \cdot 5 = 30, \quad r_{E_8} = \varphi(30) = 8 = g_{\text{car}} + N_{\text{fam}},$$

$$|R(E_8)| = rh = 240, \quad \dim E_8 = r(h+1) = 8 \cdot 31 = 248,$$

$$\text{Exp}(E_8) = (\mathbb{Z}/30)^\times, \quad \sum \text{Exp} = 120 = \text{Tr}_{S^+} X^2 = |R^+|, \quad \chi_C = \Phi_{30},$$

$$\dim S^+ = 1 + 10 + 5 = 16 = 2r, \quad A_{\text{EW}} : \mathcal{A}_H : \mathcal{A}_\Lambda = 1:5:10, \quad \mathcal{A}_H = \frac{h}{6}, \quad \mathcal{A}_\Lambda = \frac{h}{3}, \quad N_{\text{fam}} = \frac{h}{A_\Lambda} = 3,$$

$$\sum L = \frac{|R^+|}{3} = 40, \quad 10b_1 = 1 + \frac{|R^+|}{3} = 41 = 1 + h \frac{\dim S^+}{\dim \mathfrak{g}_{\text{SM}}},$$

$$\det M_\ell \propto u^{10}, \quad \frac{\rho_\Lambda}{M_{\text{Pl}}^4} \propto \left[ \frac{v_{\text{geo}}}{5\beta_{\text{rad}}^2 M_{\text{Pl}}} \right]^{10}, \quad A_{\text{EW}} = \frac{\alpha_\star^{-1}}{\sqrt{\Delta_Y}}, \quad \sqrt{\Delta_Y} = g_{\text{car}} = 5.$$

**The common decuple (= 10):**  $\binom{g_{\text{car}}}{2}$  (carrier pairs) =  $A_\Lambda$  (cosmological action) =  $u^{5+3+2}$  (lepton determinant) =  $h_{E_8}/N_{\text{fam}}$  (Coxeter per family).  $\Lambda$ , the charged-lepton determinant, the  $E_8$  Coxeter structure and the action ladder all read the *same* carrier pair-decuple. The open core is one question: can the even-flip atlas be metrised as a genuine  $E_8$  root system (the 160+80 vs 112+128 obstruction)? Its Coxeter half (Stage A) is closed on the primitive-character space by the tensor clock  $T_{30} = \text{Comp}(\Phi_5) \otimes \text{Comp}(\Phi_6)$  (**v531**); the atlas intertwiner and the metrization (Stage B) remain.

## Part II

The Coxeter–cyclotomic compiler and the explicit  $E_8$  construction

## Abstract

This short companion compresses the TFPT bridge layer to a single discrete object. The claim is that the Standard-Model family, the cosmological action ladder, the  $E_8$  counting data, the flavor budget, inflation and the cosmological constant are *not* independent coincidences but projections of one structure: the even-weight code  $C^+$  on  $g_{\text{car}} = 5$  slots, together with its  $\mathbb{Z}_{30}$  Coxeter spectrum and the single seam seed  $c_3 = 1/(8\pi)$ . The decisive new result is that  $C^+$  is precisely the half-spin weight orbit of  $D_5 = \mathfrak{so}_{10}$  ( $\text{Aut}(C^+) = W(D_5)$ , order 1920), that the family geometry  $\mathbb{P}^1 \setminus \mu_4$  is an  $A_3 = \mathfrak{su}_4$ , and that  $E_8$  is then the standard *closure algebra* of  $D_5 \times A_3$  — not an input. The fusion is exact:  $\text{disc}(D_5) = \text{disc}(A_3) = \mathbb{Z}_4$ , so the four family corners  $\mu_4$  are precisely the glue code that closes  $D_5 \oplus A_3$  into the unimodular  $E_8$  lattice. The glue is metric: the discriminant-form norms  $q(D_5) = \frac{5}{4}$  and  $q(A_3) = \frac{3}{4}$  sum to the  $E_8$  root norm 2 — and these are exactly the pre-existing TFPT constants  $\delta_2/\delta_{\text{top}}^2 = \frac{5}{4}$  and  $\xi_{\text{tree}} = \frac{3}{4}$ . The  $A_3$  side is explicit: the residue pairing on  $H^1(\mathbb{P}^1 \setminus \mu_4)$  is the  $A_3$  Cartan form, and the carrier corner rotation  $z \mapsto iz$  acts as the  $A_3$  Coxeter element (order 4, spectrum  $\{i, -1, -i\}$ ). The lift is no longer a count: the 240  $E_8$  roots are *explicitly* built from  $D_5 \oplus A_3$  plus the spinor ( $\mu_4$ ) glue, all of norm 2, Cartan determinant 1. Seven statements carry it; every lattice identity below is checked exactly, and the analytic metric is computed and characterised (a  $D_4$ -equivariant deformation), so no *lattice* structural step is left open in the  $D_5 \times A_3$  glue route (the upstream boundary kernel, EM closure and flavor Hecke saturation remain separate open problems). Master statement: *five carrier slots give  $D_5$ , four family corners give  $A_3$ , and  $\mu_4$  glues  $D_5 \times A_3$  into the  $E_8$  atlas.*

## The compression.

$$\begin{aligned} \text{TFPT} &= \underbrace{D_5 \text{ half-spin carrier}}_{C^+} + \underbrace{A_3 \text{ four-point family}}_{\mathbb{P}^1 \setminus \mu_4} + c_3 \text{ seam seed}, \\ E_8 &= \text{closure of } D_5 \times A_3 \end{aligned}$$

with  $c_3 = \frac{1}{8\pi}$ ,  $g_{\text{car}} = 5$ ,  $h := 2 \cdot 3 \cdot g_{\text{car}} = 30$ ,  $r := \varphi(h) = \varphi(30) = 8$ ,  $N_{\text{fam}} = 3$ ,  $N_{\Phi} = 1$ ,  $\dim \mathfrak{g}_{\text{SM}} = 12$ . Equivalently  $C^+ + \mathbb{Z}_{30}$  Coxeter compiler +  $c_3$  seed; the  $D_5 \times A_3$  reading is the structural lift of the same object.

## Theorem 1 — the even carrier code

Let  $E$  be the rank- $g_{\text{car}} = 5$  carrier, split  $E = E_3 \oplus E_2$  (color  $b = 3$ , weak  $s = 2$ ). The one-generation spinor packet is the even exterior algebra, equivalently the even-weight binary code

$$S^+ = \Lambda^0 E \oplus \Lambda^2 E \oplus \Lambda^4 E \cong C^+ = \{x \in \mathbb{F}_2^5 : |x| \text{ even}\}, \quad \dim S^+ = |C^+| = 2^{g_{\text{car}}-1} = 16. \quad (69)$$

The two carrier roots  $y_+ = \frac{1}{2}$ ,  $y_- = -\frac{1}{3}$  generate all hypercharges ( $Y(Q_L) = y_+ + y_- = \frac{1}{6}$ ,  $Y(u^c) = 2y_-$ ,  $Y(e^c) = 2y_+$ ,  $Y(\nu^c) = 0$ ) and the transport cusps  $\{1, \frac{2}{3}, \frac{1}{3}\} = \{2y_+, -2y_-, 2(y_+ + y_-)\}$ , with  $\gamma_{\text{car}} = \text{Tr}_E Y^2 = 3y_-^2 + 2y_+^2 = \frac{5}{6}$ . [E]

**Gauge-rank factorisation** ([[verification/v79\\_review\\_identities.py](#)]). The carrier polynomial factors into the two gauge ranks directly:

$$\boxed{6Y^2 - Y - 1 = (2Y - 1)(3Y + 1)}, \quad Y = \frac{1}{2} \ (s=2, \ SU(2)), \quad Y = -\frac{1}{3} \ (b=3, \ SU(3)),$$

i.e.  $(sY - 1)(bY + 1)$  — the polynomial splits into one linear representative of each Standard-Model non-abelian factor; the 5-slot carrier *forces* the split into  $SU(3) \times SU(2)$ .



| Hypercharge<br>[verification/v79_review_identities.py]                                                                                                                                                                                                                                                                                                                                                                                                        | Lucas–Binet<br>resonances | [E] | (mode<br>reading | [C], |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|-----|------------------|------|
| The integer carrier charges $(q_+, q_-) = (3, -2)$ (the primitive trace-free pair, $X^2 - X - 6 = (X-3)(X+2)$ , Lean <code>unique_carrier_pair</code> ) generate the Lucas $U$ -sequence                                                                                                                                                                                                                                                                      |                           |     |                  |      |
| $D_n = \frac{3^n - (-2)^n}{5} : \quad D_1, \dots, D_6 = 1, 1, 7, 13, 55, 133,$                                                                                                                                                                                                                                                                                                                                                                                |                           |     |                  |      |
| which is, <i>in order</i> , $(N_\Phi, \cdot, \text{scalaron exponent } 7, \Delta_Q=13, \text{quark numerator } 55, \dim E_7=133)$ . In particular the quark ratio reads $c_u/c_d = D_5/(N_{\text{fam}}^2 D_4) = \frac{55}{117}$ — a single recurrence of the hypercharge generator, not a free integer. [E] for the arithmetic; the per-mode physical identification is a [C] cross-domain reading (only $n \leq 6$ , the carrier-slot range, is meaningful). |                           |     |                  |      |

## Theorem 2 — Exterior–Action Duality

On  $C^+$  the flip  $F = x \oplus y$  is even, so  $|F| \in \{0, 2, 4\}$ . The three relations  $R_0, R_2, R_4$  form a symmetric association scheme (the even Hamming scheme on five bits) with

$$\begin{array}{l} \text{valencies } (1, 10, 5) = (\dim \Lambda^0 E, \dim \Lambda^2 E, \dim \Lambda^4 E), \\ \text{spectral multiplicities } (1, 5, 10) = A_{\text{EW}} : \mathcal{A}_H : \mathcal{A}_\Lambda \end{array} \quad [\text{E}]. \quad (70)$$

The eigenvalues (computed directly) are  $A_2 : \{10^{(1)}, 2^{(5)}, (-2)^{(10)}\}$  and  $A_4 : \{5^{(1)}, (-3)^{(5)}, 1^{(10)}\}$ , with  $A_2 + A_4 = K_{16} - \mathbf{1}$ . The two graphs are named objects:

$$\begin{array}{l} R_2(C^+) = \text{folded/complement, SRG}(16, 10, 6, 6) \text{ (the Clebsch complement);} \\ R_4(C^+) = \text{Clebsch graph, SRG}(16, 5, 0, 2) \end{array} \quad [\text{E}]. \quad (71)$$

**Remark.** The Standard-Model family is the *valency* side  $(1, 10, 5)$  of one scheme; the cosmological action ladder is its *spectral* side  $(1, 5, 10)$ . The action ladder is therefore not merely “also binomial” — it is the spectral dual of the even exterior family. [C]

**This duality is the MacWilliams transform** ([verification/v79\_review\_identities.py]).  $C^+$  is the even-weight code  $[5, 4, 2]$ ; its dual  $(C^+)^\perp$  is the repetition code  $[5, 1, 5] = \{00000, 11111\}$  (the uniform/vacuum word). The MacWilliams identity sends the weight enumerator of the dual to that of  $C^+$ , namely  $(1, 10, 5)$  — so the valency  $\leftrightarrow$  spectral duality *is* the coding-theoretic UV/IR duality, and the carrier grading  $\Lambda^{\text{even}}(5) = 1+10+5$  is the MacWilliams image of the repetition vacuum. [E] (code identity) / [C] (holographic reading).

**The full Bose–Mesner algebra** (machine-checkable) fixes the duality beyond eigenvalues:

$$A_2^2 = 10 \mathbf{1} + 6A_2 + 6A_4, \quad A_2 A_4 = 3A_2 + 4A_4, \quad A_4^2 = 5 \mathbf{1} + 2A_2, \quad (72)$$

with eigenmatrix and multiplicity orthogonality

$$P = \begin{pmatrix} 1 & 10 & 5 \\ 1 & 2 & -3 \\ 1 & -2 & 1 \end{pmatrix}, \quad \det P = -64, \quad \boxed{P^\top \text{diag}(1, 5, 10) P = \text{diag}(16, 160, 80)} \quad [\text{E}], \quad (73)$$

so the flip split  $240 = 160+80$  ( $|F| = 2, 4$ ) appears directly in the scheme’s eigenmatrix orthogonality.

**Carrier** =  $D_5$  (**verified**). The 16 vertices of  $C^+$  are exactly the weights of the  $D_5 = \mathfrak{so}_{10}$  half-spinor  $\frac{1}{2}(\pm 1, \dots, \pm 1)$  with an even number of minus signs. The flip-2 and flip-4 graphs are translation- and permutation-invariant, so the *affine* group  $C^+ \rtimes S_5$  (translations  $\rtimes$  coordinate permutations) acts by graph/scheme automorphisms; the group closure gives the automorphism group of the even Hamming scheme (equivalently the Clebsch graph),

$$\boxed{\text{Aut}_{\text{aff}}(C^+) = C^+ \rtimes S_5, \quad |\text{Aut}_{\text{aff}}(C^+)| = 2^{g_{\text{car}}-1} g_{\text{car}}! = 16 \cdot 120 = 1920 = |W(D_5)|} \quad [\text{E}], \quad (74)$$

and  $1920 = r |R(E_8)| = 8 \cdot 240$ . (The *linear* code automorphism group alone is only the coordinate group  $S_5$  of order 120; the factor  $2^{g_{\text{car}}-1} = 16$  is the translation/scheme part. It is the affine/scheme group that equals  $W(D_5)$ .) Edges of the Clebsch graph:  $|E(R_4)| = \frac{16 \cdot 5}{2} = 40 = |R(D_5)| = \sum_{f,j} L_{f,j}$  (the entire flavor transport budget). So the five-slot even code is the half-spin weight structure of  $\mathfrak{so}_{10}$ , the classical home of one SM family with  $\nu^c$ .

### Theorem 3 — the Coxeter closure equation selects $g_{\text{car}} = 5$

Requiring the even family to equal the action truncation through pair degree,  $\dim S^+ = 2^{g-1} = \binom{g}{0} + \binom{g}{1} + \binom{g}{2}$ , i.e.

$$\boxed{2^g - 2 = g(g+1) \iff g = 5} \quad [\mathbf{E}], \quad (75)$$

the unique positive-integer solution. At  $g = 5$  the two sides equal 30, so *carrier exhaustion*  $2^g - 2$  and the *Coxeter period*  $g(g+1)$  coincide only at five slots:  $2^{g_{\text{car}}} - 2 = 30 = h$  and  $g_{\text{car}}(g_{\text{car}} + 1) = 30$ .

### Theorem 4 — the $\mathbb{Z}_{30}$ cyclotomic Coxeter compiler

From the three atoms  $(2, 3, g_{\text{car}})$ , with  $h = 2 \cdot 3 \cdot g_{\text{car}} = 30$  and  $r = \varphi(h) = 8$ :

$$\boxed{|R(E_8)| = rh = 240, \quad \dim E_8 = r(h+1) = 8 \cdot 31 = 248, \quad h+1 = 2^{g_{\text{car}}} - 1 = 31} \quad [\mathbf{E}]. \quad (76)$$

The  $E_8$  exponents are the primitive residues modulo 30,

$$\text{Exp}(E_8) = (\mathbb{Z}/30\mathbb{Z})^\times = \{1, 7, 11, 13, 17, 19, 23, 29\}, \quad \chi_C(t) = \Phi_{30}(t) = t^8 + t^7 - t^5 - t^4 - t^3 + t + 1, \quad (77)$$

and  $|(\mathbb{Z}/30)^\times| = \varphi(30) = 8 = r$ . The local SM algebra and the root atlas are the two arithmetic projections of  $(r, h) = (8, 30)$ :

$$\Omega_{\text{adm}} = 6r = 48, \quad D_{\text{start}} = 2h = 60, \quad \gcd(6r, 2h) = \dim \mathfrak{g}_{\text{SM}} = 12, \quad \text{lcm}(6r, 2h) = rh = 240. \quad (78)$$

### Theorem 5 — the root transition atlas and the positive-edge moment

The directed even-flips of  $C^+$  are the complete graph  $K_{16}$ :

$$\boxed{\begin{aligned} |R^+(E_8)| &= \binom{\dim S^+}{2} = \binom{16}{2} = 120 = \text{Tr}_{S^+} X^2 = 5! = \sum_{m \in (\mathbb{Z}/30)^\times} m, \\ |R(E_8)| &= 2 \binom{16}{2} = 240 \end{aligned}} \quad [\mathbf{E}]. \quad (79)$$

( $X = 6Y$ .) So the positive  $E_8$  roots are the unordered transitions of one family; the second hypercharge moment, the positive-root count and the sum of the  $E_8$  exponents coincide at 120. The CP pairing  $m \leftrightarrow 30 - m$  of the exponents gives four pairs, each of sum  $h = 30$ :

$$\boxed{\text{Tr}_{S^+} X^2 = (N_{\text{fam}} + N_{\Phi}) h = 4 \cdot 30 = 120 = \dim \mathfrak{so}_{16} = \underbrace{45}_{D_5} + \underbrace{15}_{A_3} + \underbrace{60}_{(10,6)}} \quad [\mathbf{E}], \quad (80)$$

so the hypercharge moment is simultaneously four CP-paired Coxeter periods *and* the adjoint half  $\mathfrak{so}_{16} = (45, 1) \oplus (1, 15) \oplus (10, 6)$  of the  $E_8$  closure.

**How the root *origin* is now obtained — see Theorem 7.** A naive flip-size-preserving map  $R_{\text{flip}} \rightarrow \mathbb{R}^8$  cannot be  $E_8$ : the flip split  $240 = 160 + 80$  ( $|F| = 2, 4$ ) differs from the  $D_8$

split  $112 + 128$ . The correct route is *not* a direct 240-metrization but the standard branching  $E_8 \supset D_5 \times A_3$  (Theorem 7), which builds the root spaces from known  $D_5$  and  $A_3$  data. Until the  $A_3$  lattice is derived from the four punctures (Theorem 7, [O]),  $E_8$  remains a Coxeter atlas with a concrete, falsifiable lift.

## Theorem 6 — downstream readouts from $(h, r, c_3)$

All discrete budgets and the seam seed are functions of  $(h, r, c_3) = (30, 8, 1/(8\pi))$ :

$$\dim S^+ = 2r = 16, \quad h = 2(\dim S^+ - 1) = 30, \quad (81)$$

$$10b_1 = 5r + 1 = 41 = 1 + h \frac{\dim S^+}{\dim \mathfrak{g}_{\text{SM}}}, \quad I_1^{\text{EW}} = \frac{5r + 1}{6r} = \frac{41}{48}, \quad (82)$$

$$u = \frac{r}{6}c_3 + 6r c_3^4 = \frac{4}{3}c_3 + 48c_3^4, \quad \delta_2 = 4! \binom{16}{2} c_3^8 = 2880 c_3^8, \quad (83)$$

$$\xi_{\text{tree}} = \frac{6}{r} = \frac{\dim \mathfrak{g}_{\text{SM}}}{\dim S^+} = \frac{3}{4}, \quad \frac{M_{\text{scal}}^2}{M_{\text{Pl}}^2} = c_3^{\Omega_{\text{adm}} - 10b_1} = c_3^{r-1} = c_3^7. \quad (84)$$

**Two-graph product table.** With the carrier-pair graph  $K_5$  ( $|E(K_5)| = \binom{5}{2} = 10 = \mathcal{A}_\Lambda$ ) and the family-corner graph  $K_4$  on  $\mu_4$  ( $|E(K_4)| = \binom{4}{2} = 6$ ,  $|R(A_3)| = 2|E(K_4)| = 12$ ), every large budget is a product of the two graphs:

$$\boxed{40 = 10 \cdot 4, \quad 60 = 10 \cdot 6, \quad 120 = 10 \cdot 12, \quad 240 = 10 \cdot 24} \quad [\text{E}], \quad (85)$$

i.e.  $(\sum L, D_{\text{start}}, |R^+(E_8)|, |R(E_8)|) = 10 \cdot (|\mu_4|, |E(K_4)|, |R(A_3)|, |S_4|)$ . Two occupancy identities follow,

$$\boxed{10b_1 = 1 + |\mu_4| |E(K_5)| = 1 + 4 \cdot 10 = 41}, \quad (86)$$

$$\boxed{\Omega_{\text{adm}} = N_{\text{fam}} \dim S^+ = (N_{\text{fam}} + N_\Phi) |R(A_3)| = 3 \cdot 16 = 4 \cdot 12 = 48} \quad [\text{E}],$$

so the Higgs singleton is the fourth ( $\mu_4$ ) direction that closes  $N_{\text{fam}} = 3$  to **4** of  $A_3$ . **Occupancy deficit ladder.** The admissible occupancy splits cleanly,

$$\boxed{\begin{aligned} \Omega_{\text{adm}} &= \underbrace{40}_{\sum L (\text{flavor})} + \underbrace{1}_{\text{Higgs}} + \underbrace{7}_{\text{scalaron}} = 48, & \Omega_{\text{adm}} - \sum L &= 8 = h(D_5) = r(E_8), \\ \Omega_{\text{adm}} - 10b_1 &= 7 = h(D_5) - 1 \end{aligned}} \quad [\text{E}], \quad (87)$$

so the scalaron exponent  $r - 1 = 7$  is exactly the post-flavor, post-Higgs occupancy remainder,  $M_{\text{scal}}^2/\bar{M}_{\text{Pl}}^2 = c_3^{\Omega_{\text{adm}} - 10b_1} = c_3^7$ . With  $h(D_5) = 8$  and  $|R^+(A_3)| = 6$  the seed itself is a  $D_5/A_3$  ratio-product,  $u = \frac{h(D_5)}{|R^+(A_3)|} c_3 + h(D_5) |R^+(A_3)| c_3^4$ , and  $\xi = \frac{|R^+(A_3)|}{h(D_5)} (1 + |R^+(A_3)|^2 c_3^3)^{-1} = \frac{3}{4} (1 + 36c_3^3)^{-1}$ . The seed readouts and the two decuples are

$$\lambda_C^2 = u(1-u), \quad \beta_{\text{rad}} = 2c_3 u, \quad \sin^2 \theta_{13} = e^{-5/6} u, \quad \boxed{\omega_b = \frac{\lambda_C}{\mathcal{A}_\Lambda} = \frac{\lambda_C}{10} = 0.02244} \quad (\text{Planck } 0.02237), \quad (88)$$

$$\boxed{\begin{aligned} \det M_\ell &\propto u^{h/N_{\text{fam}}} = u^{10}, \\ \frac{\rho_\Lambda}{\bar{M}_{\text{Pl}}^4} &= \frac{3}{4\pi^2} e^{2\delta_{\text{ph}}} \left[ \frac{v_{\text{geo}}}{5\beta_{\text{rad}}^2 M_{\text{Pl}}} \right]^{h/N_{\text{fam}}}, \end{aligned}} \quad (89)$$

with lepton powers  $(5, 3, 2) = (g_{\text{car}}, b, s)$ :  $\sum = 10 = \mathcal{A}_\Lambda$  (the  $\Lambda$  decuple),  $\prod = 30 = h$  (the Coxeter period). The first-generation transport sums read the non-primitive  $\mathbb{Z}_{30}$  load,

$$(G_1, G_2, G_3) = (22, 13, 5) = (h - r, \dim \mathfrak{g}_{\text{SM}} + 1, g_{\text{car}}), \quad h - r = 30 - \varphi(30) = 22, \quad (90)$$

**Flavor transport is a  $D_5 \times A_3$  object.** The transport packets  $L_u=(7, 3, 0)$ ,  $L_d=(7, 5, 2)$ ,  $L_e=(8, 5, 3)$  split into a  $D_5$  budget and an  $A_3$  geometry:

$$\boxed{\sum_{f,j} L_{f,j} = 40 = |R(D_5)| \text{ (Clebsch edges)}, \quad L_{f,j} \bmod 6 \text{ lives on } C_6 = Z_6 = |R^+(A_3)|} \quad [\mathbf{E}], \quad (91)$$

the residues being the  $D_6$ -orbit of the perfect set  $\{0, 1, 3\} \subset C_6$  (cyclic distances  $\{1, 2, 3\}$ ):  $u \equiv \{0, 1, 3\}$ ,  $d \equiv 2 - \{0, 1, 3\}$ ,  $e \equiv 2 + \{0, 1, 3\}$ , with the universal first-generation  $+6$  winding. So the carrier  $D_5$  sets the total budget and the family  $A_3$  (its 6-element positive-root hexagon) sets the per-entry distances. The scalaron mass is a pure seam power  $[\mathbf{E}]$ ; the amplitude is a *conditional* readout, kept  $[\mathbf{C}]/[\mathbf{O}]$  until the scalaron variable is formally identified with the Starobinsky  $M$  parameter. That identification is now *heat-kernel grounded* (G2): the spectral-action Seeley–DeWitt coefficient  $a_4|_{R^2} \sim R^2/72$  delivers exactly the Starobinsky  $R^2$  term, with  $M^2/\bar{M}_{\text{Pl}}^2 = 6(4\pi)^2/f_0$  and the TFPT closure  $c_3^7$  fixing the cutoff moment  $f_0$  (`[verification/v36_spectral_action_g2.py]`):

$$\underbrace{\frac{M_{\text{scal}}^2}{\bar{M}_{\text{Pl}}^2} = c_3^{r-1} = c_3^7}_{[\mathbf{E}]}, \quad \underbrace{A_s = \frac{N_\star^2}{24\pi^2} \frac{M_{\text{scal}}^2}{\bar{M}_{\text{Pl}}^2} = \frac{N_\star^2 c_3^7}{24\pi^2} = 2.17 \times 10^{-9}}_{[\mathbf{C}]/[\mathbf{O}] \text{ (if } M_{\text{scal}} = M_{\text{Star}})}. \quad (92)$$

## Theorem 7 — the $D_5 \times A_3$ lift of the $E_8$ atlas

Since  $A_3 = \mathfrak{su}_4 \cong \mathfrak{so}_6 = D_3$ , one has the standard chain  $E_8 \supset \mathfrak{so}_{16} \supset \mathfrak{so}_{10} \oplus \mathfrak{so}_6 = D_5 \times A_3$ , with the exact branching (verified dimensions and root/weight counts):

$$\boxed{\mathfrak{e}_8 = (45, 1) \oplus (1, 15) \oplus (10, 6) \oplus (16, 4) \oplus (\overline{16}, \bar{4})}, \quad (93)$$

$$\underbrace{45+15+60+64+64}_{=248} = \dim E_8, \quad \underbrace{40+12+60+64+64}_{=240} = |R(E_8)|.$$

The adjoint **120** of  $\mathfrak{so}_{16}$  is  $(45, 1) \oplus (1, 15) \oplus (10, 6)$  and its spinor **128** is  $(16, 4) \oplus (\overline{16}, \bar{4})$ . The TFPT dictionary makes each summand a known object:

| piece                                     | value         | TFPT reading                                                                                           |
|-------------------------------------------|---------------|--------------------------------------------------------------------------------------------------------|
| (45, 1)                                   | $45 = 40 + 5$ | $D_5$ carrier adjoint: Clebsch edges $40 =  R(D_5) $ plus carrier rank $g_{\text{car}} = 5$            |
| (1, 15)                                   | $15 = 12 + 3$ | $A_3$ family+Higgs adjoint: $\dim \mathfrak{g}_{\text{SM}} = 12$ plus family rank $N_{\text{fam}} = 3$ |
| (10, 6)                                   | $10 \cdot 6$  | $\Lambda$ decuple $\mathcal{A}_\Lambda = \binom{5}{2} = 10$ times $Z_6$ transport $\binom{4}{2} = 6$   |
| $(16, 4) \oplus (\overline{16}, \bar{4})$ | $2 \cdot 64$  | matter half-spinor $S^+$ times the four corners $\mu_4$ , plus conjugate sheet                         |

**Proof route (replaces the old blind 240-metrization).** *Stage 1* ( $D_5$ ,  $[\mathbf{E}]$ ):  $C^+$  is the  $D_5$  half-spin orbit,  $R_4 = \text{Clebsch}$ ,  $\text{Aut} = W(D_5)$ ,  $|E(R_4)| = 40 = |R(D_5)|$  — Theorem 2. *Stage 2* ( $A_3$ ,  $[\mathbf{E}]$ ): the carrier geometry fixes  $X_f^\circ = \mathbb{P}^1 \setminus \mu_4$  with  $H_1(X_f^\circ, \mathbb{Z}) = \mathbb{Z}^3$  and a rigid  $D_4 \subset PGL_2$  generated by  $z \mapsto iz$  and  $z \mapsto 1/z$ . The logarithmic forms  $\omega_i = dz/(z - p_i)$  regular at infinity span  $H^1$  by their differences, dual under the residue pairing to the puncture loops  $\gamma_i$  with  $\text{Res}(\omega_i, \gamma_j) = \delta_{ij}$ ; hence the canonical (residue) metric on the puncture classes is  $I_4$ , and the twelve puncture-difference cycles  $\gamma_i - \gamma_j$  are exactly the  $A_3$  roots:

$$\boxed{\{\gamma_i - \gamma_j\}_{i \neq j} : \# = 12 = |R(A_3)| = \dim \mathfrak{g}_{\text{SM}}, \quad \text{rank} = 3 = N_{\text{fam}}, \quad \text{Gram} = \text{Cartan}(A_3), \quad \det = 4} \quad [\mathbf{E}], \quad (94)$$

with  $\dim A_3 = 15 = \dim S^+ - 1$ , and the  $A_3$  positive roots equal the exponent sum  $1+2+3 = 6 = Z_6 = |E(K_4)|$ . *The corner rotation realises the  $A_3$  Coxeter element.* The carrier generator  $z \mapsto iz$  permutes  $\mu_4$  as the 4-cycle  $(p_1 p_2 p_3 p_4) \in S_4 = W(A_3)$  and acts on  $H^1 = \mathbb{C}^3$  as an isometry of the Cartan form with

$$\boxed{\text{ord} = 4 = h(A_3) = |\mu_4|, \quad \text{spec} = \{i, -1, -i\}, \quad \chi(t) = \frac{t^4 - 1}{t - 1} = \Phi_4(t)\Phi_2(t)} \quad [\mathbf{E}], \quad (95)$$

i.e. the eigenvalues are the  $A_3$  exponents  $\{1, 2, 3\}$  as fourth roots of unity — the exact analogue of the  $E_8$  Coxeter element reading  $\Phi_{30}$ . The full  $D_4 = \langle z \mapsto iz, z \mapsto 1/z \rangle$  embeds in  $W(A_3) = S_4$  (order 8, index 3) by isometries. *Stage 3 (standard, [E])*: assemble  $E_8$  from the  $D_5 \times A_3$  branching — no 240-map required.

**The  $\mu_4$  glue theorem (why the four corners appear everywhere).** The two factors have equal discriminant groups, and  $E_8$  is unimodular:

$$\boxed{\text{disc}(D_5) = \text{disc}(A_3) = \mathbb{Z}_4, \quad [E_8 : D_5 \oplus A_3] = \sqrt{\frac{\det D_5 \cdot \det A_3}{\det E_8}} = \sqrt{\frac{4 \cdot 4}{1}} = 4 = |\mu_4|} \quad [\text{E}]. \quad (96)$$

So  $E_8 = (D_5 \oplus A_3) + \mu_4\text{-glue}$ : the four family corners  $\mu_4 = \mathbb{Z}_4$  are precisely the glue code that fuses the carrier and the family into the unimodular  $E_8$  lattice. The glue pairs the  $D_5$  spinor class ( $= S^+$ , the family) with the  $A_3$  class — exactly the  $(16, 4) \oplus (\overline{16}, \overline{4})$  spinor block. Hence  $\mu_4$  plays three roles at once: family corners,  $A_3$  origin, and  $E_8$  glue.

**Discriminant-norm glue lemma (and the TFPT 5/4, 3/4 identification).** The two glue generators carry the discriminant quadratic form  $q \in \mathbb{Q}/2\mathbb{Z}$ :

$$\boxed{\begin{aligned} q(D_5) &= \frac{n}{4}|_{n=5} = \frac{5}{4} \text{ (spinor class)}, & q(A_3) &= \frac{n}{n+1}|_{n=3} = \frac{3}{4} \text{ (corner class)}, \\ q(D_5) + q(A_3) &= 2 \equiv 0 \pmod{2}, & q(D_5) - q(A_3) &= \frac{1}{2} \end{aligned}} \quad [\text{E}]. \quad (97)$$

The diagonal  $\langle (s, g) \rangle \cong \mathbb{Z}_4$  is isotropic ( $k^2 \cdot 2 \equiv 0$  for all  $k$ ), so the glue closes to an *even unimodular* lattice — this is  $E_8$ .

**The two glue norms are just  $(g_{\text{car}}, N_{\text{fam}})$  over the glue index.** Because  $\text{rank } D_5 = g_{\text{car}} = 5$  (so  $q(D_5) = \frac{n}{4}|_{n=5} = g_{\text{car}}/|\mu_4|$ ) and  $\text{rank } A_3 = N_{\text{fam}} = 3$  (so  $q(A_3) = \frac{n}{n+1}|_{n=3} = N_{\text{fam}}/|\mu_4|$ , with  $n+1 = 4 = |\mu_4|$ ),

$$\boxed{q(D_5) = \frac{g_{\text{car}}}{|\mu_4|} = \frac{5}{4}, \quad q(A_3) = \frac{N_{\text{fam}}}{|\mu_4|} = \frac{3}{4}}$$

and *all four arithmetic operations* on the pair are then forced compiler atoms:

$$\begin{aligned} \underbrace{q(D_5) + q(A_3) = \frac{g_{\text{car}} + N_{\text{fam}}}{|\mu_4|} = 2}_{E_8 \text{ root norm}}, & \quad \underbrace{q(D_5) - q(A_3) = \frac{|\mathbb{Z}_2|}{|\mu_4|} = \frac{1}{2}}_{\delta \text{ (lepton transport)}}, \\ \underbrace{q(D_5)q(A_3) = \frac{g_{\text{car}}N_{\text{fam}}}{|\mu_4|^2} = \frac{15}{16}}_{\dim \mathfrak{su}(4) / \dim S^+}, & \quad \underbrace{\frac{q(D_5)}{q(A_3)} = \frac{g_{\text{car}}}{N_{\text{fam}}} = \frac{5}{3}}. \end{aligned}$$

So the distinguished lepton transport value  $\delta = \frac{1}{2} = (g_{\text{car}} - N_{\text{fam}})/|\mu_4| = |\mathbb{Z}_2|/|\mu_4|$  is the carrier-minus-family count over the glue index — a fully atomic origin, not a chosen rational (it also equals the harmonic-frame holonomy diagonal modulus, [verification/v41\_leg\_assignment.py]); and (sum, ratio) invert uniquely back to  $(\frac{5}{4}, \frac{3}{4})$ . [verification/v51\_boundary\_half\_step.py] The sum  $\frac{5}{4} + \frac{3}{4} = 2$  is literally the  $E_8$  spinor-root norm  $\frac{1}{2}(\pm)^8$  split across  $D_5 \times D_3$ :  $5 \cdot \frac{1}{4} + 3 \cdot \frac{1}{4} = 2$ . Strikingly, both norms are pre-existing TFPT constants:

$$\boxed{\frac{\delta_2}{\delta_{\text{top}}^2} = \frac{2880 c_3^8}{(48 c_3^4)^2} = \frac{5}{4} = q(D_5), \quad \xi_{\text{tree}} = \frac{\dim \mathfrak{g}_{\text{SM}}}{\dim S^+} = \frac{12}{16} = \frac{3}{4} = q(A_3)} \quad [\text{E}], \quad (98)$$

so  $\delta_2/\delta_{\text{top}}^2 + \xi_{\text{tree}} = 2$ : the two-loop/defect ratio is the  $D_5$  spinor norm and the gravitational tree normalization is the  $A_3$  corner norm, and their sum is the  $E_8$  root norm.



**The whole integer skeleton from one pair**  $(g_{\text{car}}, N_{\text{fam}}) = (5, 3)$  **[E]**  
`([verification/v53_compiler_core.py])`

Every load-bearing integer of the compiler flows from the single boundary pair  $(g_{\text{car}}, N_{\text{fam}}) = (5, 3)$  by sum, difference and squares:

$$\text{rank } E_8 = g_{\text{car}} + N_{\text{fam}} = 8, \quad |\mathbb{Z}_2| = g_{\text{car}} - N_{\text{fam}} = 2, \quad |\mu_4| = \frac{g_{\text{car}} + N_{\text{fam}}}{2} = 4,$$

and  $(N_{\text{fam}}, |\mu_4|, g_{\text{car}}) = (3, 4, 5)$  is the Pythagorean triple. This is not decorative: the grand-mass-volume exponent  $\Delta_Y = g_{\text{car}}^2 = 25$  ( $\det M_{\text{SM}} \sim (\varphi_0^{\text{ret}})^{25}$ ; document 3) splits over the sectors as a *difference of squares*,

$$\Delta_Y = g_{\text{car}}^2 = \underbrace{N_{\text{fam}}^2}_{\text{down}=9} + \underbrace{(g_{\text{car}} - N_{\text{fam}})(g_{\text{car}} + N_{\text{fam}})}_{=|\mathbb{Z}_2| \cdot \text{rank } E_8} = 9 + 16 = \dim S^+$$

i.e. the down sector carries  $N_{\text{fam}}^2$  and up+lepton carries  $|\mathbb{Z}_2| \cdot \text{rank } E_8 = \dim S^+$  (the  $K$ -row sums  $(6, 9, 10)$ ). **The anchor is the Dirac square root of this skeleton:**  $a = (1, 1, 2)$  is the *unique* 3-multiset whose elementary symmetric functions are  $(|\mu_4|, g_{\text{car}}, |\mathbb{Z}_2|) = (4, 5, 2)$ , equivalently its characteristic polynomial is  $\chi_a(t) = (t-1)^2(t-2) = t^3 - |\mu_4|t^2 + g_{\text{car}}t - |\mathbb{Z}_2|$  — the compiler atoms *are* its char-poly coefficients. Sharper still (`[verification/v491_p2_partition_corollary.py]`): the partition needs only  $e_1 = 4$  — three *positive integers* summing to 4 already force  $\{1, 1, 2\}$ , so  $g_{\text{car}} = e_2 = 5$  and  $|\mathbb{Z}_2| = e_3 = 2$  are *corollaries* of the four marks given the weight typing (integrality = Birkhoff–Grothendieck splitting exponents, positivity =  $h^0=0$ , sum = par-deg-0 “given (U)”; every dropped assumption is negative-controlled, and **AX.P2.01** stays a declared axiom — the residual is the typing, not the integer 5. That typing is itself now hardened (`[verification/v499_p2_weights_deligne_bg.py]`, **P2.TYPING.01**): given the four marks, rank 3, the cusp class and (U), Deligne + Birkhoff–Grothendieck + Mehta–Seshadri stability force  $E = \mathcal{O}(-2) \oplus \mathcal{O}(-1)^2$  outright ( $h^0=0$  is the positivity), leaving only the module identification and (U) as **[C]** residuals — and the module-identification rest is now *one* residual with the QGEO modulus rest, both reducing to the same order-4 clock = Coxeter/cusp-class carrier datum (**v503**, **QGEO.EMERGE.LIGHT.01**; markers unchanged); the clock-rigidity theorems then reduce that common residual to one alignment bit, with the clock’s order fermionically forced,  $U^2 = (-1)^F$  exactly (`[verification/v506_seam_clock_rigidity.py]`, **SEAM.CLOCK.RIGIDITY.01**); and the bit-origin theorems derive the deck’s *class* (a half-period translation of the seam double) — only its position stays the **[C]** carrier input (`[verification/v507_seam_bit_origin.py]`, **SEAM.BIT.ORIGIN.01**); the freedom theorems then type that position half as *topology* (the covering deck is free on the seam circle, the edge class is excluded), so the residual is the square-modulus datum  $\tau = i$  alone (`[verification/v510_seam_bit_freedom.py]`, **SEAM.BIT.FREEDOM.01**); and the flag-transitivity equivalence web restates that residual in discrete form — flag transitivity of the four marks ( $V_4 \rightarrow D_4$ )  $\iff \tau = i$ , with bare mark-transitivity automatic for every configuration and no local jet seeing the side bit (`[verification/v512_seam_tau_flag.py]`, **SEAM.TAU.FLAG.01**) — a residual that has since survived ten side-blind derivation attacks (**v521/v525/v529**) and carries a physical definition as the twist-class choice with a gauge-robust order parameter, in principle interferometrically readable (**v528**; it remains formal input) — and, since **v534**, the first *dynamical* selection datum: reflection positivity on the interacting mark family keeps exactly the  $\delta = \pi/2$  member alive (toy-fenced). So the theory’s only transcendental is  $\pi$  (through  $c_3 = \frac{1}{8\pi}$ ,  $8 = \text{rank } E_8$ , and the seed  $\frac{1}{6\pi}$ ,  $6 = |R^+(A_3)|$ ); everything else is this  $(5, 3)$ -generated integer skeleton.

### Boundary Carrier Selection Theorem (Theorem A) **[E]/[C]**

$E_8$  is not assumed: it is *selected* as the unique even-unimodular audit hull of the admissible boundary-carrier data. **Boundary side:**  $\mu_4 = \{1, i, -1, -i\}$  gives  $X_f = \mathbb{P}^1 \setminus \mu_4$  with  $H_1 = \mathbb{Z}^3$ , hence  $A_3$  and  $N_{\text{fam}} = 3$ . **Carrier side:** a minimal  $D$ -type carrier with a 16-dim half-spinor forces  $D_5$  (only  $n = 5$  gives  $2^{n-1} = 16$ ;  $S^+ = \Lambda^{\text{even}}\mathbb{C}^5 = 1+10+5$ ). **Glue:**  $\text{disc}(D_5) = \text{disc}(A_3) = \mathbb{Z}_4$  (isomorphic, cyclic, anti-isometric),  $q(D_5) + q(A_3) = 2$  (the  $E_8$  root norm), glue index  $4 = |\mu_4|$ , and  $E_8$  is the unique even unimodular rank-8 lattice. **Exclusion:** the four conditions

$$(i) \dim S^+ = 16, \quad (ii) N_{\text{fam}} = 3, \quad (iii) \text{cyclic } \mathbb{Z}_4 \text{ glue}, \quad (iv) q_D + q_A = 2$$

are met *only* by  $D_5 \oplus A_3$ ;  $D_8$  (non-cyclic  $\mathbb{Z}_2 \times \mathbb{Z}_2$ , no family geometry),  $E_7 \oplus A_1$  (1 family, no 16-spinor),  $E_6 \oplus A_2$  (2 families,  $\mathbb{Z}_3$ ),  $A_4 \oplus A_4$  (no  $D$ -spinor,  $\mathbb{Z}_5$ ) and  $A_8$  (single factor) each fail at least one. **Typing:** the lattice/classification facts are **[E]**; the physical selection (reflection-positive seam  $c_3 = 1/8\pi +$

minimal chiral carrier  $\Rightarrow$  exactly these inputs) is the **[C]/[O]** selection axiom (P1 analytic interface + P2 carrier interface), not a zero-input derivation. `[verification/v47_selection_theorem.py]`

**Group-level closure ( $\mu_4$  as the common centre).** The lattice glue lifts to a clean group statement, because both factors share the *same* centre and  $\mu_4$  *is* the diagonal:

$$Z(\text{Spin}(10)) = \mathbb{Z}_4, \quad Z(SU(4)) = \mathbb{Z}_4, \quad |\mu_4| = 4, \quad G_{\text{closure}} = \frac{\text{Spin}(10) \times SU(4)}{\Delta \mathbb{Z}_4} \subset E_8 \quad [\text{E}], \quad (99)$$

with  $\text{rank} = 5 + 3 = 8 = \text{rank } E_8$ . Every adjoint block is diagonal- $\mathbb{Z}_4$  neutral (generator  $\omega = i$ ; charges mod 4):  $(45, 1), (1, 15) : 0$ ;  $(10, 6) : 2+2=0$ ;  $(16, 4) : 1+3=0$ ;  $(\overline{16}, \overline{4}) : 3+1=0$ , so  $45+15+60+64+64 = 248$  are genuine reps of the quotient. This promotes  $\mu_4$  from a lattice glue index to the *common centre quotient* — the physically cleanest language for the gluing: the  $D_5$  half-spinor 16 pairs with the  $SU(4)$  fundamental 4 so the spinor charges  $\pm 1$  cancel the fundamental charges  $\mp 1$ .

**Root-length completion.** Every  $E_8$  block reaches norm 2:

| block                                          | norm split                      | reading                                 |
|------------------------------------------------|---------------------------------|-----------------------------------------|
| $(45, 1), (1, 15)$                             | 2, 2                            | $D_5, A_3$ root adjoints                |
| $(10, 6)$                                      | $1 + 1 = 2$                     | $SO(10)$ vector $\oplus$ $SO(6)$ vector |
| $(16, 4) \oplus (\overline{16}, \overline{4})$ | $\frac{5}{4} + \frac{3}{4} = 2$ | $D_5$ spinor norm + $A_3$ corner norm   |

The old question “can the 240 flip atlas be metrized to  $E_8$ ?” is now the sharp question “are the TFPT  $D_5$  and  $A_3$  metrics in standard normalization?” — and the  $5/4, 3/4$  match says yes.

**Explicit construction (the lift is now built, not counted).** Embedding  $D_5 \oplus A_3 = D_5 \oplus D_3 \subset D_8$  and adjoining the even spinor coset (the  $\mu_4$  glue), the 240 roots are produced explicitly and verified:

$$\{\pm e_i \pm e_j\}_{i < j}^8 (112) \cup \{\frac{1}{2}(\pm)^8 : \text{even}\} (128) = 240 \text{ roots, all norm 2, Cartan det} = 1 \quad [\text{E}]. \quad (100)$$

Splitting each root into its  $D_5$  (first five) and  $A_3$  (last three) parts gives exactly the branching blocks —  $(2, 0) : 40$ ,  $(0, 2) : 12$ ,  $(1, 1) : 60$ , and  $(\frac{5}{4}, \frac{3}{4}) : 128$  — and the 128 spinor roots realise  $\frac{5}{4} + \frac{3}{4} = 2$  on the nose. All 240 vectors are integer combinations of one set of 8 simple roots, so the assembled lattice *is*  $E_8$ . The blind 240-metrization is therefore eliminated:  $E_8$  is explicitly  $(D_5 \oplus A_3) + \mu_4\text{-glue}$ .

**Remark** (the metric, computed). This reframes  $E_8$  from origin to closure:  $E_8$  is the closure algebra of the  $D_5$  carrier and the  $A_3$  family geometry, glued by  $\mu_4$ . The root system, Cartan form,  $W(A_3) = S_4$  and the corner-rotation =  $A_3$  Coxeter element are exact (residue pairing + explicit carrier symmetry). The transcendental geometric metric  $G_{ij} = -\log |p_i - p_j|$  on  $\mu_4$  has been computed: it is only  $D_4$ -symmetric, with a regularisation-independent anisotropy  $-\log 2$  on the root space, hence *not* proportional to  $\text{Cartan}(A_3)$ . It need not be: the  $E_8$ -relevant  $A_3$  is the canonical *residue* pairing, and the geometric metric is a  $D_4$ -equivariant deformation that preserves the root system, Weyl group and Coxeter element. So the lattice data is exact **[E]**; the analytic metric is a *characterised* deformation, not an open gap.

**Uniform Coxeter law.** All three factors obey  $|R| = h \cdot \text{rank}$ , and the two TFPT Coxeter numbers are themselves carrier data:

$$\begin{aligned} h(D_5) &= 8 = r(E_8) = \varphi(30), & h(A_3) &= 4 = |\mu_4|, \\ h(D_5) h(A_3) &= 2^{g_{\text{car}}} = 32, & 2^{g_{\text{car}}} - 2 &= 30 = h(E_8) \end{aligned} \quad [\text{E}], \quad (101)$$

so the Coxeter closure of Theorem 3 reappears as  $h(E_8) = h(D_5)h(A_3) - 2$ , and  $h(D_5) + h(A_3) = 12 = \dim \mathfrak{g}_{\text{SM}} = |R(A_3)|$ .

## Theorem 8 — the $(1, 1, 2)$ anchor microcode and the $\varphi_0$ -ladder

The parabolic flavor bundle  $E_\bullet = \mathcal{O}(-2) \oplus \mathcal{O}(-1)^2$  on  $\mathbb{P}^1 \setminus \mu_4$  has splitting type  $a = (1, 1, 2)$ , whose power sums  $p_n(a) = 2 + 2^n$  generate the whole compiler budget:

**The anchor power compiler.**

$$(p_1, \dots, p_5) = (4, 6, 10, 18, 34) = (|\mu_4|, |R^+(A_3)|, \mathcal{A}_\Lambda, N_{\text{fam}}|R^+(A_3)|, \mathbb{Z}_6 \text{ lift}),$$

$$q(A_3) = \frac{p_2}{2p_1} = \frac{3}{4}, \quad q(D_5) = \frac{p_3}{2p_1} = \frac{5}{4}, \quad \sum L = p_1 p_3 = 40,$$

$$\Omega_{\text{adm}} = 2p_1 p_2 = 48, \quad D_{\text{start}} = p_2 p_3 = 60, \quad |R(E_8)| = 4p_2 p_3 = 240.$$

The elementary symmetric data  $\chi_a(t) = (t-1)^2(t-2)$  give  $(e_1, e_2, e_3) = (4, 5, 2) = (|\mu_4|, g_{\text{car}}, |\mathbb{Z}_2|)$ , so the anchor reconstructs the carrier rank, and  $10b_1 = e_1^2 + e_2^2 = 4^2 + 5^2 = 41$ . The seed itself is the two-term anchor form  $\varphi_0^{\text{ret}} = \frac{2p_1}{p_2} c_3 + 2p_1 p_2 c_3^4 = \frac{4}{3} c_3 + 48c_3^4$ .

**The  $\varphi_0$ -ladder.** With  $\lambda_Y = \sqrt{\varphi_0^{\text{ret}}(1 - \varphi_0^{\text{ret}})}$ , every fermion mass is the single closed form  $\hat{m}_{f,j} = \frac{v_{\text{geo}}}{\sqrt{2}} \lambda_Y^{L_{f,j}} \Lambda_{f,j}$  with  $\lambda_Y^L \Lambda = \pi c (\varphi_0^{\text{ret}})^k$ , so all nine holonomies are resolvent outputs and the masses form the  $\varphi_0^{\text{ret}}$ -order ladder  $t(0) > \{b, c, \tau\}(2) > \{s, \mu\}(3) > \{u, d\}(4) > e(5)$ .

**Secondary  $E_8$  atlas.** Beyond the main  $D_5 \times A_3$  split, the residue matrix appears in the  $E_6 \times A_2$  branching,  $248 = \|R\|_F^2 + \det R + 2(\mathbf{1}^\top R a) N_{\text{fam}} = 78 + 8 + 2 \cdot 27 \cdot 3$ , and the scalaron exponent in  $E_7 \times A_1$ ,  $112 = 7 \cdot 16$ . The seven  $E_8$  slices are charted in the companion note `e8_secondary_branching_atlas` as a falsifiable audit raster.

**The  $E_6 \times A_2$  block is the  $\tau=\omega$  face** (`[verification/v407_dn_pairings_omega.py]`, `SEAM.EQUIV.02`).  $A_2$  is the Eisenstein lattice of the family CM point  $\tau=\omega$  (Cartan  $\det = 3 = N_{\text{fam}}$ ), and the  $(d, n)$  selector's three pairings  $\{n \cdot \mathbf{1}, n \cdot a, n \cdot \sigma\} = \{2, 8, 121\}$  are exactly its family-slice atoms  $\{|Z_2|, \dim A_2 = \det R = \text{rank } E_8, \|Pl(K)\|_1^2\}$ , so deriving  $(d, n)$  (the  $R4'$  residue, `v139`) folds into the dual keystone (`v405`), leaving only “why  $11 = \|Pl(K)\|_1$ ”.

## Master statement and status

Five carrier slots give  $D_5$ , four family corners give  $A_3$ , and  $D_5 \times A_3$  gives the  $E_8$  atlas.

Equivalently:  $\mathcal{A}_\Lambda = \frac{h}{N_{\text{fam}}} = 10 = \binom{5}{2}$ ,  $h = 30 = 2 \cdot 3 \cdot 5$ ,  $(5, 3, 2) : \sum = 10, \prod = 30$ ;  $\Lambda$ ,  $\det M_\ell$ , the  $E_8$  Coxeter structure and the action ladder read the same five-slot pair-decuple. The Standard model, flavor,  $\Lambda$ , inflation and  $E_8$  are projections of the even carrier code  $C^+$  ( $= D_5$  half-spinor) on five slots together with the four-point family  $A_3$ .

| Statement                                                                                                    | Status  | Note                                   |
|--------------------------------------------------------------------------------------------------------------|---------|----------------------------------------|
| Even code $C^+$ , $\dim S^+ = 16$ , dual-root $Y$                                                            | [E]     | Theorem 1                              |
| Exterior–Action duality $(1, 10, 5) \leftrightarrow (1, 5, 10)$                                              | [E]     | scheme eigenvalues verified            |
| $R_4 = \text{Clebsch } (16, 5, 0, 2)$ , $R_2 = (16, 10, 6, 6)$ ; $\text{Aut}_{\text{aff}} = W(D_5) = 1920$   | [E]     | carrier = $D_5$ half-spinor            |
| $2^g - 2 = g(g+1) \Leftrightarrow g = 5$                                                                     | [E]     | Coxeter closure (unique)               |
| $h=30=2\cdot 3\cdot 5$ , $r=\varphi(30)=8$ , $\chi_C=\Phi_{30}$                                              | [E]/[C] | cyclotomic compiler                    |
| $ R^+  = \binom{16}{2} = \text{Tr } X^2 = 4h = 120$                                                          | [E]     | positive-edge moment                   |
| $40, 60, 120, 240 = 10 \cdot (4, 6, 12, 24)$ ; $41=1+4\cdot 10$                                              | [E]     | $K_5 \times K_4$ product table         |
| $10b_1=5r+1$ , $u$ totient, $M_{\text{scal}}^2=c_3^{r-1}$ , $\omega_b=\lambda_C/10$                          | [E]/[C] | downstream readouts                    |
| $A_s = N_*^2 c_3^7 / 24\pi^2$                                                                                | [C]/[O] | if $M_{\text{scal}} = M_{\text{Star}}$ |
| $\mathfrak{e}_8 = (45, 1) \oplus (1, 15) \oplus (10, 6) \oplus (16, 4) \oplus (\overline{16}, \overline{4})$ | [E]/[C] | $D_5 \times A_3$ lift                  |
| 240 roots built from $D_5 \oplus A_3 + \mu_4$ glue, $\det 1$                                                 | [E]     | explicit $E_8$ construction            |
| flavor $\sum L=40= R(D_5) $ , entries on $C_6= R^+(A_3) $                                                    | [E]     | transport = $D_5 \times A_3$           |
| $A_3$ from $\mathbb{P}^1 \setminus \mu_4$ : residue pairing = Cartan, $\det 4$                               | [E]     | Stage 2 (Theorem 7)                    |
| corner $z \mapsto iz = A_3$ Coxeter el.: $\{i, -1, -i\}$ , $\Phi_4 \Phi_2$                                   | [E]     | order 4 = $h(A_3) =  \mu_4 $           |
| $\text{disc}(D_5) = \text{disc}(A_3) = \mathbb{Z}_4$ , glue = $\mu_4$                                        | [E]     | $E_8 = (D_5 \oplus A_3) + \mu_4$       |
| $q(D_5)=\frac{5}{4}=\delta_2/\delta_{\text{top}}^2$ , $q(A_3)=\frac{3}{4}=\xi_{\text{tree}}$ , sum 2         | [E]     | discriminant-norm glue                 |
| $120 = \dim \mathfrak{so}_{16}$ ; $\Omega_{\text{adm}}=40+1+7$                                               | [E]     | adjoint half; deficit ladder           |
| $h(D_5)=8=r(E_8)$ , $h(A_3)=4= \mu_4 $ , $ R =h \cdot r$                                                     | [E]     | uniform Coxeter law                    |
| $\text{Exp}(E_8)$ as 41/48 budget deficits                                                                   | [O]     | exponent ledger (fingerprint)          |
| geometric metric $D_4$ -deform (anisotropy $-\log 2$ ), residue = $A_3$                                      | [E]     | metric computed/characterised          |

*Scope.* This note compresses; it does not re-derive the upstream boundary kernel (the P1 hardening chain) or the carrier rigidity theorem (Lean), and it makes no claim of a final theory. The arithmetic and the  $D_5/\text{Clebsch}/W(D_5)$ ,  $A_3/\text{residue-Cartan}/\text{Coxeter}$ , and  $\mu_4$ -glue identifications are exact, and the  $E_8$  lattice is now built explicitly from  $D_5 \oplus A_3$  plus the  $\mu_4$  glue (240 roots,  $\det 1$ ). The only residual [C] is the analytic identification of the carrier geometry's  $L^2$  cusp Hodge metric with the canonical residue metric up to scale, which does not affect the explicit root/lattice/Weyl construction. Honest non-fits (Koide  $0.6645 \neq 2/3$ ,  $\eta_B$ ,  $m_p/m_e$ ) stay outside the compiler.

## Appendix: machine-checkable $E_8 = (D_5 \oplus A_3) + \mu_4$ construction

The lift is reproducible by a short script (all entries below verified exactly).

### Cartan matrices and discriminant groups.

$$C(D_5) = \begin{pmatrix} 2 & -1 & 0 & 0 & 0 \\ -1 & 2 & -1 & 0 & 0 \\ 0 & -1 & 2 & -1 & -1 \\ 0 & 0 & -1 & 2 & 0 \\ 0 & 0 & -1 & 0 & 2 \end{pmatrix}, \quad C(A_3) = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix}; \quad \det C(D_5) = \det C(A_3) = 4.$$

Smith normal forms give  $\text{disc}(D_5) = \text{disc}(A_3) = \mathbb{Z}_4$ ; the glue index is  $[E_8 : D_5 \oplus A_3] = \sqrt{4 \cdot 4/1} = 4 = |\mu_4|$ , with discriminant-form norms  $q(D_5) = \frac{5}{4}$ ,  $q(A_3) = \frac{3}{4}$ ,  $q(D_5)+q(A_3) = 2 \equiv 0 \pmod{2}$  (even-glue). Writing  $\mu := |\mu_4| = 4$  the norms are  $q(D_5) = \frac{\mu+1}{\mu}$ ,  $q(A_3) = \frac{\mu-1}{\mu}$ , so  $q(D_5)q(A_3) = 1 - \frac{1}{\mu^2} = \frac{15}{16}$  and

$$\dim S^+(1 - q(D_5)q(A_3)) = 16 \cdot \frac{1}{16} = 1 \quad [\text{E}], \quad (102)$$

i.e. the glue product fills all  $15 = \dim S^+ - 1$  non-trivial half-spin states and the defect is exactly the singleton (the  $\nu^c/\text{Higgs}$  closure). And  $E_8$  has a tidy  $\mu_4$ -cardinal form,  $\dim E_8 = \binom{|\mu_4|^2}{2} + 2|\mu_4|^3 = \binom{16}{2} + 2 \cdot 64 = 120 + 128 = 248$  [C].

**Explicit roots.** Split  $\mathbb{R}^8 = \mathbb{R}_{(D_5)}^5 \oplus \mathbb{R}_{(A_3)}^3$ . The 240 roots are

$$\{\pm e_i \pm e_j : 1 \leq i < j \leq 8\} \text{ (112)} \cup \left\{ \frac{1}{2}(\pm 1)^8 : \text{even } \# \text{ of } - \right\} \text{ (128),}$$

all of norm 2. Splitting each by  $(\|\cdot\|_{D_5}^2, \|\cdot\|_{A_3}^2)$  gives the branching blocks

$$(2, 0): 40 [D_5], \quad (0, 2): 12 [A_3], \quad (1, 1): 60 [(10, 6)], \quad \left(\frac{5}{4}, \frac{3}{4}\right): 128 [(16, 4) \oplus (\overline{16}, \overline{4})],$$

so  $\mathfrak{e}_8 = (45, 1) \oplus (1, 15) \oplus (10, 6) \oplus (16, 4) \oplus (\overline{16}, \overline{4})$ ,  $248 = 45+15+60+64+64$ ,  $240 = 40+12+60+64+64$ .

**Simple-root Gram (unimodular check).** With the Bourbaki simple roots  $\alpha_1 = \frac{1}{2}(e_1 - e_2 - \dots - e_7 + e_8)$ ,  $\alpha_2 = e_1 + e_2$ , and  $\alpha_k = e_{k-1} - e_{k-2}$  for  $k = 3, \dots, 8$  (so  $\alpha_3 = e_2 - e_1$ ,  $\alpha_4 = e_3 - e_2$ ,  $\dots$ ,  $\alpha_8 = e_7 - e_6$ ), the Gram matrix  $G_{ij} = \langle \alpha_i, \alpha_j \rangle$  is the  $E_8$  Cartan matrix with  $\det G = 1$  and all  $G_{ii} = 2$ ; every one of the 240 roots is an integer combination of the  $\alpha_i$ . Hence the assembled lattice is  $E_8$ , not merely a 240-count. [E] (The shifted index  $e_{k-2} - e_{k-3}$  would duplicate  $\alpha_3$  and make  $G$  singular — use  $e_{k-1} - e_{k-2}$ .)



## Part III

## Appendix: ablation of the EM fixed point

The single sharpest reviewer test is: *was the closure  $F_{U(1)}(\alpha) = 0$  engineered so that  $\alpha$  comes out?* The honest answer is an ablation — vary each input and see which actually move the root. Three one-parameter scans (all others held at their compiler values) separate the *load-bearing* integers from the *fine* last-digit selector.

EM fixed point: only  $M = 41$  and  $N = 8$  land on  $\alpha^{-1}$

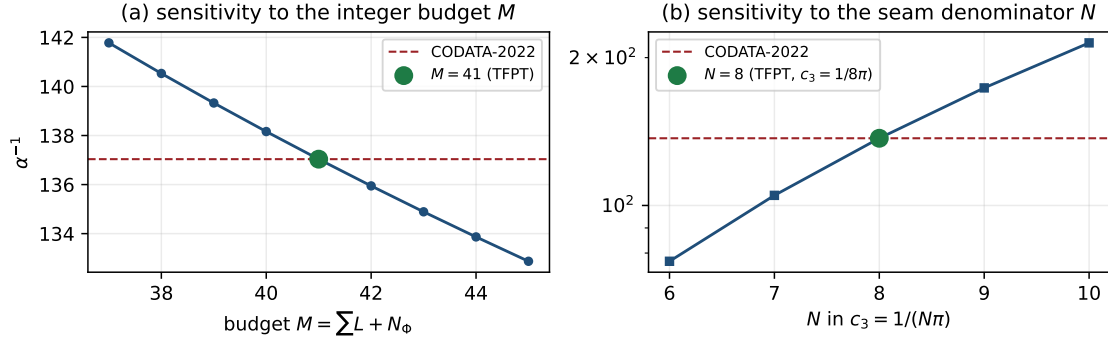


Figure 7: EM-closure ablation (data plot, `verification/make_figures.py`). **(a)**  $\alpha_\star^{-1}$  vs the integer budget  $M$ : only  $M = 41$  lands on CODATA-2022. **(b)**  $\alpha_\star^{-1}$  vs the seam denominator  $N$  in  $c_3 = 1/(N\pi)$ : only  $N = 8$  lands. Both integers are fixed elsewhere (flavor budget; rank/ $h(D_5)$ ), so the equation is not freely tunable to 137.036.

**(A1) The transport budget  $M = \sum L + N_\Phi$**  (exponent held at  $-\frac{5}{4}$ ):

| $M$                 | 38      | 39      | 40      | 41                 | 42      | 43      | 44      |
|---------------------|---------|---------|---------|--------------------|---------|---------|---------|
| $\alpha_\star^{-1}$ | 140.530 | 139.326 | 138.162 | <b>137.0359992</b> | 135.946 | 134.890 | 133.866 |

Each unit of  $M$  shifts  $\alpha_\star^{-1}$  by  $\approx 1.09$ , so *only*  $M = 41$  lands on 137.036. This is a hard lock — and  $41 = 10b_1 = \sum L + N_\Phi = 40 + 1$  is fixed by the compiler (flavor budget 40 + one Higgs). **Load-bearing.**

**(A2) The seam “8” in  $c_3 = 1/(N\pi)$**  (here  $\delta_{\text{top}} = 48c_3^4$  tracks  $c_3$ ):

| $N$                 | 6     | 7      | 8              | 9      | 10     |
|---------------------|-------|--------|----------------|--------|--------|
| $\alpha_\star^{-1}$ | 76.97 | 104.85 | <b>137.036</b> | 173.53 | 214.34 |

Only  $N = 8$  — i.e.  $c_3 = 1/(8\pi) = 1/(\text{rank } E_8 \cdot \pi)$  — reproduces 137.036. The “8” is the same one fixed by the bootstrap ( $h(D_5) = \text{rank } E_8 = \varphi(30)$ ) and the Gauss–Bonnet hardening. **Load-bearing.**

**(A3) The carrier discriminant exponent** (at  $M = 41$ ,  $c_3 = 1/(8\pi)$ ):

| exponent            | $-\frac{1}{4}$ | $-\frac{3}{4}$ | $-\frac{5}{4}$     | $-\frac{7}{4}$ | $-\frac{9}{4}$ |
|---------------------|----------------|----------------|--------------------|----------------|----------------|
| $\alpha_\star^{-1}$ | 137.035995     | 137.035997     | <b>137.0359992</b> | 137.036001     | 137.036003     |

The exponent moves only the *seventh* digit; all candidates agree to six. Here  $-\frac{5}{4} = q(D_5)$  is selected, but this is a *fine* selector, not a hard lock. **Fine (last-digit).**

**Ablation verdict — honest**

The EM fixed point is a **hard test of two integers** — the budget  $M = 41$  and the seam denominator  $8 = \text{rank } E_8$ , each decisive (a single-unit change misses by  $\sim 1$  in  $\alpha_\star^{-1}$ ) — and a **fine test of one norm**, the carrier discriminant exponent  $-\frac{5}{4} = q(D_5)$  (seventh digit). The equation is therefore *not* freely tunable to 137.036: the two integers that hit it are exactly the compiler/bootstrap numbers, fixed elsewhere. What it does *not* do is pin the carrier norm strongly — that is honest, and it is the right level of claim. [E]

**Appendix: computational verification**

The load-bearing identities of this document (marked [E], [E], [E]) are re-derived from  $\{c_3, g_{\text{car}}\}$  alone and machine-checked by the suite in the repository folder `verification/`; the inline tags `[verification/...]` point to the exact script. Relevant here:

- `v1_e8_glue.py` — the  $\mu_4$  glue ( $q(D_5)+q(A_3)=2$ ,  $\text{disc}=\mathbb{Z}_4$ , 240, 248).
- `v2_carrier_pascal.py` —  $g_{\text{car}}=5$ ,  $16=1+5+10$ ,  $N_{\text{fam}}, \Omega_{\text{adm}}, b_1$ .
- `v3_em_alpha.py` — the EM fixed point  $\alpha_\star^{-1}=137.0359992168$  (existence/uniqueness, ablation).
- `v4_flavor_matrix.py` —  $R$  with  $\det=8$ , minors  $\{2, 3, 5\}$ ,  $\sum L=40$ .
- `v7_gravity_cosmo.py` — scalaron  $c_3^7$ ,  $\Lambda/A_s$  readouts.

Run `python verification/run_all.py` (needs `mpmath`, `numpy`, `sympy`); it exits 0 iff all checks pass. Full map: `verification/README.md`.

**External works built upon**

The TFPT discrete core (the two axioms, the  $E_8$  glue, the integer skeleton) is self-contained and machine-checked. Its *bridges* to continuum/AQFT physics, to gravity and to the lattice/singularity picture build on established mathematics, listed here by topic; the body refers to these by author name.

**Modular theory and algebraic quantum field theory.**

- M. Takesaki, *Tomita's Theory of Modular Hilbert Algebras and its Applications*, Lecture Notes in Math. **128**, Springer (1970).
- J. J. Bisognano, E. H. Wichmann, *On the duality condition for a Hermitian scalar field / for quantum fields*, J. Math. Phys. **16** (1975) 985–1007; **17** (1976) 303–321.
- K. Osterwalder, R. Schrader, *Axioms for Euclidean Green's functions* I, II, Comm. Math. Phys. **31** (1973) 83–112; **42** (1975) 281–305.
- S. Doplicher, R. Haag, J. E. Roberts, *Local observables and particle statistics* I, II, Comm. Math. Phys. **23** (1971) 199–230; **35** (1974) 49–85.
- R. Haag, *Quantum field theories with composite particles and asymptotic conditions*, Phys. Rev. **112** (1958) 669–673; D. Ruelle, *On the asymptotic condition in quantum field theory*, Helv. Phys. Acta **35** (1962) 147–163.
- R. Longo, *Index of subfactors and statistics of quantum fields* I, II, Comm. Math. Phys. **126** (1989) 217–247; **130** (1990) 285–309.
- Y. Kawahigashi, R. Longo, M. Müger, *Multi-interval subfactors and modularity of representations in conformal field theory*, Comm. Math. Phys. **219** (2001) 631–669.

**Conformal nets, vertex operator algebras and lattice constructions.**

- I. B. Frenkel, V. G. Kac, *Basic representations of affine Lie algebras and dual resonance models*, Invent. Math. **62** (1980) 23–66; G. Segal, *Unitary representations of some infinite-dimensional groups*, Comm. Math. Phys. **80** (1981) 301–342.
- T. J. Osborne, A. Stottmeister, *Conformal field theory from lattice fermions*, Comm. Math. Phys. **398** (2023) 219–289, [arXiv:2107.13834](https://arxiv.org/abs/2107.13834), [doi:10.1007/s00220-022-04521-8](https://doi.org/10.1007/s00220-022-04521-8). (This is the seam scaling-limit theorem the TFPT continuum leg cites — the Koo–Saleur  $\rightarrow$  Virasoro convergence,  $c = \frac{1}{2}$ , and the smeared WZW-current Theorem 5.1. The internal TFPT code-token “MMST” and the legacy claim-ids `SEAM.MMST.*` / filenames `v*_mmst_*` denote *this* Osborne–Stottmeister theorem.)

- V. Morinelli, G. Morsella, A. Stottmeister, Y. Tanimoto, *Scaling limits of lattice quantum fields by wavelets*, Comm. Math. Phys. **387** (2021) 299–360, [arXiv:2010.11121](#), [doi:10.1007/s00220-021-04152-5](#); and *Operator-algebraic renormalization and wavelets*, Phys. Rev. Lett. **127** (2021) 230601. (The operator-algebraic (wavelet/Wilson–Kadanoff) renormalization *framework* the Osborne–Stottmeister theorem above builds on.)
- M. S. Adamo, Y. Moriwaki, Y. Tanimoto, *Osterwalder–Schrader axioms for unitary full vertex operator algebras*, [arXiv:2407.18222](#) (2024).
- M. S. Adamo, L. Giorgetti, Y. Tanimoto, *Rational and non-rational two-dimensional conformal field theories arising from lattices*, [arXiv:2506.01008](#) (2025); *Representations of conformal nets associated with infinite-dimensional groups*, [arXiv:2508.07109](#) (2025).
- P. Naaijken, D. Penneys, B. Wallick, *(commuting-projector reflection positivity / LTO-RP)*, [arXiv:2605.10693](#).

### Lattices and quadratic forms.

- J. H. Conway, N. J. A. Sloane, *Sphere Packings, Lattices and Groups*, 3rd ed., Grundlehren der math. Wiss. **290**, Springer (1999).
- C. L. Siegel, *Über die analytische Theorie der quadratischen Formen* (the Minkowski–Siegel mass formula), Ann. of Math. **36** (1935) 527–606.

### Singularities and geometry.

- E. Brieskorn, *Beispiele zur Differentialtopologie von Singularitäten*, Invent. Math. **2** (1966) 1–14.
- J. Milnor, *Singular Points of Complex Hypersurfaces*, Ann. of Math. Studies **61**, Princeton Univ. Press (1968).

### Spectral geometry, gravity and thermal time.

- A. H. Chamseddine, A. Connes, *The spectral action principle*, Comm. Math. Phys. **186** (1997) 731–750.
- A. Connes, C. Rovelli, *Von Neumann algebra automorphisms and time–thermodynamics relation in general covariant quantum theories*, Class. Quantum Grav. **11** (1994) 2899–2918.
- R. M. Wald, *Black hole entropy is the Noether charge*, Phys. Rev. D **48** (1993) R3427–R3431.
- H. Casini, M. Huerta, R. C. Myers, *Towards a derivation of holographic entanglement entropy*, JHEP **05** (2011) 036, [arXiv:1102.0440](#).